

Leopard 2

The **Leopard 2** is a third generation German main battle tank (MBT). Developed by Krauss-Maffei in the 1970s, the tank entered service in 1979 and replaced the earlier Leopard 1 as the main battle tank of the West German army. Various iterations of the Leopard 2 continue to be operated by the armed forces of Germany, as well as 13 other European countries, and several non-European countries, including Canada, Chile, Indonesia, and Singapore. Some operating countries have licensed the Leopard 2 design for local production and domestic development.

There are two main development tranches of the Leopard 2. The first encompasses tanks produced up to the Leopard 2A4 standard and are characterised by their vertically faced turret armour. The second tranche, from Leopard 2A5 onwards, has an angled, arrow-shaped, turret appliqué armour, together with other improvements. The main armament of all Leopard 2 tanks is a smoothbore 120 mm cannon made by Rheinmetall. This is operated with a digital fire control system, laser rangefinder, and advanced night vision and sighting equipment. The tank is powered by a V12 twin-turbo diesel engine made by MTU Friedrichshafen.

In the 1990s, the Leopard 2 was used by the German Army on peacekeeping operations in Kosovo. In the 2000s, Dutch, Danish and Canadian forces deployed their Leopard 2 tanks in the War in Afghanistan as part of their contribution to the International Security Assistance Force. In the 2010s, Turkish Leopard 2 tanks saw action in Syria. In 2023, Ukrainian Leopard 2 tanks saw action in the Russo-Ukrainian War.^[10]

History

Development

Even as the Leopard 1 was just entering service, the West German military was interested in producing an improved tank in the next decade. This resulted in the start of the MBT-70 development in cooperation with the United States beginning in 1963.^[11] However

Leopard 2	
 A Leopard 2A7 of the German <i>Bundeswehr</i> in 2015	
Type	Main battle tank
Place of origin	 West Germany
Service history	
In service	1979–present ^[1]
Used by	See <i>Operators</i>
Wars	War in Afghanistan Syrian Civil War Russo-Ukrainian War
Production history	
Designer	 Krauss-Maffei
Designed	1970s
Manufacturer	 KraussMaffei Maschinenbau Kiel Rheinmetall ^[2] Santa Bárbara Sistemas Hägglunds AB Contraves Hellenic Vehicle Industry
Unit cost	2A6 (secondhand): US\$5.74 million (FY 2007) ^[3] 2A7+: €13–15 million (FY 2019) ^[4]
Produced	1979–present

already in 1967 it became questionable whether the MBT-70 would enter service at any time in the foreseeable future. Therefore, the German government issued the order to research future upgrade options for the Leopard 1 to the German company Porsche in 1967.^[12]

This study was named *vergoldeter Leopard* (Gilded Leopard) and focused on incorporating advanced technology into the Leopard design. The projected upgrades added an autoloader, a coaxial autocannon and an independent commander's periscope.^[13] The anti-air machine gun could be operated from inside the vehicle and a TV surveillance camera was mounted on an extendable mast. The shape of the turret and hull was optimised using cast steel armour, while the suspension, transmission, and engine exhaust vents were improved.^[14]

Prototype development

Following the end of the Gilded Leopard study in 1967, the West German government decided to focus on the *Experimentalentwicklung* (experimental development) in a feasibility study and to develop new components for upgrading the Leopard 1 and for use on a future main battle tank programme.^[13] At first 25 million DM were invested, but after the industry came to the conclusion that with such a low budget the development of the two projected testbeds was not possible, a total of 30 to 32 million DM was invested. The experimental development was contracted to the company Krauss-Maffei, but with the obligation to cooperate with Porsche for the development of the chassis and with Wegmann for the development of the turret.^[15]

Two prototypes with different components were built with the aim of improving the conception of Leopard 1 in such a way that it would match the firepower requirements of the MBT-70. A high first-hit probability at ranges of 2,000 metres (6,600 ft) and the ability to accurately engage targets on the move using a computerised fire control system were the main goals of the experimental development. The resulting vehicles were nicknamed *Keiler* ("tusker"). Two prototypes (ET 01 and ET 02) of the Keiler were built in 1969 and 1970, both of them being powered by the MB 872 engine.^[15]

The MBT-70 was a revolutionary design, but after large cost overruns and technological problems, Germany withdrew from the project in 1969. After unsuccessful

No. built	3600 ^[5]
Variants	See <i>Variants</i>
Specifications	
Mass	2A6: 62.3 tonnes (68.7 short tons) 2A7V: 66.5 tonnes (73.3 short tons)
Length	2A6: 9.97 metres (32.7 feet) (gun forward)
Width	2A6: 3.75 m (12.3 ft)
Height	2A6: 3 m (9.8 ft)
Crew	4
Armor	2A6: 3rd generation composite; including high-hardness steel, tungsten and plastic filler with ceramic component.
Main armament	1 × Rh-120 L/44 120 mm or Rh-120 L/55 120mm <u>Rheinmetall Rh-120</u> ^[1] (42 rounds)
Secondary armament	2 × 7.62 mm <u>MG3A1</u> ^[1] or 2 × 7.62 mm <u>FN MAG</u> (4,750 rounds)
Engine	<u>MTU MB 873 Ka-501 liquid-cooled V12 twin-turbo diesel engine</u> 1100 kW at 2600 rpm
Power/weight	2A6: 17.8 kW/t (24.2 PS/t) 2A7V: 16.6 kW/t (22.6 PS/t)
Transmission	Renk HSWL 354
Suspension	<u>Torsion bar suspension</u>
Fuel capacity	1,200 litres (264 imperial gallons; 317 US gallons) ^[6]
Operational range	Road: 340 km (210 mi) Cross country: 220 km (140 mi) Average: 280 km (170 mi) ^[7]
Maximum speed	70 km/h (43 mph) ^{[8][9]}

attempts at saving the MBT-70 by conceptual

changes in order to eliminate the biggest issue—the driver being seated in the turret—it became clear in late 1969 that Germany would stop the bi-national development.^[14] The assistant secretary of the military procurement division of the German Ministry of Defence suggested reusing as many technologies developed for the MBT-70 as possible in a further programme, which was nicknamed *Eber* ("boar") due to his being named Eberhardt. The *Eber* used a modified MBT-70 turret and hull, with the driver being seated in the hull. Only a wooden mock-up was made.



A Leopard 2 PT15 with 105 mm smoothbore gun

One year later, a choice was made to continue the development based on the earlier *Keiler* project of the late 1960s, instead of finishing the development of the *Eber*. In 1971, the name of the design was determined as *Leopard 2* with the original Leopard retroactively becoming the Leopard 1, and Paul-Werner Krapke became the project officer of the Leopard 2 program.^[16] Originally two versions were projected: the gun-armed Leopard 2K and the Leopard 2FK, which would be armed with the XM150 gun/launcher weapon of the MBT-70.^[17]



A Leopard 2 prototype (1983)

In 1971, 17 prototypes were ordered but only 16 hulls were built as the production of hull PT12 was cancelled. Ten were ordered initially before another seven were ordered. The 17 turrets were designated T1 to T17, and the hulls were designated PT1 to PT11 and PT13 to PT17. To test a larger number of components and concepts, each prototype was fitted with components not found on the other prototypes. Ten of the turrets were equipped with 105 mm smoothbore guns and the other seven prototypes were equipped with a 120 mm smoothbore gun.^{[17][18]}



The Leopard 2 T14 mod. with the modified turret housing composite armour

Hulls PT11 and PT17 were fitted with a hydropneumatic suspension based on the MBT-70 design.^[17] The running gears of these two hulls had only six road wheels. Different types of auxiliary power units (APUs) were mounted in the prototypes. All turrets were equipped with a machine gun for air defence, except the turret mounted on PT11, where a 20 mm remotely operated autocannon was mounted. With the exception of hulls PT07, PT09, PT15, and PT17, all prototypes used the MB 873 engine. The road wheels were taken from the MBT-70 and the return rollers from the Leopard 1.^[17] The prototypes were designed with a projected weight of MLC50, which equals approximately 47.5 tonnes (46.7 long tons; 52.4 short tons). The welded turret utilised spaced armour formed by two steel plates.^[19] The prototypes were equipped with an EMES-12 optical rangefinder and fire control system, which later was adopted on the Leopard 1A4.

In mid-1973 a new turret was designed by Wegmann saving 1.5 tonnes (1.7 short tons) in weight.^[20] It was nicknamed the *Spitzmaus-Turm* (shrew turret) due to the highly sloped front. This design was only possible with the new EMES-13 optical rangefinder, which required a base length of only 350 millimetres (14 in) instead of the previous 1,720 millimetres (68 in).^[19] Based on experiences in the Yom Kippur War, a higher level of protection than the prototypes' heavily sloped spaced armour was demanded in late 1973 and the *Spitzmaus-Turm* was never produced.^[21]

The weight limit was increased from MLC50 to MLC60, which equals approximately 55 tonnes (54 long tons; 61 short tons). The T14 turret was modified to test a new armour configuration, taking on a blockier-looking appearance as a result of using vertical modules of spaced multilayer armour. It was also used to test the new EMES-13 optical rangefinder. The modified T14 turret was designated *T14 mod.*^[21] and was fitted with a fully electric turret drive and stabilization system, which was developed jointly by General Electric and AEG Telefunken.

American evaluation of Leopard 2AV and XM1 Abrams

In July 1973 German Federal Minister of Defence Georg Leber and his US counterpart James R. Schlesinger agreed upon a higher degree of standardisation in main battle tanks being favourable to NATO. By integrating components already fully developed by German companies for the Leopard 2, the costs of the XM1 Abrams, U.S. prototype tank developed after the MBT-70, could be reduced. A German commission was sent to the US to evaluate the harmonisation of components between the XM1 and Leopard 2.^[22] However, under American law it was not possible for a public bidder to interfere in a procurement tender after a contract with intention of profits and deadline was awarded to private sector companies.^[22]

As a result, the modification of the Leopard 2 prototypes in order to meet the US Army requirements was investigated. Following a number of further talks, a memorandum of understanding (MOU) was signed on 11 December 1974 between Germany and the US, which declared that a modified version of the Leopard 2 should be trialed by the US against their XM1 prototypes,^[23] after the Americans had bought and investigated prototype PT07 in 1973.^[24] The MOU obliged the Federal Republic of Germany to send a complete prototype, a hull, a vehicle for ballistic tests and a number of special ballistic parts to the US, where they would be put through US testing procedures for no additional costs.^[25]

The Leopard 2AV (*austere version*) was based on the experiences of the previous Leopard 2 development. It was created in order to meet the US requirements and the latest protection requirements of the German MoD. The T14 mod turret was used as the base for the Leopard 2AV's turret, but meeting the required level of protection for the hull required several attempts until the final ballistic trials on 23 to 26 June 1976.^[26] Following the US' preference of laser rangefinders, the turret of prototype PT19 was fitted with a laser rangefinder developed together with the American company Hughes.^[27]

In comparison with the earlier Leopard 2 prototypes, the fire control system was simplified by replacing the EMES-12 optical rangefinder and removing the crosswind sensor, the air pressure and temperature sensors, the powder temperature sensor, the PERI R12 commander sight with IR searchlight, the short-range grenade launcher for use against infantry, the retractable searchlight, the spotlight, the retractable passive night vision sight, the APU and the mechanical loading assistant.^[25]

Due to the design and production of the Leopard 2AV taking more time than expected, the shipment to the US and the US evaluation was delayed. It was not possible to test the Leopard 2AV before 1 September 1976.^[26] Despite the German wish that the Leopard 2AV and the XM1 prototypes would be evaluated at the same time, the US Army decided not to wait for the Leopard 2AV and tested the XM1 prototypes from Chrysler and General Motors beforehand.^{[22][28]}



Leopard 2 prototype tested at the Yuma Proving Ground, September 1975

Two new prototype hulls and three turrets were shipped to the US: PT20 mounting a 105 mm rifled L7 gun and a Hughes fire control system, PT19 with the same fire control system but able to swap out the gun for the 120 mm Rheinmetall smoothbore gun, and the PT21 fitted with the Krupp Atlas Elektronik EMES-13 fire control system and the 120 mm Rheinmetall gun.^[24] The Leopard 2AV fully met the US requirements.^[29] A study made by the American FMC Corporation showed that it was possible to produce the Leopard 2AV under licence in America without exceeding the cost limits set by the US Army.^[29] Before the trials were finished, it was decided that instead of the US Army possibly adopting the Leopard 2AV, the focus was shifted to the possibilities of common components between the two tanks. FMC, after having acquired the licenses for the production of the Leopard 2AV, decided not to submit a technical proposal, as they saw little to no chance for the US Army adopting a vehicle not developed in the US.^[28]

The US Army evaluation showed that on the XM1 a larger portion of the tank's surface was covered by special armour than on the Leopard 2AV.^[28] Differences in armour protection were attributed to the different perceptions of the expected threats and the haste in which the Leopard 2AV was designed to accommodate special armour.^[28] On mobility trials the Leopard 2AV performed equal to better than the XM1 prototypes. The AGT-1500 turbine engine proved to consume about 50% more fuel^[30] and the Diehl tracks had a higher endurance, while the tracks used on the XM1 prototypes failed to meet the Army's requirements.^[29] The heat signature of the MTU diesel engine was much lower.^[30] The fire control system and the sights of the Leopard 2 were considered to be better and the 120 mm gun proved to be superior.^[28] The projected production costs for one XM1 tank were \$728,000 in 1976, and the costs for one Leopard 2AV were \$56,000 higher.^[28]

After the American evaluation of the Leopard 2AV and the US Army's decision to opt for the XM1 Abrams, both American and German sources blamed the other side. According to American literature, it was discovered that the Leopard 2AV prototype used for mobility trials was underweight.^[nb 1]

In Germany, the test conditions were criticised for being unrealistic and favouring the XM1. Instead of using actual performance data, the calculated hypothetical acceleration was used.^[30] The XM1 was found to have a slightly higher rate of fire despite having internal layouts similar to the Leopard 2AV because the XM1 prototypes were manned by professional crews, while the Leopard 2AV had to be manned by conscripts in order to prove that the Leopard 2AV was not too complicated.^[30] Firing on the move was demonstrated on flat tracks, which nullified the better stabilization systems of the Leopard 2AV.^[30]

Series production

The decision to put the Leopard 2 tank in production for the German army was made after a study was undertaken, which showed that adopting the Leopard 2 model would result in a greater combat potential of the German army than producing more Leopard 1A4 tanks or developing an improved version of the Leopard 1A4 with 105/120 mm smoothbore gun, improved armour protection, a new fire control system and a 890 kilowatts (1,190 hp) or 1,100 kilowatts (1,500 hp) engine.^[31] Various changes were applied to the Leopard 2 design before the series production started in 1979.^{[32][33]} The engine, transmission, and suspension were slightly modified and improved. The ballistic protection of the turret and hull was improved and weak spots were eliminated.^[34]



Leopard 2 tanks during a manoeuvre in 1986

The turret bustle containing the ready ammunition racks and the hydraulic system was separated from the crew compartment and fitted with blowout panels. The development of several new components was introduced to the Leopard 2 during the Leopard 2AV development and after the US testing was completed.

For the series version, the Hughes-designed laser rangefinder made with US Common Modules was chosen over the passive EMES-13 rangefinder. The EMES-13 system was considered to be the superior solution, but the Hughes system was cheaper and fully developed.^[34]

The German company Krupp-Atlas-Elektronik acquired the licence of the Hughes design and modified it to meet the needs of the German army.^[34] The modified rangefinder received the designation EMES-15. The installation of the US AGT-1500 turbine engine in the Leopard 2 was tested by MaK.^[29] The AGT-1500 was from the United States and required deep modifications to the Leopard 2's chassis. However, driving tests at the WTD 41 revealed a number of drawbacks such as high fuel consumption and the poor performance of the transmission including the brakes.^[29] This project was thus terminated.

In January 1977 Germany ordered a small pre-series of three hulls and two turrets which were delivered in 1978. These vehicles had increased armour protection on the front of the hull. One of the hulls was fitted with the earlier T21 turret and was used by the German army school in Munster for troop trials until 1979.^[35] In September 1977, 1800 Leopard 2 tanks were ordered, to be produced in five batches. The main contractor was Krauss-Maffei, but Maschinenbau Kiel (MaK) was awarded a contract for producing 45% of the tanks. The first batch consisted of 380 tanks. The delivery of six tanks was scheduled for 1979, 114 for 1980, 180 for 1981, and 300 tanks each following year.^[36]

The first series-production tank was delivered on 25 October 1979. By 1982, all of the first batch of 380 Leopard 2 tanks had been completed. 209 were built by Krauss-Maffei (chassis no. 10001 to 10210) and 171 by MaK (chassis no. 20001 to 20172). The first production tanks were fitted with the PzB-200 image intensifier due to production shortages of the new thermal night-sight system, which was later retrofitted to the earlier models. After the original five batches, three further batches of Leopard 2 tanks were ordered, increasing the number of Leopard 2 tanks ordered by Germany to a total of 2125.^[37] The sixth batch was ordered in June 1987 and consisted of 150 tanks, which were produced between January 1988 and May 1989. The seventh batch of 100 tanks was produced between May 1988 and April 1990. The last batch for the German army totalling 75 tanks was produced from January 1991 to March 1992.^[37]

During its production run during the Cold War, 16 Leopard 2 tanks were being produced per month. The vehicles were produced at a slower rate in the following decades, however KMW still retained the capacity to return to such manufacturing levels should they need to be made again at a higher rate and supply chains are able to deliver sufficient materials.^[38]

Further improvements

While previous models only varied in detail, the Leopard 2A4 introduced a digital ballistic computer and an improved fire extinguishing system. Starting with the sixth batch, tanks were fitted with an improved armour array and new side skirts. In 1984 the German military procurement agency stated a number of requirements for a future Leopard 2 upgrade. In 1989, the *Kampfwertsteigerung* (combat potential improvement) programme was initiated in Germany with the delivery of first prototypes. The official military requirements were published in March 1990.^[39]



Dutch Leopard 2A4 tanks, 1997

The KWS programme was projected to consist of three stages. The first stage replaced the Rheinmetall 120 mm L/44 gun barrel and the corresponding gun mount with a longer barrelled and more lethal L/55 version.^[39] This stage was adopted in the form of 225 Leopard 2A6 tanks, starting in 2001 and lasting until 2005.^[40] Stage 2 focused on improvements of armour protection and survivability: it was adopted in the form of the Leopard 2A5, starting in 1995. The base armour of the tank was exchanged and additional

armour modules were installed at the turret. The first batch of 225 Leopard 2 tanks was upgraded to Leopard 2A5 configuration between 1995 and 1998; a second batch of 125 followed from 1999 to 2002.^[41]

The third stage was the planned replacement of the Leopard 2 turret by a new turret fitted with a 140 mm NPzK tank gun, an autoloader, and the IFIS battlefield management system.^[39] The ballistic protection at the hull was to be improved.^[39] Originally a total requirement for 650 Leopard 2 tanks with KWS 3 was projected.^[40] It was never finalised, but the 140 mm NPzK tank gun was tested on an older prototype. In 1995 it was decided to cancel due to changes in the political environment. The funds were redirected to the Neue Gepanzerte Plattformen (New Armoured Platforms) project of the German army. The Leopard 2A6M was developed with a kit providing enhanced protection against mines that can detonate below the hull (like mines with bending wire triggers) and explosively formed penetrator mines.^[40] The weight of the Leopard 2A6M is 62.5 tonnes.^[42]



The Leopard 2A5 can be recognized for its angular spaced armour on the turret cheeks and 44-calibre 120 mm main gun.

The latest version of the tank is the Leopard 2A7, which entered service in an initial batch of 20 tanks in 2014.^[43] Already before the first Leopard 2A7 tank was handed over to the German Army, plans for upgrades were made.^[44] At this time an "extensive" increase in combat value, while retaining the original mobility of the Leopard 2, was planned.^[44] The optics of the tank will also be improved.^[44]



A German Army Leopard 2A6, assigned to the 104th Panzer Battalion conducting high-speed manoeuvres

In April 2015, *Welt am Sonntag* claimed that tungsten (wolfram) rounds used in Leopard 2 cannot penetrate the Russian T-90 or the modernized version of the T-80. They also stated that the German military will develop a new improved round, but it will be exclusively developed for the Leopard 2A7.^[45]

In 2015 Rheinmetall disclosed that it was developing a new 130 mm smoothbore gun for the Leopard 2 tank and its successor. This gun will offer a 50% increase in performance and penetration.^[46] Marketing for the new gun was slated to begin in 2016.

Replacement

The Leopard 2 first entered service in 1979, and its service life is anticipated to end around 2030. In May 2015, the German Ministry of Defence announced plans to develop a tank jointly with France as a successor to both the Leopard 2 and Leclerc tanks. Technologies and concepts will be investigated to determine what capabilities are needed in a future tank.^[47] Deployment of the new tank, titled Main Ground Combat System (MGCS), will be preceded by incremental upgrades to the Leopard 2, including a new digital turret core system and situational awareness system and an active protection system (APS).^[48]

A short-term lethality increase will come from a higher pressure 120 mm gun firing new ammunition, expected to deliver 20 percent better performance than the L/55. Mid-term efforts will focus on a Rheinmetall 130 mm cannon concept offering 50 percent better armour penetration. With the Russian T-14 Armata being equipped with the Afghanit, an active protection system designed to mitigate the effectiveness of ATGM, more importance is being placed on direct-fire weapons.^[48]

Design

Protection

The Leopard 2 uses spaced multilayer armour throughout the design.^[49] The armour consists of a combination of steel plates of different hardness, elastic materials, and other non-metallic materials.^{[50][51][52]} Steel plates with high hardness and high ductility are used. The armour is a result of extensive research about the formation and penetration mechanism of shaped charge jets.^[53] The Leopard 2's armour might be based on the British Burlington armour, which had already been demonstrated to West Germany in 1970.^[54]

Later, in the mid-1970s, full details about Burlington were handed over to the West German government. The frontal arc of the Leopard 2's armour is designed to withstand large caliber kinetic energy penetrators and shaped charge projectiles.^[52] During the 1980s, it was estimated that the Leopard 2's front would resist 125 mm armour-piercing fin-stabilized discarding sabot (APFSDS) rounds fired from a distance of 1,500 m.^{[53][55]}

The Leopard 2A4's armour has a maximum physical thickness of 800 millimetres (31 in) based on unofficial measurements and estimates made by former conscripts and professional soldiers of the German army.^[56] On the Leopard 2A5 and subsequent models, the thickness is increased by the wedge-shaped armour module to 1,500 millimetres (59 in).^[56]

The side and the rear of the tank protect against heavy machine guns, medium caliber rounds, and older types of tank ammunition. The side of the hull is covered by armour skirts to increase protection against projectiles and rocket-propelled grenades (RPGs). The frontal third of the hull sides is covered by heavy ballistic skirts, while the rest of the hull sides is covered by steel-reinforced rubber skirts. For increased protection against mines, the sides of the hull floor are sloped by 45° and the floor is reinforced with corrugations.^[50]

Secondary protection

The Leopard 2's design follows the concept of compartmentation. Possible sources of fire or explosions have been moved away from the crew.^[52] In the turret, the ammunition and the hydraulics are located in compartments separated from the crew. In case of a detonation,



Arrowhead-shaped armour module of the Leopard 2A5



The turret and hull sides of the Leopard 2A7+ are fitted with additional armour modules.



The Leopard 2SG is fitted with AMAP composite armour.

the blow-off panels on the compartment roofs will direct the explosion and fire away from the crew. The crew is also protected against nuclear, biological and chemical (NBC) threats, as the Leopard 2 is equipped with a Dräger NBC overpressurization system, which provides up to 4 hPA over-pressure inside the vehicle.^[57]

Two groups of four Wegmann 76 mm smoke mortars are mounted on either side of the turret and can be electrically fired either as single rounds or in salvos of four. They are mounted on most Leopard 2 models, with the exception of Dutch Leopard 2s, which are equipped instead with a Dutch-designed smoke mortar system with six barrels on each side.^[57] Swedish Stridsvagn 122 utilises French GALIX smoke dispensers, similar to the system found on the French Leclerc.^[58]

The Leopard 2 is equipped with a fire protection system. Four 9 kg halon fire extinguisher bottles are installed on the right behind the driver's station. The bottles are connected to pipes and hoses and are activated automatically by the fire detection system when temperatures rise above 82 °C (180 °F) inside the fighting compartment, or manually via a control panel in the driver's compartment.^[57] An extra 2.5 kg halon fire extinguisher is stored on the floor beneath the main gun.

Armour upgrades

Following Leopard 2's introduction into service in 1979, the armour has been gradually improved over the years. A modified version of spaced multilayer armour was introduced beginning with the 97th vehicle of the 6th production batch.^[59] The same batch also introduced an improved type of heavy ballistic skirts.

The Leopard 2A5 upgrade focused on increased armour protection. While upgrading a Leopard 2 tank to the Leopard 2A5 configuration, the roof covering the armour modules is cut open and new armour modules are inserted.^{[60][61]} New additional armour modules made of laminated armour cover the frontal arc of the turret. They have a distinctive arrowhead shape and improve protection against both kinetic penetrators and shaped charges.^{[61][62]} The side skirts also incorporate improved armour protection.^[62] A 25 mm-thick spall liner reduces the danger of crew injuries in case of armour penetration.^{[57][61]}

The Leopard 2A7 features the latest generation of passive armour and belly armour providing protection against mines and IEDs.^[63] The Leopard 2A7 is fitted with adapters for mounting additional armour modules or protection systems against RPGs.^[64]

For urban combat, the Leopard 2 can be fitted with different packages of modular armour. The Leopard 2A4M CAN, Leopard 2 PSO (Peace Support Operations) and the Leopard 2A7 can mount thick modules of composite armour along the flanks of the turret and hull, while slat armour can be adapted at the vehicle's rear. The armour modules provide protection against the RPG-7, which depending on the warhead can penetrate between 280 millimetres (11 in) and 600 millimetres (24 in) of steel armour.^[65] The Leopard 2A6M CAN increases protection against rocket-propelled grenades (RPGs) by including additional slat armour.^[66]



Ammunition storage in a Leopard 2A4



Indonesian Leopard 2RI of the 1st Cavalry Battalion with AMAP composite armour

Additional armour packages have been developed by a number of different companies. IBD Deisenroth has developed upgrades with MEXAS and Advanced Modular Armor Protection (AMAP) composite armour, the latter is being used on Singaporean and Indonesian Leopard 2 tanks. RUAG has developed an armour upgrade utilizing their SidePRO-ATR composite armour. This upgrade was first presented on the IAV 2013.

The Leopard 2A4M and 2A6M add an additional mine protection plate for the belly, which increases protection against mines and IEDs.^[57]

On 22 February 2021, the German Defence Ministry agreed to acquire Trophy, an active protection system of Israeli design. 17 German Army tanks will be fitted with the system, with integration planned to be completed in 2023.^[67]

Armour protection estimates

Estimated levels of protection for the Leopard 2 range from 590 to 690 mm RHAe on the turret, 600 mm RHAe on the glacis and lower front hull on the Leopard 2A4, to 920–940 mm RHAe on the turret, 620 mm RHAe on the glacis and lower front hull on the Leopard 2A6 against kinetic projectiles.^[57]

According to a description page hosted by the Federation of American Scientists, the armour of the Leopard 2A4 is believed to provide protection equivalent to 700 mm armour steel (RHA) against kinetic energy penetrators and 1000 mm RHA against shaped charge warheads.^[68]

Armament

Primary

The primary armament for production versions of the Leopard 2 is the Rheinmetall 120 mm smoothbore gun—the same gun later adapted for use on the M1 Abrams—in either the L/44 variant (found on all production Leopard 2s until the A5), or the L/55 variant (as found on the Leopard 2A6 and subsequent models).^[57] Ammunition for the gun comprises 27 rounds stored in a special magazine in the forward section of the hull, to the left of the driver's station, with an additional 15 rounds stored in the left side of the turret bustle, which is separated from the fighting compartment by an electrically operated door.^[57]

If the ammunition storage area is hit, a blow-off panel in the turret roof would direct an explosion upwards away from the crew compartment.^[57] The gun is fully stabilised, and can fire a variety of types of rounds, such as the German DM43 APFSDS-T anti-tank round, which is said to be able to penetrate 560 millimeters (22 in)^[69] of steel armour at a range of 2,000 metres (2,200 yd),^[70] and the German DM12 high-explosive anti-tank (HEAT).^[71]

For the L/55 gun, a newer APFSDS-T round was introduced to take advantage of the longer barrel, the DM-53, which is said to be able to penetrate 750 mm of RHAe armour at a range of 2,000 meters.^[57] The bore evacuator and the gun's thermal sleeve of the A4 and A5, designed to regulate the temperature of the barrel, are fabricated from glass-reinforced plastic. The barrel has a chrome lining to increase barrel life.^[72] The main gun is capable of power elevating from +20° to −9°.^[73]



A view through the panoramic PERI R17 sight

Rheinmetall has developed an upgrade for Leopard 2 tanks to give them the ability to fire the Israeli LAHAT anti-tank guided missile through the main gun. The missile can engage targets out to a range of 6,000 metres (20,000 ft).^[74]

Secondary

The Leopard 2 is equipped with two machine guns, one mounted co-axially, the other on an anti-aircraft mount. German models use the MG 3 7.62 mm machine gun; Dutch and Singaporean models use FN MAG 7.62 mm machine guns; Swiss models use Swiss MG 87 7.5 mm machine guns.^[57] 4,750 rounds of machine gun ammunition are carried on board the Leopard 2. More recent variants such as the Leopard 2A7+ are capable of mounting a Remotely-Controlled Weapons Station fitted with a Browning M2HB Heavy Machine Gun, near the commander's hatch.

Fire control

The standard fire control system found on the Leopard 2 is the German EMES 15 fire control system with a dual magnification stabilised primary sight. The primary sight has an integrated neodymium yttrium aluminium garnet Nd:YAG laser rangefinder and a 120 element Mercury cadmium telluride, HgCdTe (also known as CMT) Zeiss thermographic camera, both of which are linked to the tank's fire control computer.^[75] A backup 8x auxiliary telescope FERO-Z18 is mounted coaxially for the gunner.^[57]

The commander has an independent periscope, the Rheinmetall/Zeiss PERI-R 17 A2. This is a stabilised panoramic periscope sight designed for day/night observation and target identification. It provides an all round view with a traverse of 360°. The thermal image from the commander's periscope is displayed on a monitor inside the tank. Initial production tanks were not equipped with a thermal sight, due to the sight not being ready, and instead temporarily substituted the PZB 200 low light TV system (LLTV).^[57]

The fire control suite is capable of providing up to three range values in four seconds. The range data is transmitted to the fire control computer and is used to calculate the firing solution. Because the laser rangefinder is integrated into the gunner's primary sight, the gunner is able to read the digital range measurement directly. The maximum range of the laser rangefinder is up to 10,000 m with a measuring accuracy within 10 m at this range.^[75] The combined system allows the Leopard 2 to engage moving targets at ranges of up to 5,000 meters whilst itself being on the move over rough terrain.

Propulsion

The Leopard 2 is propelled by the MTU MB 873 Ka-501 engine. It provides 1,500 PS (1.1 MW) at 2600 RPM and 4,700 N·m (3,500 lb·ft) of torque at 1600–1700 RPM. The MTU MB 873 Ka-501 is a four-stroke, 47.7 litre, 90° V-block 12-cylinder, twin-turbocharged and intercooled, liquid-cooled diesel engine (with multi-fuel capability). It has an estimated fuel consumption rate of around 300 litres per 100 km on roads and 500 litres per 100 km across the country, and is coupled to the Renk HSWL 354 gear and brake system.^{[57][75]}

The Renk HSWL 354 transmission has four forward and two reverse gears, with a torque converter and is completely automatic, with the driver selecting the range.^[57] The Leopard 2 has four fuel tanks, which have a total capacity of approximately 1,160 litres, giving a maximum road range of about 500 km.^[57] The propulsion pack is



The Leopard 2's MB 873 Ka-501 V12 engine

capable of driving the tank to a top road speed of 68 km/h (limited to 50 km/h during peacetime by law), and a top reverse speed of 31 km/h.^[76] The power pack can be changed in the field in 35 minutes.^[57] The engine and transmission are separated from the crew compartment through a fireproof bulkhead.^[75] An enhanced version of the EuroPowerPack, with a 1,650 PS (1.2 MW) MTU MT883 engine has been trialled by the Leopard 2.^[75]



German Leopard 2A4 with turret snorkel, 2010

The Leopard 2 has a torsion bar suspension and has advanced friction dampers. The running gear consists of seven dual rubber-tired road wheels and four return rollers per side, with the idler wheel at the front and drive sprocket at the rear.^[57] The tracks are Diehl 570F tracks, with rubber-bushed end connectors, which have removable rubber pads and use 82 links on each track. For use in icy ground, up to 18 rubber pads can be replaced by the same number of grousers, which are stored in the vehicle's bow when not in use.^[57] The upper part of the tracks are covered with side skirts.^[73]

The Leopard 2 can drive through water 4 meters (13 ft) deep using a snorkel or 1.2 meters (3 ft 11 in) without any preparation. It can climb vertical obstacles over one metre high.

The German Army has prioritised mobility in the Leopard 2, which has made it one of the fastest MBTs in the world.^[77]

Exports

Germany has fielded about 2,125 Leopard 2 main battle tanks in various versions, but many were sold following German reunification. The Leopard 2 became very popular in the 1990s, when the shrinking German army offered many of its redundant Leopard 2s at a reduced price. It became successful enough in Europe that the manufacturer started calling it the *Euro Leopard*, despite France, Britain, and Italy all operating their own MBTs. With further non-European orders, the name "*Global-Leopard*" is now used instead.^[78] Leopard 2 tanks have also been resold by original export customers, although reexport has always been conditional on consent from the German government who control the platform's export licence. Other countries have bought newly manufactured vehicles or have produced them locally under licence.

Europe

The Netherlands ordered 445 Leopard 2 tanks on 2 March 1979, after examining the results of the Leopard 2AV in the United States.^[79] It became the first export customer of the Leopard 2 and the vehicles were delivered between July 1981 and July 1986. Dutch Leopard 2 tanks have been subsequently exported to Austria, Canada, Norway, and Portugal. Leopard 2 tanks remaining in Dutch service have been upgraded to 2A5 and 2A6 standards.



A Leopard 2A6 in Dutch service

The Swiss Army decided to purchase Leopard 2 tanks over the M1A1 Abrams after trialling both tanks between August 1981 and June 1982. The Swiss decision was made in August 1983 and the funding was approved by the government in 1984.^[80] Thirty-five of the tanks were delivered by Kraus-Maffei by June 1987. Eidgenössische Konstruktionswerkstätte Thun started license production of 345 additional vehicles in December 1987.

After investigating the option of a locally developed replacement for the Strv 103 tank, Sweden decided to buy a foreign tank model. The Leopard 2 Improved (Leopard 2A5 prototype) won the competition against the M1A2 Abrams and the French Leclerc. The Swedish military also evaluated the Soviet T-80U tank, but separately from the other tanks. After intensive tests from January to June 1994, the Swedish military opted for the Leopard 2.^[81] The Swedish military found that the Leopard 2 Improved met their military demands by 90%.^[81] The M1A2 met the Swedish requirements by 86%, whereas the Leclerc met 63%. In June 1994 Sweden ordered the production of 120 modified Leopard 2A5, to be known as Stridsvagn 122 (Strv 122) in Swedish service. Strv 122 features Swedish-developed appliqué armour, a new command system, and improved electronics.^[82] Of the 120 Strv 122, 29 were manufactured in Germany by Krauss-Maffei Wegmann while the other 91 were manufactured by the Swedish firms Bofors and Hägglunds.^[83] The first Stridsvagn 122 was delivered in December 1996. These remain in Swedish service and have received periodic upgrades.



A Stridsvagn 122 in 2019. This is a Swedish derivative of the Leopard 2A5.

Sweden also leased and later bought a total of 160 Leopard 2A4 tanks in 1994 and 1995, known in Swedish service as Stridsvagn 121 (Strv 121). The first Strv 121 was delivered in February 1994.^[81] The Strv 121 fleet was mothballed by 2006.^[84] Sweden has retained a number of Strv 121 tanks as training aids; 6 Strv 121 have since been converted into AEV 3 Kodiak armoured engineering vehicles, and a further 6 have been converted into Leguan armoured bridgelayers.^{[85][86]}



Stridsvagn 121 (Swedish Leopard 2A4)

Denmark bought 51 ex-German Leopard 2A4 tanks after the Danish military school, the Hærens Kampskole, recommended basing the adoption of a new tank on the Swedish army trials. The first tanks were delivered in 1998, but the upgrade to Leopard 2A5 level was already decided the next year.^[87] In 2004-2006 the Danish army bought another six ex-German Leopard 2 tanks.^[88]

In 1998, Greece held a competition to determine the main battle tank for the Hellenic Army. The Leopard 2 Improved managed to outperform the Challenger 2E, Leclerc, M1A2 Abrams, T-80U, and T-84 and was chosen by the Greek officials. In March 2003 Greece ordered 170 Leopard 2 tanks, of which 140 were locally assembled.^{[89][90]} Greece also bought 183 Leopard 2A4 and 150 Leopard 1 tanks.^[91]



A Greek Leopard 2A6 on a training exercise in 2021

Spain initially leased 109 Leopard 2A4 tanks, after Krauss-Maffei withdrew from the Lince development, a special lighter version of the Leopard 2 developed together with Santa Bárbara Sistemas. Before the end of the Lince tank, Spain had already rejected the M1A1 Abrams and the Vickers Valiant. After deciding to purchase the leased tanks, Santa Bárbara Sistemas acquired the licence to locally produce 219 Leopard 2A6 tanks for the Spanish army.^[92]

Poland received 128 Leopard 2A4 tanks from German army stocks in 2002. In 2013 Poland ordered a further 119 ex-German Leopard 2s. Finland bought 124 used Leopard 2A4 tanks and six armoured bridge-layer Leopard 2L tanks from Germany in 2002 and 2003. The tanks served as replacements for the old

Soviet-made T-55 and T-72M1. The Netherlands resold 114 of their tanks (and one turret) to Austria, 80 to Canada in 2007,^[93] 52 to Norway, 37 to Portugal and 100 to Finland.

In December 2018, Hungary placed an order for 44 Leopard 2A7+s and 12 second-hand 2A4s. The order coincided with the procurement of 24 Panzerhaubitze 2000, and was expected to replace Hungary's current fleet of T-72 tanks "no sooner than 2020".^{[94][95]}

In February 2023, the Norwegian Prime Minister, Jonas Gahr Støre, announced that Norway would be ordering 54 new Leopard 2A7 tanks at a cost of Nkr 19.7 billion with a further option for 18 vehicles to be delivered at a later date. The first Leopard 2A7s are due to be delivered by 2026, and operational by 2031. The Norwegian government had been weighing up either the Leopard 2A7 or the South Korean K2 Black Panther as a replacement for its aging Leopard 2A4 fleet.^{[96][97]} The tanks will be designated Leopard 2A8 NOR.^[98]

Beyond Europe

In 2005, Turkey ordered 298 Leopard 2 tanks from German army stocks.^[99] The Leopard 2 was chosen in 2001 after successfully competing one year earlier in the Turkish army trials against the T-84 Yatagan, Leclerc and a version of the M1A2 Abrams fitted with a German MTU diesel engine. Turkey wanted to buy 1,000 Leopard 2 tanks in 1999, but the German government rejected the deal.

Singapore bought 96 Leopard 2 tanks from Germany in 2006.^[100] Chile bought 172 ex-German Leopard 2A4 tanks and 273 Marder 1A3 IFVs in 2007.

Indonesia ordered 103 Leopard 2 tanks and 42 Marder 1A3 IFVs in 2013.^[101] At first the export of heavy weapons to Indonesia was not allowed by the German government, due to the questionable human rights record of Indonesia. 61 of the 103 Leopard 2 tanks will be upgraded by Rheinmetall to the Leopard 2RI standard, based on Rheinmetall's *Revolution* modular upgrade concept.^[102]

Qatar ordered 62 Leopard 2A7 tanks and 24 Panzerhaubitze 2000s in 2013.^[103] The delivery of the tanks started in late 2015 and the first tanks were displayed on a military parade in December 2015.^[104]



An Indonesian Army Leopard 2A4+ of the 8th Cavalry Battalion

Failed exports

Saudi Arabia has shown interest in buying the Leopard 2 since the 1980s. Due to political circumstances and the questionable situation of human rights in Saudi Arabia, no deal was made. Saudi Arabia renewed its intention of buying Leopard 2 tanks in 2011.^[105] While earlier news reports suggested an interest in buying about 200 tanks, later reports revealed an increased order of 600 to 800 tanks.^[105] The German government at first approved the deal, but cancelled it later due to human rights concerns and Saudi Arabia's military intervention in the 2011 Bahraini uprising.^{[105][106]}

The Leopard 2 was tested by the United Kingdom. In 1989 the Leopard 2 was evaluated as a possible replacement for the Challenger 1 tank.^[107] Ultimately the British armed forces decided to adopt the locally made Challenger 2.

The Australian Army evaluated ex-Swiss Army Leopard 2s as a replacement for its Leopard 1AS tanks in 2003, but selected the M1A1 AIM instead due to easier logistics. More modern versions of the Leopard 2 or M1 Abrams, such as the Leopard 2A6, were not considered due to their higher price.^[108]

Transfer to Ukraine

On 2 July 2023 Ukraine had reportedly received 54 Leopard 2A4 from NATO-member countries.^[109]

Political discussions

Since April 2022, in the wake of the Russian invasion, the Ukrainian government has requested that their allies donate Western-made main battle tanks. Poland, Finland and others have all announced a willingness to contribute Leopard 2 tanks from their stocks, with around 100 tanks from various states ready to be transferred to Ukraine.^[110] However, when Germany exported the tanks to these countries, it had made reexport conditional on a German government permit, as based on the *Kriegswaffenkontrollgesetz* and *Außenwirtschaftsgesetz*. Until 22 January 2023, it was unclear whether such a consent would be provided with Germany determined to avoid any perception of escalating the conflict, and wary of being labelled an aggressor.^[111] Germany has also been keen to extract an American commitment to provide its own M1 Abrams tanks, before sending German-made Leopard 2 tanks to Ukraine.^[112]

The matter was discussed at a conference of defence ministers from NATO members and allies, but no agreement was reached.^{[113][114]} The German position was heavily criticized by some of the other European governments with Poland threatening to unilaterally export their Leopards if permission from Berlin was not given.^[115] The Polish and Ukrainian governments then announced that Ukrainian soldiers would start training on Polish Leopard 2 tanks in Poland.^[116]

On 22 January 2023, the German minister for foreign affairs, Annalena Baerbock, told French media that Germany "would not stand in the way" of any Polish decision to send Leopard 2 tanks to Ukraine but noted that "for the moment the question has not been asked."^[117] Two days later, Poland made an official request for permission to transfer Polish Leopards to Ukraine.^[118]

On 24 January *Der Spiegel* reported that the German Chancellor, Olaf Scholz, would the following day, officially announce the transfer of an undisclosed number of German Leopard 2 tanks to Ukraine, beginning with 14 Leopard 2A6s from the active Bundeswehr inventory.^[119] The German government would also give permission to any country seeking to reexport Leopard 2 tanks to Ukraine.^[120] These reports were formally confirmed by the German government on 25 January.^[121]

It was also reported that Boris Pistorius, the German defence minister, was encouraging Leopard-operating states to start training Ukrainian personnel in their use.^[111]

The German decision to provide and approve the provision of Leopard 2 tanks to Ukraine came after U.S. President Joe Biden decided to provide Ukraine with M1 Abrams tanks, as Germany insisted on moving in coordination with its allies. The decision to provide M1 Abrams tanks had previously been opposed by the Pentagon.^[122]

On 15 February 2023, Pistorius expressed his frustration at Germany's allies in their announcement of any donations of Leopard 2 tanks to Ukraine, saying it had "not been exactly breathtaking, to put it mildly." Since the change in German position on exports, only Poland had matched the German promise to provide 14 tanks - the two largest donations from a single country.^[123]

Germany

On 25 January 2023, the German government confirmed that it would make 14 Leopard 2A6 (a company-strength number) available for Ukraine and would give authorisation to European partners to reexport their vehicles.^[121] The goal is for European Leopard 2 operators to provide two battalions worth, or 88 tanks, to Ukraine.^[124]

According to the German government, the tanks will be delivered to Ukraine after training Ukrainian service personnel. This process would take up to three months. Some tanks are held by the Bundeswehr, others by the Leopard manufacturer Rheinmetall. The latter has said they could be ready to be dispatched by March 2023. However, they also warned that some vehicles held in long-term storage would require extensive refurbishment and updating before they could be considered suitable for combat in Ukraine.^{[120][111]} On 26 January, Pistorius, the German minister of defense, stated that the tanks would be delivered in late March or early April.^[125] On 24 February 2023, the German Department of Defence confirmed Germany would increase its Leopard 2A6 contribution from 14 to 18, thus providing 1 full tank battalion together with the contributions of Portugal and Sweden.^[126]

On 27 March 2023, Germany delivered 18 Leopard 2A6 tanks to Ukraine with two accompanying armoured recovery vehicles and necessary spare parts.^[127]

On 28 March 2023, the security board of the Swiss National Council recommended that 25 Leopard 2 in storage should be sold to the German manufacturer Krauss-Maffei-Wegmann so that Germany can replace some of its own tanks now serving in Ukraine. As of this day, Switzerland keeps 96 Leopard 2 in storage. As the tanks' sale requires decommissioning, the Bundeswehr waits for approval by the Swiss parliament.^[128]

On 27 September 2023, the Swiss parliament has voted to decommission 25 Leopard 2 tanks, allowing them to be sold back to Germany. Germany has given assurances that ex-Swiss tanks won't be sent to Ukraine.^[129]

Poland

Poland welcomed the German decision to allow the transfer of Leopard 2 tanks to Ukraine, with the Polish government announcing on 25 January it would transfer 14 Leopards to Ukraine from a total Polish stock of 247, in addition to a further 50–60 Soviet-era tanks.^[130] On 24 February 2023, the first 4 Leopard 2A4 tanks from Polish stocks were delivered to Ukraine. These are the first Leopard 2 tanks to be operated by Ukraine.^[131]

Canada

On 26 January 2023, the Canadian Defence Minister, Anita Anand, announced that Canada would be sending four Leopard 2A4 tanks to Ukraine, with the potential for more to follow. These vehicles were reportedly 'combat ready.' Canada will also provide appropriate training to Ukrainian forces who will go on to operate these tanks.^[132] The first Canadian Leopard 2 tanks had arrived in Poland on 5 February, to

allow Ukrainian troops to begin training.^[133] On 24 February, the Canadian Prime Minister, Justin Trudeau, announced that Canada would be sending a further four Leopard 2A4 tanks to Ukraine along with 1 armoured recovery vehicle (ARV) and 5,000 rounds of ammunition.^[134]

Czech Republic

On 20 January 2023 the Czech Defense Ministry issued a statement that the media reports indicating the Czech Republic and Slovakia were willing to give up their Leopard 2 tanks for Ukraine were false, and that rumors that the Czech Republic would receive American Abrams tanks in exchange for Leopards were also false.^[135] As of December 2022, both countries had only received 1 Leopard 2A4 each from Rheinmetall as part of Germany's *Ringtausch* program in exchange for them delivering older Soviet equipment to Ukraine, while the remaining Leopard 2A4s are still being refurbished (expected to be delivered throughout 2023).^[136]

Denmark

While it was reported in the international press that Denmark had signalled that it could contribute a number of tanks,^[137] there was no official indication from the Danish Government that any Danish Leopard 2's would be sent to Ukraine. Of the Danish inventory of 44 tanks, 14 were deployed to Estonia as part of NATO EFP, with a further number receiving upgrades at the German manufacturers.^[138] Some Danish opposition politicians voiced support for a transfer of Danish tanks.^[139] Unlike those considered by other countries, the Danish Leopards are of the new 2A7 standard, which would represent some logistic and mechanical challenges compared to the other variants.^[140] Denmark instead announced on 7 February that it was jointly purchasing 100-178 Leopard 1A5 tanks with Germany and the Netherlands.^[141] Denmark also announced on 20 April 2023 that they would jointly finance with the Netherlands the purchase and refurbishing of 14 Leopard 2A4 for Ukraine from private industry stocks.^[142]

Finland

Finland had suggested before the German decision on reexport, that it could supply Ukraine with a limited number of Leopard 2 tanks. The Finnish President, Sauli Niinistö, warned that "the number of tanks [to be sent to Ukraine] cannot be large, since Finland borders on Russia and is not part of NATO."^[143] On 23 February Finland announced it will give 3 Leopard 2R mine clearing vehicles to Ukraine. Finnish total stock of this type will return the same (6) with planned purchase of 3 replacement vehicles from Germany.^[144]

Greece

On 25 January 2023 it was reported that Greece was one of the countries that had indicated it would supply Leopard 2 tanks to Ukraine.^[145] On 29 January 2023 it was erroneously reported that approximately 14 Leopard 2 tanks would be provided, though it was unclear if it would be the 2A4 or 2A6 HEL variant.^[146] However, on 31 January, Greek Prime Minister Kyriakos Mitsotakis clarified that Greece would not supply any Leopard tanks to Ukraine, arguing "they are absolutely necessary for our defense posture."^[147] Leaked

Pentagon documents revealed that Greece was preparing to donate 5 of its Leopard 2 tanks to Ukraine, though Greek defense minister Nikolaos Panayiotopoulos reiterated later in April his government's opposition to tank deliveries to Ukraine, making confirmation of the delivery uncertain.^[148]

Netherlands

The Netherlands was considering Ukrainian requests for Leopard 2 tanks, with Prime Minister Mark Rutte suggesting that the Dutch government could purchase tanks from other countries and donate them.^[149] There were also some discussions about Netherlands purchasing the 18 Leopard 2A6 tanks they lease from Germany. No refusal was made from Germany to sell them as no formal request to buy them was made; both countries agreed that they were critical to the operations of the joint German-Dutch military unit. However, the Netherlands agreed to supply ammunition for Leopard tanks and announced it had not abandoned its intention to contribute to the tank initiative, while also noting it was helping to purchase Leopard 1A5 tanks with Denmark and Germany.^[150] The Netherlands later announced it was jointly financing with Denmark the purchase and refurbishing of 14 Leopard 2A4 for Ukraine from private industry stocks, with delivery expected in early 2024.^[151]

Norway

On 25 January 2023, Norway pledged to donate spare Leopards to Ukraine with reports speculating that between four and eight of its 36 Leopard 2A4 tanks would be sent.^{[137][152][153]} Norwegian defence minister Bjørn Arild Gram said in an interview with Norwegian public television NRK: "Norway and the government support the donation of battle tanks to Ukraine. Norway will take part," but did not specify how many would be sent.^[154] On 14 February 2023, Gram announced that 8 Norwegian Leopard 2 tanks and 4 Leopard 1 derived "special purpose vehicles" would be handed over to Ukraine. The special purpose vehicles turned out to be a mix of 3 NM189 AEVs and 1 NM217 ARV.^{[155][156]} In February 2023, Norway ordered 54 Leopard 2A7 tanks (later changed to 2A8 variant) to be delivered from 2026, with a further option for 18 vehicles if necessary.

Portugal

When the German government changed its position on the reexport of Leopard 2 tanks, it was reported that the Portuguese government was preparing to send four of their Leopard 2A6 tanks to Ukraine.^[157] On 4 February, Prime Minister António Costa confirmed that Portugal would send Leopards to Ukraine, but did not confirm the number of vehicles set to be delivered. It is understood that Portugal is working with Germany to obtain the necessary parts to repair inoperable tanks in its inventory of 37 Leopard 2 tanks, but it has been widely reported by local media that most are inoperable. Costa expressed hopes that Portuguese tanks would be delivered by March 2023.^[158] On 14 February, it was reported that Portugal had earmarked 3 Leopard 2A6 tanks for donation to Ukraine.^[123] All 3 were confirmed donated on March 27 together with the 18 German Leopard 2A6. Leaked Pentagon documents revealed that Portugal may be donating an additional 3 Leopard 2A6 tanks to Ukraine.^[159]

Slovakia

On 25 January 2023, Slovakia announced it was not considering the transfer of any of its Leopard 2A4 tanks to Ukraine, but were willing to provide T-72 tanks if other allies replaced them with other tanks. The reason cited for this was that Slovakia had only received 1 of the 15 Leopard tanks it was promised as part of Germany's *Ringtausch* program, which it was getting in exchange for supplying Ukraine with 30 Soviet-era infantry fighting vehicles.^[160]

Spain

On 4 June 2022 it was reported that Spain was considering providing 40 of its 53 retired Leopard 2A4 tanks stored in Zaragoza.^[161] On 8 June German Chancellor Olaf Scholz denied receiving any formal request from Spain for an export permit for Leopard 2 tanks to Ukraine.^[162] However it was later confirmed on 10 June by the Spanish defense minister they were considering the transfer.^[163] On 12 June, anonymous government sources told Der Spiegel that the Scholz government blocked any transfer of Leopard tanks to Ukraine, arguing this "would constitute a departure" from an informal NATO decision not to provide Western tanks at that time.^[164] By 2 August, Spain officially backtracked on its decision, arguing the tanks at Zaragoza were in "deplorable condition" and that it could not send any.^[165]

The Spanish government was later reported to be considering to send an unconfirmed number of Spanish Leopard 2E tanks to Ukraine in January 2023 following a meeting of the Ukraine Defense Contact Group.^[166] On 14 February the Spanish defence minister Margarita Robles confirmed Spain was providing training to 55 Ukrainian tank crew members and technicians on the Leopard 2.^[167] On 22 February, Robles also confirmed that Spain was in the process of preparing six Leopard 2A4 tanks for delivery to Ukraine by late March or early April.^[168] On 5 February 2023, it was reported that at least 11 of the 53 tanks stored at Zaragoza had been drawn out of storage to be refurbished and upgraded to have "the same lethality" as modern versions of the Leopard 2 before being sent to Ukraine. Five were sent to the Santa Bárbara Sistemas plant in Alcalá de Guadaira and six had already been moved to Seville for diagnostics and repairs. The Ministry of Defence had not published the contracts for the retrofits out of secrecy. It was unclear however if ammunition would be supplied with the tanks as Spain only has 9 rounds for each of its Leopard 2E tanks.^[169] On 23 February Spanish Prime Minister Pedro Sánchez confirmed during a visit to Kyiv that Spain was sending at least 10 Leopard 2 tanks to Ukraine.^[170]

Sweden

Sweden began considering Ukrainian requests for Leopard 2 tanks in January 2023 and did not rule out contributing its Strv 122 at a later stage.^[171] On 24 February, the Swedish minister of defence Pål Jonson announced that Sweden would be providing 10 Leopard 2A5 tanks to Ukraine.^[172]

Combat history

KFOR

Starting on 12 June 1999, 28 Leopard 2A5 tanks were deployed to Kosovo by the German Army as part of the Kosovo Force (KFOR). The vehicles of Panzerbataillon 33 and 214 were sent from Macedonia to Prizren. They were used for patrols, protecting checkpoints and bases as well as part of the show of force. On 13 June 1999, two members of the Serbian paramilitary started firing from inside a Fiat 125p car at one of the checkpoints in Prizren and both were killed by return fire. A Leopard 2A5 was located at the checkpoint, but it could not participate in the fighting as it was only partially crewed.^[173]

On 26 June 1999, a Leopard 2A5 fired four warning shots above the town of Orahovac.^[174] From late 2000 to early 2001, the tanks were replaced by the Leopard 2A4 model. Leopard 2A4s were deployed to Macedonia in 2001 as part of the NATO intervention. The tanks served to protect Bundeswehr logistic sites in Macedonia. Until their return in 2004, the Leopard 2 tanks were stationed at the Austrian-Swiss camp "Casablanca".^[174]



A German Leopard 2A4 being unloaded from a SLT 50 Elefant tank transporter in Kosovo, July 2002

IFOR/SFOR

The Dutch contingent in Bosnia-Herzegovina operated Leopard 2 tanks.^[175] Dutch Leopard 2A4s and Leopard 2A5s were stationed at the NLD bases at Bugojno, Novi Travnik, Sisava, Knezevo, Maslovare and Suica.

ISAF/OEF

In October 2003, Canada was planning to replace its Leopard C2s with wheeled Stryker Mobile Gun Systems. However, operational experience in Afghanistan and in particular during Operation Medusa, convinced the Canadian military of the usefulness of maintaining a tank fleet.^[176] Leopard C2s were deployed to Kandahar in December 2006,^[177] but they were by then almost 30 years old, and were nearing the end of their operational life. The Canadian government decided to borrow 20 Leopard 2A6s and three armoured recovery vehicles from Germany for rapid deployment to Afghanistan. In late August 2007, the first Leopard 2s were airlifted into Afghanistan to equip Lord Strathcona's Horse (Royal Canadians).^[178]

In an assault on 2 November 2007, a Leopard 2A6M hit an improvised explosive device (IED) and survived without casualties: "My crew stumbled upon an IED (improvised explosive device) and made history as the first (crew) to test the (Leopard 2A6) M-package. It worked as it should." wrote a Canadian officer in an email to German defence officials.^[179] Canadian Chief of the Defence Staff General Rick Hillier denied reports that a Leopard 2 tank that was struck by an IED was a write-off, insisting that the tank has been repaired and is once again in use. "The Taliban have been engaged with some of the new Leopard 2 tanks in several ambushes" and that as a result the Taliban "learned some very harsh lessons" and lost the battle in question "very quickly and very violently."^[180]

In October 2007, Denmark deployed Leopard 2A5 DKs in support of operations in southern Afghanistan. The Danish tank unit, drawn from the first battalion of the Jydske Dragonregiment (Jutland Dragoons Regiment),^[181] was equipped with three tanks and one M113 armored personnel carrier, with an armoured recovery vehicle and another tank kept in reserve.^[182] The Danish version of the Leopard 2A5 is fitted with Swedish-made Barracuda camouflage mats that limit the absorption of solar heat, thus reducing infrared signature and interior temperature.^[181] It also has a conventional driver's seat bolted on the floor of the tank, whereas in the Canadian 2A6M (as part of the mine-protection package) the driver's seat has been replaced by a "dynamic safety seat",^[183] which is a parachute-harness like arrangement that the driver wears around his hip. This way, the driver does not have any contact with the hull except on the pedals and is out of the shockwave area of exploding land mines or IEDs.

In January 2008, Danish tanks halted a flanking manoeuvre by Taliban forces near the Helmand River by providing gunfire in support of Danish and British infantry from elevated positions.^[184] On 26 February 2008, a Danish Leopard 2 was hit by an explosive device, damaging one track. No one was injured and the tank returned to camp on its own for repairs.^[185] The first fatality suffered by a crew operating a Leopard 2 happened on 25 July 2008. A Danish Leopard 2A5 hit an IED in Helmand Province. The vehicle was able

to continue 200 metres (656 ft) before it halted.^[186] Three members of the four-man crew were able to escape even though wounded, but the driver was stuck inside. Onsite treatment by Danish medics could not save him. The vehicle was towed to Forward Operating Base (FOB) Attal and then later to FOB Armadillo for investigation and possible redeployment. During the same contact with Taliban forces, a second tank was caught in an explosion but none of the crew were wounded.^[186]

Beginning on 7 December 2008, Leopard 2 tanks took part in Operation Red Dagger, firing 31 rounds in support of Coalition troops as they recaptured Nad Ali District. A press release from the British Ministry of Defence praised the tank's fire accuracy and mobility, claiming the Leopard 2 was a decisive factor in the coalition's success.^[187]

Turkish intervention in Syria

Turkey operates 354 Leopard 2A4 tanks. Initially using other tank types including upgraded M60s, in December 2016 Turkey deployed a number of Leopard 2A4s to the Syrian Civil War against Islamic State of Iraq and the Levant (ISIS) as part of Operation Euphrates Shield. Initially, three of the Turkish Leopard 2A4s operating in Syria were destroyed or damaged by ISIS using anti-tank missile systems (possibly Fagot or Konkurs anti-tank guided missiles obtained from Syrian or Iraqi Army captured stocks). In mid-December 2016, two 2A4 tanks were captured by ISIS near al-Bab city in Syria during Euphrates Shield operations. Amaq News Agency posted video of vehicles claimed to be captured Leopard 2A4s.^[188]

By late December 2016, ISIS had captured or incapacitated 10 Leopard 2A4s. These were damaged by anti-tank weapons, IEDs, or other unknown causes.^{[189][190]} Additional ISIS propaganda images and video depicting several completely destroyed Leopards, some with their turrets blown off, were published in January 2017.^[191] Tanks which suffered the worst damage may have been destroyed by airstrikes in order to prevent capture but sources generally state that the damage was caused solely with anti-tank missiles or car bombs driven by suicide bombers.^{[192][193]}

In January 2017, the German newspaper *Die Welt* reported that ISIL fighters used 9M133 Kornet anti-tank missiles to destroy six Leopard 2 tanks used by the Turkish military in Syria.^[194]

At least eight Leopard 2 MBT have been destroyed according to photographic reports.^[195]

Turkey also confirmed the use of Leopard 2A4 tanks during the Turkish military operation in Afrin to the German government. These tanks were designed during the Cold War to fight against Soviet tanks in Europe, not counterinsurgencies against guerrillas, where the primary risk is improvised explosive devices and anti-tank missiles. These tanks were retired from German usage when sold to Turkey.^{[196][197]}

There is a belief that the Turkish purchase of Leopard 2A4s from Germany was subject to the condition that they were not to be used against Kurdish separatists. Prior to 2016, the Leopard 2A4s were kept in northern Turkey. Once the German government discovered that the Leopard tanks were being used against Kurdish forces, planned upgrades to make them “less vulnerable to explosives” were halted. Ultimately Turkey was forced to upgrade the Leopard 2A4s with domestic components, including a possible replacement of the original turret with that of the Turkish Altay main battle tank.^{[196][198][199][200]}

Russo-Ukrainian War

In April 2023, during training of a Ukrainian crew a Leopard 2A4 was damaged with its turret being completely dislodged from the hull. The crew was reported as fine.^[201] In July 2023, a Leopard 2A4 was modified to carry Kontakt-1 ERA on the sides of the hull and the turret. It was also shown with rubber screens attached to the turret and gun mantel.^{[109][202][203][204]}

Beginning in June 2023, Ukraine deployed Leopard 2s during the [Russo-Ukrainian War](#) against Russia in preparation for their [2023 counteroffensive](#). During Ukrainian attacks in [Zaporizhzhia Oblast](#) on 8 June 2023, a Russian artillery strike on a Ukrainian vehicle column at Novopokrovka destroyed at least one Leopard 2A4 tank.^[205] Additionally, more Russian drone footage from the same day and days after showed 4 Leopard 2A6 tanks and 3 Leopard 2Rs damaged or destroyed south the road of [Mala Tokmachka](#).^[206]

On 28 August, [Forbes](#) claimed that over the course of 13 weeks of the Ukrainian counteroffensive, five Leopard 2 tanks (two 2A4s and three 2A6s) had been destroyed with 10 more damaged. According to Ukrainian soldier Oleksandr Solonko, their ability to withstand damage is superior to Soviet tanks due to its design that stores ammunition separately from the crew to explode outward if struck, enabling crewmembers to survive hits that would be fatal in a T-72 that stows ammunition under the turret. Damaged Leopard 2s can be repaired in Poland or Germany.^{[207][208]}

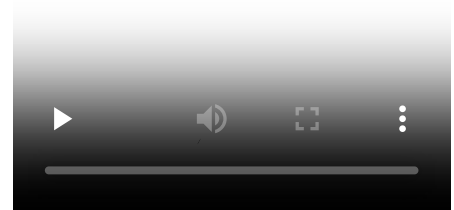
On August 28 2023, Russia's Defense Minister [Sergey Shoigu](#) claimed that Russian forces had destroyed or damaged all 16 Leopard 2A4 tanks supplied to Ukraine by Poland and Portugal.^[209]

As of 25 September 2023, according to the [Oryx](#) blog, at least seven Ukrainian Leopard 2s were destroyed by photos and videos (3 Leopard 2A6, 2 Leopard 2A4, 1 Leopard 2A4V, 1 Leopard 2R); While another 13 Ukrainian Leopard 2s of various models were confirmed damaged to various degrees, with four of them being classified as abandoned by their crews: 4 Leopard 2A4, 6 Leopard 2A6 with 3 of them being abandoned, 1 Stridsvagn 122 that is also abandoned, 2 Leopard 2R.^[210]

Variants

Leopard 2

The baseline Leopard 2, sometimes informally called the "A0" to differentiate it from later versions, was the first series manufactured version. The vehicles were manufactured from October 1979 until March 1982, altogether 380 vehicles. 209 were built by Krauss Maffei and 171 by MaK. The basic equipment consisted of electrical-hydraulic stabiliser WNA-H22, a fire control computer, a laser rangefinder, a wind sensor, a general-purpose telescope EMES 15, a panorama periscope PERI R17, the gunner's primary sight FERO Z18, on the turret roof as well as a computer-controlled tank testing set RPP 1–8. 200 of the vehicles had a low-light enhancer (PZB 200) instead of thermal imaging. Two chassis served as driver training vehicles.



Two [German Army](#) Leopard 2s demonstrate their [deep-wading](#) capabilities.

Leopard 2A1

Minor modifications and the installation of the gunner's thermal sight^[1] were worked into the second batch of 450 vehicles Leopard 2, designated the A1. Krauss-Maffei built 248 (Chassis Nr. 10211 to 10458) and Mak built 202 (Chassis Nr. 20173 to 20347). Deliveries of the 2A1 models started in March 1982 and ended in November 1983. The two most notable changes were the modification of the ammunition racks to be identical to those in the M1A1 Abrams, and redesigned fuel filters that reduce refuelling time.

A third batch of 300 Leopard 2, 165 by Krauss-Maffei (Chassis Nr. 10459 to 10623) and 135 by MaK (Chassis Nr. 20375 to 20509.), was built between November 1983 and November 1984. This batch included more minor changes that were later retrofitted to the earlier 2A1s.

Leopard 2A2

This designation was given to upgraded vehicles of the first batch of Leopard 2s, brought up to the standard of the second and third batches. This modernisation gradually replaced the original PZB 200 sights in the first batch with thermal sights for the EMES 15 as they became available. The upgrade included the fitting of filler openings and caps to the forward hull fuel tanks to allow separate refuelling. There was an addition of a deflector plate for the periscope and a large coverplate to protect the existing NBC protection system. The tank was given new five metre towing cables with a different position. The programme began in 1984 and ended in 1987. The third, fourth and fifth batches, which were produced during this period, had the same features. The modernised first batch can be recognised by the circular plate covering the hole where the crosswind sensor for the fire control system was removed.^[211]

Leopard 2A3

The fourth batch of 300 vehicles, 165 by Krauss-Maffei (Chassis Nr. 10624 to 10788) and 135 by Mak (Chassis Nr. 20510 to 20644), was delivered between December 1984 and December 1985. The main change was the addition of the SEM80/90 digital radio sets (also being fitted to the Leopard 1 at the same time), and the ammunition reloading hatches being welded shut. Even with these minor changes the new batch was known as the 2A3.

Leopard 2A4

The most widespread version of the Leopard 2 family, the 2A4 models included more substantial changes, including an automated fire and explosion suppression system, an all-digital fire control system able to handle new ammunition types, and an improved turret with flat titanium/tungsten armour. The Leopard 2s were manufactured in eight batches between 1985 and 1992. All the older models were upgraded to 2A4 standard. Until 1994, Germany operated a total of 2,125 2A4s (695 newly built and the rest modified older versions), while the Netherlands had an additional 445 tanks. The 2A4 was licensed and manufactured in Switzerland as the **Panzer 87 "Leopard"** or **Pz 87**. This version included Swiss-built 7.5 mm MG 87 machine guns and communications equipment and featured an improved NBC protection system. Switzerland operated 380 Pz 87 tanks.



Early Dutch Leopard 2 tanks in 1986

After 2000, Germany and the Netherlands found themselves with large stocks of tanks that they had no need for after the Cold War. These tanks were sold to NATO or friendly armies around the world. Among these buyers of the surplus tanks were Turkey (purchasing 354 vehicles), Greece (183), Sweden (160), Chile (140), Finland (139), Poland (128), Austria (114), Spain (108), Canada (107), Indonesia (103), Singapore (96), Norway (52), Denmark (51), and Portugal (37).^[212]

The *Pz 87WE* (*WertErhaltung*) is a planned Swiss modification and upgrade of the Pz 87.^[213] The modification significantly improves protection through the addition of the Leopard 2A6M's mine protection kit, thicker armour on the front glacis, and a turret equipped with a Swiss-developed armour package using titanium alloy. The turret roof armour is improved and the smoke grenade launchers are redesigned. Further

improvements enhance survivability and combat capabilities, such as a turret electric drive similar to the Leopard 2A5, a driver rearview camera, an independent weapons station for the loader, and enhanced command and control systems. The fire control system is upgraded, using the Carl Zeiss Optronics GmbH PERI-R17A2 fire control system. A remote weapons station containing a fully stabilised Mg 64 0.50 in (12.7 mm) machine gun is fitted to the tank.

The Swiss company RUAG offered an upgrade package for **Pz 87** main battle tanks to meet a possible requirement of the Swiss Army for improved protection. It is fitted with a new modular composite armour package. Armour modules can be tailored to provide enhanced protection against a specific threat. Damaged modules can be easily replaced in field conditions. Upgraded tanks are less vulnerable to direct hits, anti-tank missiles, RPG rounds, mines, and IEDs. The vehicle received new sensors such as a sniper detection sensor and a laser warning receiver.^{[214][215]} In 2021, the two prototypes with add-on armour were handed to the Militär- und Festungsmuseum Full-Reuenthal.^[216]

The Indonesian Army operates the *Leopard 2 Republic Indonesia* or *Leopard 2RI*, an upgraded variant based on Rheinmetall's MBT Revolution upgrade for the Leopard 2A4 tank. The upgrade includes AMAP armour from IBD Deisenroth and Rheinmetall Chempro, improved fire control systems, and battlefield management and situational awareness systems.^{[217][218]}

The *Pz 87-140*^[219] is an experimental variant of the Swiss Pz 87 with a 140 mm gun and additional armour, which was later used on the newer production variants.

The *Leopard 2A4CHL* is the upgraded Chilean version of the Leopard 2A4 ordered by Chile in 2007. Upgrades include new electronics, sighting, and information systems meant to elevate the Leopard 2A4's networking capability to be equal to that of the Leopard 2A6, a new suspension system, and the upgrading of the tank main gun to the L/55 smoothbore cannon used on the Leopard 2A6. Other upgrades are remote weapon stations over the gunner and commander hatches fitted with the MG3 and HK GMG. The Leopard 2A4CHL has improved roof and side turret armour and can be uplinked with Chile's battlefield control network.

The *Leopard 2A4M CAN* is the upgraded Canadian version of the Leopard 2A4 acquired from the Royal Netherlands Army surplus. The Leopard 2A4M CAN is specially designed for the war in Afghanistan, based on experience gained by Leopard 2 operators. The first 20 were delivered in October 2010. Five were deployed to Afghanistan at the end of 2010 and operated until July 2011, when combat operations stopped.^[220] Though originally planned to be up-gunned to the L/55 for consistency with the 2A6M CAN, the longer barrelled guns (optimised for tank-vs-tank warfare) were found to be less than ideal in Afghanistan, therefore it was decided to retain the L/44. Only small areas of slat armour were added, in contrast with the fully caged 2A6M CANs. The protection of the Leopard 2A4M CAN has been further augmented by the addition of applique armour resembling that found on the most recent Leopard 2A7+ variant, but modified to fit the turret configuration of the 2A4.^[221] Of the 2A4s acquired, 11 were converted for training use (9 A4s, 2 A4Ms). In February 2011, Canada bought 12 2A4s/Pz 87 from Switzerland for the 'Force Mobility Enhancement'



An Austrian Leopard 2A4 gunner's sights



Indonesian Leopard 2RIs during a parade



A Leopard 2A4 in Canadian Army configuration, including Saab Barracuda thermal armour

project which, along with the remaining unused ex-Dutch tanks, saw 18 converted to Armoured Engineering Vehicles and 4 converted to Armoured Recovery Vehicles.^[222] Canada has also purchased 15 2A4s from Germany as Logistic Stock Vehicles (for spare parts).



A Leopard 2A4 of the Austrian Bundesheer

The *Leopard 2NG (Next Generation)* is a privately funded Turkish upgrade by ASELSAN that includes the application of AMAP, upgraded optics, completely overhauled turret mechanics, and a new fire control system on the work since 1995 and to be delivered by late 2011, which is intended to be used on the new Altay MBT. It was developed without an order from the Turkish Army, but might meet the requirements for the modernization of the Turkish Leopard 2A4s.^[223] The old powerpack and the L/44 gun barrel are kept, but the combat weight is increased to 65 tonnes.^[223] According to Turkish news sources, Finland was interested in getting the Turkish upgrade package to modernise their fleet of Leopard 2A4s. However, in 2015 Finland purchased 120 2A6 vehicles from the Netherlands.^{[224][225]}

The Leopard 2 hull was used for the Vickers Mk 7 main battle tank, which featured a British-designed turret, where some of the innovations later were incorporated into the Challenger 2 design.

In December 2015, Bumar-Labedy signed an agreement with German Rheinmetall Landsysteme GmbH concerning the technological support of the Polish modernization program for Leopard 2A4 tanks. The company will design, document, and execute six prototypes. The first upgraded Leopard 2PLs have arrived in Poland in June 2020, with all 142 tanks to be delivered by 2023.^[226] The upgrades include third generation night vision systems (production of the Warsaw PCO), new additional armour modules and anti-splash lining, removal of flammable components (turret drive system and main propulsion system), installation of the new fire protection system, modernization of the tank's integrated monitoring and testing equipment, the possibility of using new types of ammunition (programmable DM-11 and DM-63), and an auxiliary generator set (APU). Construction of all 142 units will be completed by the end of 2020.

Turkey is planning to modernize its Leopard 2A4 MBTs as *Leopard 2A4TR* with the T1 Modernization Package. According to the Defense Industry Presidency, Leopard 2A4 tanks will be modernized with; Explosive Reactive Armour (ERA), T1 Reactive-Passive Armour, High Ballistic Strength Cage Armour, Hollow Modular Add-on Armour, Close Range Surveillance System (YAMGÖZ), Laser Warning Receiver System (LIAS), SARP Remote Controlled Weapon System (UKSS), PULAT Active Protection System (AKS), a new power distribution unit, ASELSAN Driver Surveillance System (ADİS) and voice alert system integrations. The modernization programme is to be completed in 2 batches. The programme will start with 84 Leopard 2A4 tanks in the first batch and the remaining tanks will be modernized within the 2nd batch. A total of 334 tanks (including prototypes) are planned to be upgraded with the T1 modernization programme. The new modernized Leopard 2A4 that was presented at the BMC factory was fitted with an Altay turret, which is armed with one 120 mm 55 caliber smoothbore gun designed and manufactured by the Turkish Company MKE, based on a technology transfer from Hyundai Rotem of South Korea. A remote-controlled weapon station is mounted on the top of the turret which is armed with a 12.7mm caliber heavy machine gun.^{[227][228][229][230][231][232]}

Leopard 2A4V

On 1 September 2023, video on social media showed Ukrainian crews with 30 Leopard 2A4 upgraded with Kontakt ERA and T-80U style rubber screens, denoted Leopard 2A4V.^[233] This followed previous sightings in Ukraine of a Canadian supplied Leopard 2A4 upgraded with Kontakt-1 ERA.^{[109][204]}

Leopard 2 Marksman

Finland has modernised its Marksman SPAAG vehicles by replacing the original T-55AM chassis with a newer Leopard 2A4 chassis.^[234] The upgraded Marksman vehicles were scheduled to enter service with the Finnish Army in 2016.^[235] The new Leopard 2 chassis greatly improves mobility compared to the older T-55AM chassis, both on- and off-road. The Leopard 2 chassis is larger, providing a more stable firing platform for the Marksman turret to operate from.^[236]



A Leopard 2 Marksman of the Finnish Army

Leopard 2 Imp

"Leopard 2 Improved" was a prototype-series for enhancing the A4, introducing a wedge-shaped, spaced add-on armour to the turret front and the frontal area of the sides. These spaced armour modules defeat a hollow charge prior to reaching the base armour, and cause kinetic-energy penetrators to change direction, eroding them in the process. It does not form a shot-trap, since it does not deflect the penetrators outwards to hit the hull or turret ring. The gun mantlet was redesigned to accept the new armour.^[58]

Leopard 2A5

The Leopard 2 Imp was developed into the A5. There were some improvements in the main armour composition. The interior received spall liners to reduce fragments if the armour is penetrated. The frontal "heavy" third of the side skirts was replaced with a stronger type. The commander's sight was moved to a new position behind the hatch and it received an independent thermal channel. The gunner's sight was moved to the turret roof as opposed to the cavity in the front armour in previous models. A heavier sliding driver's hatch was fitted.^[237]



Leopard 2A5 of the Polish Land Forces, 2017

Turret controls went all-electric, increasing reliability and crew safety, as well as weight savings. The gun braking system was improved to prepare for the later mounting of the new L/55 gun tube and to enable firing of more powerful ammunition, such as the DM53 APFSDS. The first A5s were handed over to the German army tank school in 1995 and started to enter regular service with Panzerbataillon 33 in December the same year.^[237]

The Leopard 2A5 DK is a variant of the Leopard 2A5 similar to the Leopard 2A6 with some small modifications, used by the Danish Army.^[238]

Stridsvagn 122

Stridsvagn 122 is a Swedish Army tank based on the Leopard 2 Improved. 120 units were built, 91 of which were licence-produced in Sweden. The tank features increased armour on the turret top and front hull, and improved command-, control- and fire-control systems. Externally, it can be distinguished from the

Leopard 2A5 by the French GALIX smoke dispensers, different storage bins, and the much thicker crew hatches.^[58] The Strv 122B+ Evolution, a variant equipped with modular AMAP composite armour from IBD Deisenroth, has increased 360° protection against threats like EFPs, RPGs and IEDs.^[239] The width of 4 metres (13 ft) has been kept, while the weight increases by only 350 kilograms (770 lb).^[239]



A Stridsvagn 122 of the Swedish Army

Leopard 2-140

In the early 1990s, Rheinmetall began development of a 140 mm smoothbore cannon for use in future tank designs. The new gun was intended to counter new Soviet tank developments, especially since the next generation of Soviet main battle tanks were rumoured to be armed with a 135 mm or 152 mm cannon. The new 140 mm cannon was part of a modernisation programme for the Leopard 2 known as the KWS III.^[240] Test firing of the new 140 mm cannon was conducted. Results showed that the gun had high penetration values, and had a muzzle velocity of around 2,000 metres a second, with potential to be increased further. However, the 140 mm rounds were too heavy for the tank crew to handle effectively.

The KWS III upgrade was to feature a new turret. This new turret was equipped with the planned 140 mm cannon and an autoloader. The introduction of an autoloader reduced the tank's crew to three members, as a dedicated loader was no longer needed. The gun's 32 rounds of ammunition were stored separate from the crew in a large compartment occupying the entire rear of the turret, in order to increase crew survivability in the event of a cook off. The turntable-style turret had the gun offset to the left side, due to the autoloader's lateral feeding of ammunition into the cannon breech. The turret was powered by an electro-hydraulic drive and featured an IFIS battlefield management system. The crew was protected by an armoured capsule and ballistic protection for the hull was to be improved. The planned protection level of the KWS III upgrade was to be equal to or better than the Leopard 2A5.^{[39][240]}

A total of 650 Leopard 2 KWS III tanks were originally projected to be purchased.^[40] However, in 1995, the KWS III programme was cancelled due to changes in the political environment.

Despite this, development continued on the 140 mm cannon, with Rheinmetall coordinating with the British Royal Ordnance and French GIAT companies. The 140 mm cannon was fitted to an old Leopard 2 prototype with the turret T19.^[241] Counterweights were added to the rear of the turret to balance the increased weight of the 140 mm cannon. The modified Leopard 2 was not equipped with any other KWS III upgrades apart from the new gun. Live fire testing showed mixed results, where the 140 mm cannon showed superior penetrating power compared to the existing 120 mm cannon, but demonstrated poorer handling characteristics.^[240] The lack of the autoloader on the prototype further hampered performance.^[240]

Leopard 2A6

The Leopard 2A6 includes the addition of the Rheinmetall 120 mm L/55 smoothbore gun and other changes. All German tank battalions of the "crisis intervention forces" are equipped with the A6. Canada purchased twenty Leopard 2A6s from the Netherlands. These were delivered in 2007.^[242] Portugal purchased 37 Leopard 2A6s from the Dutch in 2007, with delivery in 2008. In January 2014, Finland purchased 100 L2A6s, as well as munitions, simulators, and a ten-year supply of reserve parts from the Netherlands. The tanks were delivered in batches between 2015 and 2019.^[243]

The Leopard 2A6A1 is a command version of the Leopard 2A6, stemming from the KWS I programme. The vehicle includes additional radios (with the amount depending on the level of command) to accommodate the communications for section, platoon, troop, company, squadron or battalion commanders.^[244] For section leaders, the vehicle is fitted with 1x SEM 80 and 1x SEM 90 VHF radios. For platoon or troop commanders, the vehicle is fitted with 1x SEM 80 and 1x SEM 90 VHF radios along with an 800m field telephone cable drum fitted at the rear of the vehicle. For company, squadron or battalion commanders the vehicle is fitted with 2x SEM 80 or 1x SEM 93 and 1x SEM 90 VHF radios along with an 800m field telephone cable drum, for either fitted at the rear of the vehicle.^[244]



A German Leopard 2A6M with turret reversed

The Leopard 2A6M is a version of the 2A6 with enhanced mine protection under the chassis, and internal enhancements to improve crew survivability.^[245] In the summer of 2007, Canada borrowed 20 A6Ms from Germany for deployment to Afghanistan.



A Hellenic Army Leopard 2A6HEL

The Leopard 2 Hel is a derivative of the 2A6 that was ordered by the Greek Army in 2003 – the "Hel" stands for "Hellenic". The 170 tanks were delivered between 2006 and 2009. A total of 140 were built in Greece by ELBO, which delivered the first units in late 2006.^[246]

The Leopard 2A6M CAN is a Canadian variant of the Leopard 2A6M. Significant modifications include distinctive black boxes mounted on the rear of the turret bustle,^[247] and stand-off slat armour.^[248] The first tanks configured in this variant were 20 loaned from the German Bundeswehr in an effort to increase firepower and protection given to Canadian troops operating in the south of Afghanistan. The loaned tanks retain their German MG3 machine guns, the ex-Dutch tanks are also expected to retain their FN MAG machine guns due to commonality with Canadian stocks of C6 GPMG, itself a variant of the FN MAG.^[249]

Due to the loaned status of the first twenty tanks, the air conditioning unit originally could not be installed, as only minimal changes could be made. The crew wore cooling vests instead, and the turret's electric drive generates less heat than the hydraulic drive of the older Leopard C2. The loaned German tanks will be kept by the Canadian Forces and may be further upgraded, while ex-Dutch Leopard 2A6s were modified to German Leopard 2A6M specifications and used as restitution for the loaned tanks.^[250] Canadian Leopard 2s in Afghanistan were later fitted with air conditioning units, a much-needed commodity in the scorching desert of Afghanistan, and Saab's Barracuda camouflage mats, which reduce solar loading by 50 percent.^[66]

The Leopard 2A6TR was the Turkish variant during the Turkish Army tank procurement project in 2000. The version was based on 2A6EX. The project was dropped in favor of developing the indigenous Altay (main battle tank).^[251]

Leopard 2E

The Leopard 2E is a derivative of the 2A6, with greater armour protection,^[252] developed under a programme of co-production between the defence industries of Spain and Germany. The programme was developed within the frame of collaboration decided in 1995 between the Defence Ministries of both countries, which also included the cession of use by period of five years of 108 Leopard 2A4 from the

German Army to the Spanish Army. The session was extended up to 2016, and after that those tanks will be the sole property of the Spanish Army, as has been made public in January 2006, then having been paid a total of €15,124,014 in ten yearly installments, giving the Spanish co-ownership from 2006.

In 1998, the Spanish government agreed to locally build 219 tanks of the Leopard 2E line, 16 recovery tanks of Leopard 2ER (Buffalo), and 4 training vehicles. They chose Santa Bárbara Sistemas as the main contractor. The programme, with a budget of €1,939.4 million, includes integrated logistical support, training courses for crew instructors and maintenance engineers, and driving, turret, maintenance, aiming, and shooting simulators. Deliveries of the first batch began in 2004.



Spanish Leopard 2E

Leopard 2PL

The Leopard 2PL is a Polish modernized version of the Leopard 2A4, carried out in cooperation with Rheinmetall and the Polish Armaments Group (Polish: *Polska Grupa Zbrojeniowa*, PGZ). The Leopard 2PL MBT is primarily tasked with assault, maintaining territory, and supporting mechanized and motorized subdivisions with its onboard weapon systems in all weather conditions during the day and night. The main upgrades when compared to the Leopard 2A4 include modernization of the commander's and gunner's sight, additional ballistic modules on the turret, replacement of the hydraulic stabilization system with a new electric system, and new fire extinguishing and fire suppression systems.^{[253][254]}



A Leopard 2PL

The upgrade included a new commander's control and monitoring system, the installation of an auxiliary power unit (APU), a new turret stowage compartment for crew equipment, the modernization of its main gun to use new types of programmable ammunition, and the integration of day/night rear camera for drivers. Included are customized towing vehicles due to the increased weight of the upgraded tank.^{[253][254]} The upgraded 2PL version is in service with the Polish Land Forces. Of the Leopard 2A4s from the first (128) and the second (14) batches, 24 have been upgraded to Leopard 2PL standard.^[255] The rest will be upgraded to the 2PLM1 standard.^[256]

Leopard 2 PSO

The new Leopard 2 PSO (Peace Support Operations) variant is designed especially for urban warfare, which had been encountered in peacekeeping operations with increasing frequency. The Leopard 2 PSO is equipped with more effective all-around protection, a secondary weapons station, improved reconnaissance ability, an obstacle clearance blade (OCB), a shorter gun barrel for manoeuvring on urban streets at the expense of fire range, non-lethal armament, close-range surveillance ability through camera systems,



A Leopard 2 PSO at Eurosatory 2006

a searchlight and further changes to improve its perseverance and mobility in built-up non-wide open areas. These features are similar to the Tank Urban Survival Kit for the American M1A2 Abrams.

Leopard 2A7

The Leopard 2A7 vehicle is not intended to be optimized for combat in urban terrain, thus it is fundamentally different from the KMW variant, the: 2A7+ (see below). A total of 20 vehicles were provided for converting. It involves former Dutch A6NL models returned by Canada to Germany. The original upgrade to A6M has been extended in coordination with Canada and includes a crew-compartment cooling-system from the Leopard 2 A6M-HEL series, a new 20 kW auxiliary power unit based on the Steyr Motors M12 TCA UI engine,^[257] the Saab Barracuda Mobile Camouflage System (MCS) with Heat-Transfer Reduction (HTR CoolCam) system,^[258] a field trial proven combat management and information system (IFIS: *Integriertes Führungs- und Informationssystem*), onboard network optimization with ultracapacitors in the chassis and turret, a SOTAS IP digital intercom system, a renewal of the fire suppression system in the crew compartment, and the retrofitting of Attica thermal imaging module in the commander optics.^[43]

The weapon system is adapted for firing HE ammunition. It is fitted for, but not with, additional passive side protection armour. The first Leopard 2A7 was handed over to the German Army in Munich in December 2014. 14 vehicles were produced for Tank Battalion 203, 4 went to the Armoured Corps Training Centre and one vehicle went to the Technical School for Land Systems and School for Technology of the Army. The last tank remains as a reference vehicle at KMW.^[43]

The Danish Armed Forces received its first Leopard 2A7 main battle tanks upgraded in Germany from the Leopard 2A5DK version at the Dragoon Barracks in Holstebro. The Danish Army will receive a total of 44 Leopard 2A7 vehicles by 2022.^[259]

Siemon T. Wezeman, senior researcher at SIPRI's arms transfers and military expenditure programme, stated that information from the UN Register of Conventional Arms 2016, indicated that some Leopard 2A7s were transferred to Singapore after 2014. SIPRI reported that the Singapore Army probably acquired a total of 45 Leopard 2A7s between 2016 and 2019, but the Singapore's Ministry of Defence denied having acquired the 2A7 version, presumably to minimise anxiety among her neighbors.^{[260][261][262]}

In February 2023, Norway ordered 54 Leopard 2A7 tanks to be delivered from 2026, with a further option for 18 vehicles if necessary.^[263] This order was later changed into a newly specified 2A8 NOR subvariant of the 2A8.

Leopard 2A7+

The Leopard 2A7+ was first shown to the public during the Eurosatory 2010, featuring the label "Developed by KMW – tested and qualified by the German Ministry of Defence". The Leopard 2A7+ has been tested by the Bundeswehr under the name *UrbOp* (urban operations).

The Leopard 2A7+ is designed to operate both in low-intensity and high-intensity conflicts.^[264] The tank's protection has been increased by modular armour. The frontal protection has been improved with a dual kit on the turret and hull front, while 360° protection against RPGs and mine protection increase the survivability of the tank in urban operations.^[264] The modular armour's system components were first used



A Leopard 2A7+ at Eurosatory 2010

by Canada in Afghanistan.^[265] It can fire programmable High Explosive munitions. The turret-mounted MG3 has been replaced with a stabilised FLW 200 remotely controlled weapon station. Mobility, sustainability, and situational awareness have also been improved.^[264]

In December 2018, Hungary ordered 44 2A7+s, making them the second operator of the improved version, after Qatar.^{[94][95]}

Leopard 2A8

To replace the 18 Leopard 2A6 tanks handed over to the Armed Forces of Ukraine, the German Ministry of Defence is expected to purchase a batch of at least 18 Leopard 2A8 tanks (with option for several hundred).^[266] The plans still have to be approved by the German parliament due to the contract exceeding a value of €25 million.

Italian government confirmed during a Parliament audition that Italy intends to purchase up to 250 Leopard 2A8 tanks over 2023-2025 period.^[267]

The Leopard 2A8 will be based on the current Leopard 2A7HU production model but feature additional improvements; among them the integration of the Trophy APS and an all-round situational awareness system with sensor-fusion capability.^[266]

The Dutch MoD is planning on procuring up to 52 Leopard 2A8 tanks in addition to their 18 leased Leopard 2A6 tanks.^[268] This planned acquisition is part of the nation's effort to rebuild their army after severe budget cuts of the 21st century.

Leopard 2A8 NOR

The Norwegian armed forces' order of 54 Leopard 2A7NOs was changed into a new 2A8 NOR variant by the middle of June 2023. A variety of Norwegian changes and alterations necessitated the renaming of the original 2A7NO into a unique new 2A8 NOR variant. This subvariant mainly differs from the German 2A8 by integrating Norwegian-designed and manufactured Kongsberg ICS/CORTEX battlefield management systems and a Protector RWS.^[269]

Engineering and driver training tanks

Bergepanzer BPz3 Büffel (Gr. Buffalo)

The BPz3 armoured recovery vehicle includes both a bulldozer and a crane with integral winch, allowing it to approach damaged vehicles, even over rough and fought-over terrain, and tow them to safety. It is equipped with a machine gun for local self-defence, a smoke grenade launcher, and NBC protection. Like the tank, it is powered by a 1,500 PS (1,479 hp, 1,103 kW) diesel engine. It is in service with Germany (where it is also designated *Büffel* or *Bergepanzer 3* for Salvage Tank 3), the Netherlands (who co-developed it and calls it *Buffel*), Canada, Greece, Singapore (where it is called *L2-ARV* locally), Spain (where it is called *Leopard 2ER Búfalo*), Sweden (in modified form as the *Bgbv 120*) and Switzerland (*BPz3*).



A BPz3 "Büffel", German Army

Panzerschnellbrücke 2

This vehicle, created by MAN Mobile Bridges GmbH, is an armoured vehicle-launched bridge developed from the Leopard 2 tank chassis. It is designed to carry a folding mobile bridge, which it can "launch" across a river. Once emplaced, the bridge is sturdy enough to support most vehicles, even other Leopard tanks. When the crossing is complete, the bridge layer simply hooks up to the bridge and re-stows it.

Panzerschnellbrücke Leguan

This modular system combines a bridge module created by MAN Mobile Bridges GmbH with a tank chassis. The Bundeswehr is testing the Leguan on Leopard 2 chassis.^[270]



A Panzerschnellbrücke Leguan folding mobile bridge on a Leopard 2 chassis, demonstrated by the German Army

AEV 3 Kodiak

The AEV 3 Kodiak is a combat engineering vehicle conversion of the Leopard 2 used by the Netherlands, Singapore, Sweden, and Switzerland. It is equipped with a bulldozer blade, excavator arm, and dual capstan winches. In lieu of a turret, a Remote Weapon Station or other armaments can be fitted. It is built on the Leopard 2 chassis with a built-up forward superstructure. The vehicle is used primarily for the clearance of obstacles, including minefields. The Dutch version has additional bomblet protection for the crew compartments. Spain may procure 24 examples for the Spanish Army from converted Leopard 2A4 hulls (one vehicle has been trialled in Spain)^[271] and the type will be offered to Germany.



The AEV 3 Kodiak has an articulated excavator arm and bulldozer blade among its adaptations for obstruction removal.

Driver Training Tank (Fahrschulpanzer)

The Leopard 2 Driver Training Tank, as the name implies, is a non-combatant Leopard 2 for instructing soldiers in the finer points of handling the tank. The turret is supplanted by a weighted and fixed observation cab with forward and side-facing windows and a dummy gun. The instructor rides in this cab, with override controls for critical systems, and space is provided for two other students to observe.

Leopard 2R

A heavy mine breaching vehicle developed by Patria for the Finnish Army, based on the Leopard 2A4. Six vehicles were converted. The vehicles are equipped with a mine plough or a dozer blade, and an automated marking system.^{[272][273]} All were donated to Ukraine in 2023.

Leopard 2L

An armoured vehicle-launched bridge developed by KMW and Patria for the Finnish Army. Ten Finnish 2A4 tanks were re-built to carry the LEGUAN bridge.^{[272][273]}

WISENT 2

A multi-purpose, Leopard 2–based Armoured Support Vehicle developed by Flensburger Fahrzeugbau. The vehicle's modular design allows it to be converted quickly from an

Armoured Recovery Vehicle (ARV) to an Armoured Engineer Vehicle (AEV) in less than five hours.^[274]

Technical data

Technical data^[275]

Description	Leopard 2A4	Leopard 2A5	Leopard 2A6/A6M
Crew:	4		
Engine:	MTU MB 873 Ka-501 12-cylinder twin-turbocharged diesel engine		
Displacement:	Bore × stroke: 170 × 175 mm, 47,666 cm ³ displacement		
Power output:	1,500 PS (1,479 hp, 1,103 kW), rpm: 2,600/min		
Torque output:	4,700 Nm (3,466 lb-ft), rpm: 1,600–1,700/min		
Transmission:	Hydro-mechanical control, reversing and steering gear HSWL 354 with combined hydrodynamic-mechanical service brake, 4 forward, 2 reverse		
Suspension system:	Torsion bar spring mounted support roller drive with hydraulic dampers		
Length Turret forward:	9.67 m		10.97 m
Width:	3.7 m	3.76 m	
Height:	2.79 m	3.03 m	
Ground clearance:	0.54 m		
Wading depth without preparation:	1.2 m		
Wading depth with snorkel:	4 m		
Trench passability:	3 m		
Climbing ability:	1.1 m		
Empty weight:	52 t	57.3 t	57.6 t A6M 60.2 t
Combat weight:	55.15 t	59.5 t	A6 59.9 t (maximum mass; 61.7 t), A6M 62.5 t
Maximum speed:	68 km/h; backwards 31 km/h		
Fuel capacity:	1,160 liters (limited to 900 liters when not in battle)		
Fuel consumption and operating range:	Road: ca. 340 L/100 km, ca. 340 km Terrain: ca. 530 L/100 km, ca. 220 km Average: ca. 410 L/100 km, ca. 280 km Static test: 12.5 L/h, 72–93 hours (with 900–1,160 liters capacity)		
Rotation time (360°):	10 seconds		
Armament:	<u>Rheinmetall 120 mm smoothbore gun L/44</u> and 2 machine guns		<u>Rheinmetall 120 mm smoothbore gun L/55</u> and 2 machine guns
Turret weight:	16 t	21 t	
Turret rotation time:		360° in 9 seconds (electric)	

Operators

Current operators



Leopard 2 operators as of February 2023

Country	Type	Quantity (Estimated)	Origin	Notes
 <u>Austria</u>	2A4	56	 <u>West Germany</u>	The Austrian Army initially acquired 114 Leopard 2A4s from surplus Dutch stocks plus one turret in 1996. In 2006 the number of tanks was reduced to two tank battalions, with 40 of the Leopard 2A4s sold back to KMW in 2011 (Canada lost the bid to purchase them). In 2017–19, a third tank battalion was added and the number of active duty tanks was increased to 56. ^[276] In 2021 it was announced that Austria would modernize its tank fleet to the 2A7 standard.
 <u>Canada</u>	2A4 CAN	34 <i>(used for training)</i>	 <u>West Germany</u>	Acquired 80 Leopard 2A4 from the Netherlands in 2009. An additional 15 Leopard 2A4 tanks were purchased from the German Army for spare parts. 20 modified into 2A4M CAN and 18 converted into AEVs, leaving 42 remaining in active service by 2022 primarily for training. 8 delivered to Ukraine by April 2023. ^[277] It was reported in January 2023 that the majority of Canadian Leopard 2 tanks were in a state of disrepair, with as few as 15 gun tanks, or only 20%, operational and ready for use. ^{[278][279]}

	2A4M CAN	20	 <u>West Germany</u>	Modified from some of the Dutch 2A4s acquired in 2009; shorter turret compared to the 2A6M CAN and slightly different armor configuration.
	2A6M CAN	20	 <u>Germany</u>	20 2A6 initially leased from the German Army beginning in mid-2007 to support the Canadian deployment in Afghanistan. First tank received on 2 August 2007, ^{[245][280]} and arriving in Afghanistan on 16 August 2007. ^[247] Leased vehicles later returned and replaced with 20 tanks purchased from the Netherlands. ^[242]
	BPz3 Büffel ARV	11	 <u>Germany</u>  <u>Switzerland</u>	2 <i>Bergepanzer 3 Büffel</i> purchased from Bundeswehr for use in Afghanistan. ^[281] 12 surplus Swiss Pz 87 were later purchased in 2011 for conversion to ARVs. ^[282] Canada announced it would provide 1 of its 12 ARV's to Ukraine. ^[283]
	Wisent 2 "Ram" AEV	18	 <u>Germany</u>  <u>Canada</u>	FFG Canada received the contract in 2014 to convert 18 of Canada's Dutch Leopard 2A4s into Wisent 2 AEVs. Vehicles were converted domestically in New Brunswick and all 18 were delivered by 2018. ^[284]
 <u>Chile</u>	2A4CHL	132	 <u>West Germany</u>	Acquired in 2007, upgraded to the Leopard 2A4CHL standard (plus 8 to be used as spares). ^{[285][286]}
 <u>Czech Republic</u>	2A4	14 (3 delivered)	 <u>West Germany</u>	In July 2016, officers of the Army of the Czech Republic visited a Spanish military base at Zaragoza where Spanish Leopards 2A4 are stored. The Czech Republic was interested in replacing their domestically produced T-72M4CZ and aging T-72M/T-72M1. No official agreement was signed. ^[287] Later, the Czech Army unofficially announced that the Spanish Leopards were in too poor a condition to be purchased. ^[288] In May 2022 the Czech Ministry of Defence announced it will

				get 15 Leopards 2A4 from Germany as an exchange for Czech tanks that will be given to Ukraine to help defend against Russian invasion and may purchase up to 50 modern 2A7+ variants later. ^{[289][290]} The first Leopard 2A4 was delivered on 21 December 2022. The total number delivered will be 14 pieces of L2A4 and 1 piece of Bpz3 Büffel. ^[136]
	Bpz3 Büffel ARV	1 (pending delivery)	 <u>West Germany</u>	To be delivered as part of "Ringtausch" package.
 <u>Denmark</u>	2A7 DK	44	 <u>Germany</u>	The Royal Danish Army received its first Leopard 2 tanks from surplus German stocks in 1998. ^[87] These were upgraded from 2A4 standard to 2A5DK (equal to Leopard 2A6 minus the L/55 gun) in 2004–2006, at which point the army operated 57 Leopards. ^[88] In 2023, they had 44, which all were upgraded to A7 standard with Danish modifications between 2019 and 2023. ^{[291][292][293]}
	Leguan AVLB	10	 <u>Germany</u>	7 Leopard 2A5DK converted to bridge layers. ^[294] 10 owned as of 2021. ^[295]
	Leopard 2 ARV	10	 <u>Germany</u>	Based on Leopard 2, from 2025 ^[296]
	Leopard 2 MC/AEV	6	 <u>Germany</u>	Based on Leopard 2, from 2025 ^[296]
 <u>Finland</u>	2A4	139 (+27 for spare parts. All 2A4s in reserve)	 <u>West Germany</u>	The Finnish Army originally bought 124 2A4s from surplus German stocks in 2003. ^[297] Of these, 12 were converted into bridge-laying and combat engineering tanks and 12 were disassembled for use as spares, leaving 100 operational tanks. ^{[298][299]} In 2009, the Finnish Army bought 15 more German surplus Leopard 2A4s for spare parts of existing fleet, bringing the total number of Finnish Leopard 2A4 tanks to 139. ^[300]
	2A6	100	 <u>Germany</u>	In January 2014, Finland agreed with the

				Netherlands to purchase 100 used Leopard 2A6NL tanks for approximately €200 million. ^[301] In 2015, most of the Leopard 2A4s were moved into reserve, with some converted into Marksman AA vehicles, bridge-laying, and mine-clearing tanks. ^[302]
	<u>Leopard 2 Marksman</u>	7	 <u>West Germany</u>  <u>Finland</u>  <u>United Kingdom</u>	Seven Leopard 2A4 chassis were used to replaced the T-55AM chassis in 2015 for the Marksman SPAAG.
	Leguan AVLB "Siltapanssarivaunu Leopard 2L"	10	 <u>West Germany</u>  <u>Finland</u>	Six Leopard 2A4s domestically converted into bridge layers in 2008. See <i>2A4 and 2A6 notes</i> . 10 bridgelayers owned as of 2021. ^[295]
 <u>Germany</u>	2A4	N/A	 <u>West Germany</u>	Retired from service. Remaining Leopard 2A4s have been converted to support vehicles, are in long-term storage, or with KMW and Rheinmetall for resale.
	2A5	19	 <u>Germany</u>	Primarily used for training purposes. See <i>2A7V notes</i> .
	2A6	165	 <u>Germany</u>	183 in Bundeswehr stocks as of 2021. ^[295] 18 given to Ukraine from Bundeswehr stocks.
	2A7	57	 <u>Germany</u>	In 2021 Germany had 103 of the 2A7. ^[295] Many have since been upgraded to the 2A7V model. See <i>2A7V notes</i> .
	2A7V	53	 <u>Germany</u>	As of May 2022, the German Army has a total of 312 Leopard 2s with 99 of them being repaired by the armaments industry. Of these Leopard 2s, 53 are the 2A7V version and 19 the A5 version, although the A5 tanks are only used by the German Army to represent enemy tanks in the army's combat training center and have been retired from combat. ^{[303][304][305]} This is to be increased to 328 A6, A6M and A7 tanks in 2026. ^[306]
	2A8	18 on order	 <u>Germany</u>	In 2023 Germany ordered 18 Leopard 2A8 with an

		105 option		option of another 105 from KMW. Deliveries of the 18 Leopard 2A8 replacing the 18 Leopard 2A6 Germany sent to Ukraine begin in 2025. ^[307]
	BPz3 Büffel ARV	75	 <u>West Germany</u>	75 in army stocks as of 2021. ^[295]
	PiPz 3 Kodiak AEV	44	 <u>Germany</u>	44 in army stocks as of 2021. ^[295]
	Leguan AVLB	31	 <u>Germany</u>	31 in army stocks as of 2021. ^[295]
 <u>Greece</u>	2A4	183	 <u>West Germany</u>	In May 2022 it was reported that Greece was in discussions with KMW to upgrade all 183 to the 2A7 HEL standard. ^[308]
	2A6 HEL	170	 <u>Greece</u>	Manufactured domestically in Greece.
	BPz3 Büffel ARV	25	 <u>West Germany</u>	25 operated as of 2021. ^[295]
 <u>Hungary</u>	2A4	12	 <u>West Germany</u>	All twelve A4s arrived as of December 2020. ^[309] See 2A7 notes
	2A7 HU	44	 <u>Germany</u>  <u>West Germany</u>	A deal for 44 Leopard 2A7+ and 12 second hand Leopard 2A4 was signed in December 2018. ^{[94][95]}
	Wisent 2 ARV	6	 <u>Germany</u>	Six owned as of 2021. ^[295]
	BPz3 Büffel ARV	9	 <u>West Germany</u>	Nine owned as of 2021. ^[295]
	Leguan AVLB	3	 <u>Germany</u>	Three owned as of 2021. ^[295]
 <u>Indonesia</u>	2A4+	42	 <u>West Germany</u>  <u>Switzerland</u>	Indonesia sought and obtained approval for the purchase of 103 refurbished Leopard 2A4 (Panzer 87s) tanks from Bundeswehr surplus stocks in 2013. ^[310] In September 2013, the Indonesian Army received the first two Leopard 2A4 tanks. The Leopard 2s have been modified to suit Indonesia's tropical climate, and have been internally renamed as Leopard 2RI (RI for "Republic of Indonesia"). Indonesian Leopard 2s were originally Panzer 87s, ^[311] and use FN MAG, with some using M240C/D

				as coaxial/pintle mounted machine gun. ^[312]
	2RI	61	 <u>West Germany</u>  <u>Switzerland</u>	About 62 of Leopard 2A4 were upgraded to Revolution standard by Rheinmetall. ^[310]
	BPz3 Büffel ARV	4	 <u>West Germany</u>	Purchased with refurbished Leopard 2A4+ tanks. ^[310]
	PiPz 3 Kodiak AEV	3	 <u>West Germany</u>	Purchased with refurbished Leopard 2A4+ tanks. ^[310]
	Leguan AVLB	3	 <u>West Germany</u>	Purchased with refurbished Leopard 2A4+ tanks. ^[310]
 <u>Netherlands</u>	2A6	18 (leased)	 <u>West Germany</u>  <u>Germany</u>	The Royal Netherlands Army operated 445 Leopard 2s. 330 of these were updated to 2A5 standard in 1993, and later, 188 of these were converted to 2A6 standard. Many Leopard 2s were sold after the end of the Cold War. In April 2011, the Dutch Ministry of Defence announced that the last remaining tank division would be disbanded and the remaining Leopard tanks sold due to large budget cuts.^[313] In May 2011, the last tank fired the final shot at the Bergen-Hohne Training Area. They were due to be delivered to the Indonesian Army, which planned to purchase the entire Dutch stock of Leopard 2A6s.^[314] The deal was scrapped after opposition from the Dutch Parliament.^[315] The Dutch Army offered its formerly operated Leopard 2A6s for comparative tests to be conducted by the Peruvian Army for possible acquisition.^[316] By September 2013, the Leopard 2A6 had been disqualified by Peru due to logistical complexities.^[317] The Leopard 2s were eventually sold to Finland in a deal signed in January 2014 for €200 million with deliveries to start from 2015 to 2019.^[318] In September 2015, the Dutch government published that the army would have 16 tanks put out of storage and 18 are currently leased

				from Germany for the joint German-Dutch 414 Tank Battalion. One Leopard 2 tank is exhibited at the military museum. ^[319]
	BPz3 Büffel ARV	25	 <u>West Germany</u>	Operates 25 ARVs as of 2021. ^[295]
	PiPz AEV	10	 <u>Germany</u>	Operates 10 AEVs as of 2021. ^[295]
	Leguan AVLB	10	 <u>Germany</u>	Operates 10 as of 2021. ^[295]
 <u>Norway</u>	2A4NO	38 (28 active)	 <u>West Germany</u>	The Norwegian Army initially purchased 52 ex-Dutch Leopard 2A4s, but a number of them have since been cannibalized for spares and/or been rebuilt into support variants. Only 36 of these tanks are still active and operational as of 2023; 8 were donated to Ukraine. ^{[320][321]}
	2A8 NOR (initially 2A7NO)	54 (pending delivery)	 <u>Germany</u>	On 18 February 2023 it was reported the Norwegian Government had placed an order for 54 new Leopard 2A7NO tanks from KMW. ^[320] Later, in the middle of June, it was announced that the 2A7 order had been upgraded to a newly specified 2A8 NOR variant. The 2A8 NOR builds on the German 2A8 model with integrated EuroTrophy APS, while having certain Norway-specific features such as Kongsberg ICT/CORTEX battlefield management systems and Kongsberg Protector RWS. ^[322]
	Wisent 2 ARV	6+3	 <u>Germany</u>	In May 2015, it was announced that the Norwegian Army had ordered 6 Wisent 2 in the ARV configuration. ^[323] 3 more ARVs were ordered in June 2023. ^[324]
	Wisent 2 AEV	6+8	 <u>Germany</u>	A second order of 6 Wisent 2 was announced in September 2018, but these were delivered in the AEV configuration. ^[325] 8 more AEVs were ordered in June 2023. ^[324]
	Leguan AVLB	6	 <u>Germany</u>	In March 2019, Norway signed an agreement with Krauss-Maffei Wegmann

				for the procurement of 6 new LEGUAN bridgelayers, with deliveries to begin in summer 2022. ^{[326][327]}
 <u>Poland</u>	2A4	86	 <u>West Germany</u>	The first batch of 128 Leopard 2A4s (produced between 1985 and 1987) as well as 49 other armoured vehicles (like Bergepanzer 2 ARVs and M113 family APCs) and 151 trucks and 4x4s was transferred to Poland in 2002 and 2003 for 100 million PLN. They are used by the 10th Armoured Cavalry Brigade based in Świętoszów. An additional 14 were purchased with Leopard 2A5s in 2013. ^[328] See <i>2A4PL notes</i> 14 of Poland's 100 2A4s are being donated to Ukraine; training began in early February 2023 and the first four were delivered on 24 February. ^[329]
	2A5	105	 <u>West Germany</u>  <u>Germany</u>	105 Leopard 2A5s were purchased with 14 more 2A4s and other equipment by Poland in 2013 for an estimated 180 – 200 million euro. The last tank was delivered on 30 November 2015. ^[328]
	<u>2PL</u>	45	 <u>West Germany</u>  <u>Germany</u>  <u>Poland</u>	128 Leopard 2A4 were scheduled to be upgraded to Leopard 2PL standard, in a contract worth 2,415M PLN that was signed in December 2015. In 2018 a follow-on deal to upgrade a second batch of 14 Leopard 2 A4s was signed. ^[330] As of July 2022, 45 have been converted as of February 2023. ^{[329][331]} The Leopard 2PL will introduce new sighting equipment for the gunner, commander and driver, an increased protection level, an upgraded gun, an upgraded fire suppression system and the installation of an auxiliary power unit. ^{[332][333][253]}
 <u>Portugal</u>	2A6	34	 <u>Germany</u>	The Portuguese Army operates 37 ex-Dutch Leopard 2A6s, acquired in 2008 for €80 million. Also bought 1 for training and 1

				for spares in A4 version. ^[334] The Army, in 2010, planned to acquire 18 more units, leaving a total of 55 Leopard 2A6s, but the Ministry of National Defence denied the purchase of additional units due to budget cuts. ^[335] The <i>Military Programing Law</i> signed in 2019, provides for the modernization of all Leopard 2A6s from 2026 to 2030, and may also receive an active protection system. ^{[336][337]} 3 donated to Ukraine in 2023.
	Leopard 2 Fahrschulpanzer	1	 Germany	One unit acquired in 2008 to train Leopard 2A6 new drivers. ^[334]
 <u>Qatar</u>	2A7+	62	 Germany	Qatar signed a contract for 62 Leopard 2A7+ tanks in April 2013. ^[338] Deliveries commenced in late 2014/early 2015 ^[339] and were completed in 2018. ^[340] The first units were displayed at Qatar's annual national day parade in December 2015. ^[341]
	Wisent 2 ARV	6	 Germany	Six owned as of 2021. ^[295]
 <u>Singapore</u>	2SG	158	 West Germany  Germany	The Singapore Army acquired 96 ex-German Leopard 2A4s, including 30 spare tanks. A number were upgraded with additional AMAP composite armour in 2010 by IBD Deisenroth and ST Kinetics and renamed Leopard 2SG in October 2010. By 2019, it was reported that Singapore had taken delivery of 158 Leopard 2A4s and 45 Leopard 2A7s. However, the Singapore Government has denied that it received deliveries of Leopard 2A7s. The latest export numbers indicate that Singapore to have received more than 224SG Leo 2 tanks so far. ^{[342][343][262][344][345]}
	2A7	45	 Germany	See 2SG notes
	BPz3 Büffel ARV	20	 West Germany	20 owned as of 2022. ^[346]

	PiPz 3 Kodiak AEV	14	 Germany  Switzerland	14 owned as of 2022. ^[347]
	Leguan AVLB	10	 Germany	10 owned as of 2013. ^[348]
 Slovakia	2A4	15 (3 delivered)	 West Germany	In August 2022 the Slovak Ministry of Defence announced it will get 15 Leopards 2A4 from Germany in an exchange for its 30 tracked BMP-1 infantry fighting vehicles from reserve that will be given to Ukraine to help defend against the Russian invasion. Germany's tank package includes ammunition, training, and spare parts. The plans for future modernisations or the purchase of modern versions of the Leopard 2 were not revealed at the time.^{[349][350]} The first Leopard 2A4 was delivered on 19 December 2022.^[351] The third Leopard 2A4 was delivered on 2 June 2023.^[352]
 Spain	2A4	98 (55 active)	 West Germany	108 leased from Germany for training purposes in 1995, and purchased them in 1998. As of 2023, 55 are in active service with Montesa" n°3 and Alcántara n°10 Armoured Cavalry Regiments^[353] The remaining 53 have been stored in Zaragoza. 11 Leopard 2 A4 have been removed from the Zaragoza storage facility to be refurbished to "have the same lethality as modern Leopard 2".^[354] Spain is considering sending 40 of the 53 stored at Zaragoza, though they require extensive repairs due to a flood in 1997; the remaining 13 are turretless or unserviceable.^[355] At least 10 have been committed to Ukraine.^{[356][170]} On 5 April, Defence Minister Margarita Robles announced that Spain would send 6 Leopard 2A4s to Ukraine by the second half of April.^[357]

	2E	219	 <u>Spain</u>	Santa Bárbara Sistemas manufactured 60% of the contract under license. 219 main battle tanks + 4 training tanks.
	BPz3 Büffel ARV	16	 <u>West Germany</u>	16 owned as of 2021 ^[358]
 <u>Sweden</u>	Stridsvagn 122 (2A5 equivalent)	110 (42 active)	 <u>West Germany</u>  <u>Sweden</u>	In August 1994, 160 ex-German Leopard 2A4s were leased and received minor modifications. They were used under the designation Stridsvagn 121 until the reorganisation of the <u>Swedish Army</u> in 2000 when they were placed in storage until the lease contract expired in 2011. 20 Strv 121 tanks were purchased for conversion into engineering and bridge-laying vehicles, and the 140 remaining tanks were returned to Germany. Sweden has acquired 120 Leopard 2 Improved tanks, upgrading them as the Stridsvagn 122, with 42 Strv 122 tanks remaining in active service. ^{[359][360][361]} Sweden confirmed it was delivering 10 of its 120 to Ukraine.
	BPz3 Büffel ARV	14	 <u>West Germany</u>	14 owned as of 2021. ^[295]
	PiPz 3 Kodiak AEV	6	 <u>Germany</u>  <u>Switzerland</u>	Six owned as of 2021. ^[295]
	Leguan AVLB	10	 <u>Germany</u>	10 owned as of 2021. ^[295]
	Pz 87 WE	134	 <u>Switzerland</u>	Modernized to the Pz 87 WE standard in 2011. See <i>Pz 87 notes</i> for further details.
 <u>Switzerland</u>	Pz 87	96 (in storage)	 <u>West Germany</u>  <u>Germany</u>  <u>Switzerland</u>	The Swiss Army purchased 380 Leopard 2A4s designated <i>Panzer 87</i> . 35 were manufactured by KMW and Rheinmetall, while 345 were domestically made in Switzerland. 42 were later sold back to Rheinmetall, ^[362] and 12 were sold to the <u>Canadian Army</u> and were transformed in <i>L2-ARV (BPz3 Büffel ARV)</i> .

As of 2023, 134 upgraded Pz 87 WE tanks are active while the remaining 96 Pz 87s were decommissioned and placed in long term storage,^[363] and 49 Pz 87 have been transformed in engineering variants since 2011 (see below).

On 3 March 2023 it was reported that Germany sought to purchase an unknown number of Switzerland's 96 reserve tanks to backfill those given to Ukraine by other allies.^[364]

Rheinmetall CEO Armin Papperger later confirmed in an interview the company sought to purchase 36 of these tanks.^[365] Switzerland's Security Commission voted to fully decommission 25 of the tanks, enabling their re-sale back to German manufacturer KMW pending approval from parliament.^[366]

BPz3 Büffel ARV	25	 <u>West Germany</u>	From 2004 ^[367]
PiPz 3 Kodiak AEV	12	 <u>Germany</u>  <u>Switzerland</u>	12 Pz 87 transformed in Kodiak variants, and with it several mission modules were purchased for more flexibility: ▪ 12 modules for engineering ▪ 6 modules for demining operations ▪ 4 mine protection sets

				[368]
	Leguan AVLB	12	 <u>Germany</u>	From 2018 [363]
 <u>Turkey</u>	2A4	250*	 <u>West Germany</u>	354 originally purchased from Germany. At least 8 visually confirmed destroyed in Syria; another 13 damaged or captured by ISIS. See <i>Turkish intervention in Syria</i> above for details. Others are being upgraded to the 2A4TR model. See <i>2A4TR notes</i> .
	2A4TR	84	 <u>West Germany</u>  <u>Turkey</u>	84 units are undergoing modernization and will be renamed as Leopard 2A4TR. The modernization program was awarded to Roketsan and the first tank was re-delivered to the Turkish Army in February 2021. The tanks undergoing modernization are being equipped with new ERA panels, fire control systems, and a full overhaul of the power transmission systems.
 <u>Ukraine</u>	2A4	≥ 90 (40 delivered: 34 active, 2 destroyed, 4 damaged). [369][370]	 <u>West Germany</u>	As of 23 February 2023, confirmed transfers to Ukraine from army stockpiles include 14 2A4 from Poland, 8 2A4NO from Norway, 8 2A4CAN from Canada, 10 2A4 from Spain, an unknown number of battle tanks from Finland. The Netherlands agreed to help procure and supply ammunition for the tanks. Ukrainian tank operators began training on the vehicles in February 2023 and they are expected to arrive in Ukraine in March. The first 4 Leopard 2A4 tanks from Polish stocks were delivered to Ukraine on 24 February 2023. [371] The remaining 10 arrived after 7 March. The 8 Norwegian tanks were delivered on 20 March 2022. [372] All 8 Canadian Leopard 2A4CAN were delivered by 18 April 2023. [277] Spain delivered its first 6 in late April, and the remaining four were to arrive in Poland by June 19 and transferred to Ukraine. [373] An additional

			50 Leopard 2A4 tanks are being prepared for Ukraine by Rheinmetall, with 30 refurbished as of 4 March 2023.^{[374][375]} The remaining 20 tanks could be ready by the end of 2023 or early 2024, according to a company spokesperson.^[376] On 20 April 2023 the Netherlands and Denmark announced the joint procurement of 14 Leopard 2A4 from German defense industry stocks for Ukraine, with delivery scheduled for Q1 2024.^[151]
Stridsvagn 122 (2A5 equivalent)	10	 Germany  Sweden	On 24 February 2023, the Swedish Minister of Defence Pål Jonson announced that Sweden would be providing 10 Leopard 2A5 tanks to Ukraine.^[172] These are presumably domestically manufactured Stridsvagn 122 variant (2A5 equivalent) as they are the only Leopards operated by Sweden. Delivered by July 2023.
2A6	21 (12 active, 3 destroyed, 6 damaged) ^[369]	 Germany	Ukraine was originally provided 14 from Germany and an additional 3 from Portugal. Training on the 2A6 began in Germany in February 2023 and they are expected to arrive in Ukraine in March. Germany's Defense Ministry announced it would send an additional 4 Leopard 2A6 on 24 February to ensure one full tank battalion.^[377] Germany delivered all 18 tanks promised as of 27 March 2023.^[378] Portugal's Ministry of Defense confirmed that the 3 Leopard 2A6 pledged were also delivered to Ukraine.^[379] The leaked Pentagon documents revealed that Greece may be donating 5 Leopard 2A6HEL and Portugal an additional 3 2A6 tanks to Ukraine, though these deliveries have not been officially confirmed by either government.^{[159][380]}

	BPz3 Büffel ARV	3 (2 active, 1 damaged) ^[369]	 <u>West Germany</u>	On 24 February 2023, Canada confirmed it would provide 1 of its ARVs with its next delivery of four Leopard 2 tanks. ^[283] Canada delivered its BPz3 on 18 March 2023. ^[381] Germany confirmed it was providing 2; both delivered by 27 March. ^[382]
	2R Minebreacher "Raivauspanssarivaunu Leopard 2R"	6 (3 active, 1 destroyed, 2 damaged) ^[369]	 <u>Germany</u>  <u>Finland</u>	Finland first announced it was providing 3 Leopard 2R heavy mine clearing vehicles and an undisclosed amount of main battle tanks. ^[383] An additional 3 of the 2R mine clearers were later promised. ^[384] 3 were reportedly abandoned in battle in June 2023. ^[385]
 <u>United Arab Emirates</u>	Wisent 2 ARV	4	 <u>Germany</u>	First ordered in 2016. ^[386] Four owned as of 2021. ^[295]

Potential operators

-  Italy: In March 2023 it was reported that the Italian Army is considering the purchase of up to 250 Leopard 2A7 tanks for roughly €8 billion. ^[387] In July of the same year, Italy's Defence undersecretary Isabella Rauti said that her government is planning to buy no less than 125 Leopard 2A8 MBT and support variants, with initial funds worth 4 billion euros (\$4.5 billion) from 2024 (inside the multi-year defence budget of 2023-2025) to purchase the new tanks and upgrade other military hardware; there's also the option to invest the whole sum to acquire a total of around 250 of such vehicles. This investment would carry through to 2037. ^{[388][389]}
-  Lithuania: In July 2023, Lithuanian MoD sent a letter of intent to purchase Leopard 2 from German KMW. ^[390] It was previously reported that the country seeks to acquire 54 tanks. ^[391] German media suggested that Lithuania is considering the latest Leopard 2A8 variant. ^[392]

Failed bids

-  Bulgaria: The Bulgarian Ministry of Defence has been interested in purchasing a minimum of 24 units of the Leopard 2A6 variant. The deal has received support from a military budget increase to 2.0% of GDP by 2016 in the new Bulgarian budget, and a military pledge to spend €2.2 billion on new armaments for the Bulgarian military. ^[393]
-  Croatia: The Croatian Ministry of Defence is looking to replace its M-84A4 tanks currently in use. Negotiations with Germany over the purchase of stored Leopard 2A5 tanks were underway in 2014 and 2015. The Milanović Government was negotiating the purchase of up to 50 tanks with spare parts and a support package including spare engines, tank transporters, and engineering vehicles valued at 875 million kunas. The purchase would have coincided with the purchase of PzH 2000 howitzers. Due to budget constraints, at least for the time being, the vehicles were not purchased. Interest still remains high, as Croatia has no plans to modernise or maintain its M-84A4 tanks beyond 2020. ^[394]

-  Peru: The Dutch Army offered its formerly operated Leopard 2A6s for comparative tests to be conducted by the Peruvian Army for possible acquisition.^[316] By September 2013, the Leopard 2A6 had been disqualified by Peru due to logistical complexities.^[317]
-  Romania: As part of a modernization program, since 2019 the Romanian Army has been considering the acquisition of Leopard 2 tanks.^[395] However, in March 2023 Romania's Ministry of Defense announced they had opted to purchase around 54 M1 Abrams tanks from the United States.^[396]
-  Saudi Arabia: The Saudi Arabian government sought to buy Leopard 2A7s (total of 600–800 desired). In early July 2011, the German press reported that the *Bundessicherheitsrat* (Federal Security Council) approved the sale by KMW of more than 200 units of the 2A7+ tanks to Saudi Arabia.^{[397][398]} This news was met with criticism both inside and outside of Germany, because of the autocratic nature of the Saudi Arabian state and its involvement in repressing popular protests in the neighboring country of Bahrain.^[399] Criticism also came from within the Chancellor Angela Merkel's government coalition,^[400] and, later from within KMW.^[401] In June 2012, reports surfaced that Saudi Arabia had raised the number of tanks it was interested into 600–800. A contract was never finalized, and the issue was debated both in the German public and in Germany's federal parliament.^[402] In April 2014, a German newspaper reported that the deal for Leopard 2 tanks for Saudi Arabia was likely to be cancelled due to opposition from the Federal Economy Minister at that time, Social Democrat Sigmar Gabriel.^[403] In 2015, Germany blocked the sale of Leopard 2 tanks to Saudi Arabia.^[404] In 2016, the Saudi Arabian ambassador to Germany stated that Saudi Arabia was no longer interested in purchasing Leopard tanks and suggested that the issue was being exploited for internal political gain.^{[405][406]}

See also

- Leopard 1
- List of main battle tanks by generation
- Panther KF51
- Tanks in the German Army

Notes

1. In *King of the Killing Zone* Orr Kelly reported that after knocking on one Leopard 2AV prototype, a hollow sound was discovered by Colonel Robert J. Sunell. When confronting the German Colonel Kettman, he admitted that PT19 (the mobility test bed) was not fitted with any special armour. Orr Kelly claims that the Leopard 2AV would have weighed 64 short tons instead of 59.6 short tons and as a result, the performance data of the mobility was invalidated. However, the weight value claimed by Orr Kelly does not match the actual weight of a Leopard 2 fitted with special armour nor does it match the weight mentioned in US documents from the evaluation.

References

1. Gelbart, Marsh (1996). *Tanks: main battle and light tanks*. Brassey's UK Ltd. pp. 109–110. ISBN 978-1-85753-168-8. OCLC 36171829 (<https://www.worldcat.org/oclc/36171829>).
2. Majumdar, Dave (18 February 2016). "Get Ready, Russia: This European Power Has Plans For a Lethal New Tank" (<https://nationalinterest.org/blog/the-buzz/get-ready-russia-european-power-has-plans-lethal-new-tank-15251>). The National Interest.

3. "Canada Acquires 120 Leopard 2 Tanks from German, Dutch Surplus" (https://web.archive.org/web/20091231060606/http://defense-update.com/newscast/0407/news/160407_leo2.htm). *Defense update*. 16 April 2007. Archived from the original (http://defense-update.com/newscast/0407/news/160407_leo2.htm) on 31 December 2009. Retrieved 12 October 2010.
4. "Hungarian Leopard MBTS Unveiled. What was the cost? [ANALYSIS]" (<https://defence24.com/hungarian-leopard-mbts-unveiled-what-was-the-cost-analysis>). 7 July 2020.
5. "Deutsche Panzer für Katar" (<https://www.tagesspiegel.de/wirtschaft/deutsche-panzer-fuer-katar/8371316.html>). *Tagesspiegel Online* (in German). 19 June 2013. Retrieved 26 October 2019.
6. Foss, Christopher F. (2002), *Jane's Tanks and Combat Vehicles Recognition Guide*, New York: HarperCollins, p. 32
7. "Weapon systems and heavy equipment in the Bundeswehr" (<https://www.bundeswehr.de/resource/blob/5031824/7b51f1e2b8e19099dacc615881c591a4/waffensysteme-und-grossgeraet-data.pdf>) (PDF). Retrieved 9 June 2022.
8. "Die Leistungsmerkmale des LEOPARD 2 A6M" (<https://www.kmweg.com/systems-products/tracked-vehicles/main-battle-tank/leopard-2-a6/>). Retrieved 30 June 2021.
9. "LEOPARD 2 A6 – KMW" (<https://www.kmweg.com/systems-products/tracked-vehicles/main-battle-tank/leopard-2-a6/>). kmweg.com. Retrieved 1 July 2021.
10. "The Russians Just Destroyed Their First Ukrainian Leopard 2 Tank" (<https://www.forbes.com/sites/davidaxe/2023/06/08/the-russians-just-destroyed-their-first-ukrainian-leopard-2-tank/?sh=6ec7def175c9>). *Forbes*. 9 June 2023. Retrieved 10 June 2023.
11. Jerchel, p. 3.
12. Krapke, p. 19.
13. Krapke, p. 20.
14. Krapke, p. 22.
15. Jerchel, p. 5.
16. Hilmes, Rolf. "50 Jahre gepanzerte Kampftruppen" (<https://www.panzertruppe.com/detailansicht/id-50-jahre-fahrzeuge-der-gepanzerten-kampftruppen.html>). *Panzertruppe*. Archived (<https://web.archive.org/web/20180712001316/https://www.panzertruppe.com/detailansicht/id-50-jahre-fahrzeuge-der-gepanzerten-kampftruppen.html>) from the original on 12 July 2018. Retrieved 22 February 2018.
17. Jerchel, p. 6.
18. Hilmes, Rolf (1983). *Kampfpanzer Entwicklungen der Nachkriegszeit* (in German). Soldat Und Technik. p. 27. ISBN 9783524890012.
19. Krapke, p. 27.
20. Jerchel, p. 7.
21. Krapke, p. 30.
22. Krapke, Paul-Werner (1984). *Das Waffensystem Kampfpanzer Leopard 2* (in German). Allgemeine schweizerische Militärzeitschrift.
23. Krapke, p. 32.
24. Jerchel, p. 8.
25. Krapke, p. 33.
26. Krapke, p. 34.
27. Krapke, pp. 37 and 38.
28. Staats, Elmer B. (28 November 1977). *Department Of Defense Consideration Of West Germany's Leopard As The Army's New Main Battle Tank*. United States: U.S. Government Accountability Office.
29. Krapke, p. 35.

30. "Orientalische Sitten" (<http://www.spiegel.de/spiegel/print/d-40941854.html>). Der Spiegel. 14 March 1977. Archived (<https://web.archive.org/web/20160209170007/http://www.spiegel.de/spiegel/print/d-40941854.html>) from the original on 9 February 2016. Retrieved 16 December 2015.
31. Krapke, p. 31.
32. Krapke, p. 41.
33. Krapke, pp. 54 and 55.
34. Krapke, p. 38.
35. Jerchel, pp. 10 and 11.
36. Krapke, p. 39.
37. Jerchel, pp. 20 and 21.
38. Norway wants to buy dozens of new Leopard 2 tanks (<https://www.defensenews.com/global/europe/2023/02/03/norway-wants-to-buy-dozens-of-new-leopard-2-tanks/>). *Defense News*. 3 February 2023.
39. Hilmes, Rolf (30 November 2007). *Kampfpanzer heute und morgen: Konzepte – Systeme – Technologien* (in German). Germany: Motorbuch Verlag. p. 68. ISBN 978-3613027930.
40. Hilmes, Rolf (30 November 2007). *Kampfpanzer heute und morgen: Konzepte – Systeme – Technologien* (in German). Germany: Motorbuch Verlag. p. 70. ISBN 978-3613027930.
41. Hilmes, Rolf (30 November 2007). *Kampfpanzer heute und morgen: Konzepte – Systeme – Technologien* (in German). Germany: Motorbuch Verlag. p. 69. ISBN 978-3613027930.
42. Hilmes, Rolf (30 November 2007). *Kampfpanzer heute und morgen: Konzepte – Systeme – Technologien* (in German). Germany: Motorbuch Verlag. p. 71. ISBN 978-3613027930.
43. "Germany receives first Leopard 2A7 MBT" (<https://www.janes.com/article/46904/germany-receives-first-leopard-2a7-mbt>). IHS Jane's Defence Weekly. Archived (<https://web.archive.org/web/20150501043922/http://www.janes.com/article/46904/germany-receives-first-leopard-2a7-mbt>) from the original on 1 May 2015. Retrieved 22 December 2014.
44. Alessa Weber; Patricia Franke (12 December 2014). "Neue "Leos" für das Heer" (http://www.bundeswehr.de/portal/a/bwde/!ut/p/c4/NYvBDolwEET_qEsTE9GbwMWb4aDihSx0QxpL2yxbufjxtgdnknd5M_CCXI8fu6DY4NHBE4bZnqddTbshhW9J5BxtCtM2GuJROMVI8CjHPJiDJykU8mlzF0YJrGJgccUk5myUNTBUmV0sfpHf0_9rW3vh7rurk0PcV0vP2ckU98!). *bundeswehr.de* (in German). Bundeswehr. Archived (https://web.archive.org/web/20160406164307/http://www.bundeswehr.de/portal/a/bwde/!ut/p/c4/NYvBDolwEET_qEsTE9GbwMWb4aDihSx0QxpL2yxbufjxtgdnknd5M_CCXI8fu6DY4NHBE4bZnqddTbshhW9J5BxtCtM2GuJROMVI8CjHPJiDJykU8mlzF0YJrGJgccUk5myUNTBUmV0sfpHf0_9rW3vh7rurk0PcV0vP2ckU98!) from the original on 6 April 2016. Retrieved 29 March 2016.
45. Jungholt, Thorsten (26 April 2015). "Bundeswehr-Kampfpanzer Leopard 2 fehlt wirksame Munition" (<https://www.welt.de/politik/deutschland/article140083741/Bundeswehr-Kampfpanzer-fehlt-wirksame-Munition.html>). *DIE WELT*. Retrieved 30 December 2020.
46. "Why 130 mm and not a bigger calibre?" (<https://www.edrmagazine.eu/why-130-mm-and-not-a-bigger-calibre>). *EDR online*. 5 March 2021. Retrieved 28 March 2023.
47. Germany, France to jointly develop 'Leopard 3' tank (<http://www.dw.de/germany-france-to-jointly-develop-leopard-3-tank/a-18468270>) Archived (<https://web.archive.org/web/20150525214308/http://www.dw.de/germany-france-to-jointly-develop-leopard-3-tank/a-18468270>) 25 May 2015 at the *Wayback Machine* – DeutscheWelle.de, 22 May 2015.
48. Get Ready, Russia: This European Power Has Plans For a Lethal New Tank (<http://nationalinterest.org/blog/the-buzz/get-ready-russia-european-power-has-plans-lethal-new-tank-15251>) Archived (<https://web.archive.org/web/20160607182023/http://nationalinterest.org/blog/the-buzz/get-ready-russia-european-power-has-plans-lethal-new-tank-15251>) 7 June 2016 at the *Wayback Machine* – Nationalinterest.org, 18 February 2016.

49. Foss, Christopher F. (2005). *Jane's Armour and Artillery 2005–2006*. Jane's IHS. p. 36. ISBN 978-0710626868.
50. Krapke, Paul-Werner (1984). *Das Waffensystem Kampfpanzer Leopard 2* (in German). Allgemeine schweizerische Militärzeitschrift ASMZ Nr. 5/1984. p. 240.
51. Hilmes, Rolf (1983). *Kampfpanzer Entwicklungen der Nachkriegszeit* (in German). Report Verlag GmbH. p. 76.
52. Krapke, Paul-Werner (2004). *Leopard 2 sein Werden und seine Leistung* (in German). Books on Demand GmbH. p. 60. ISBN 978-3833414251.
53. Krapke, Paul-Werner (2004). *Leopard 2 sein Werden und seine Leistung*. Books on Demand GmbH. p. 59. ISBN 978-3833414251.
54. *Chieftain tanks: engine problems and improvements; main battle tank armour development (Burlington) proposed collaboration with Germany (DEFE 13/1369)* (<http://discovery.nationalarchives.gov.uk/details/r/C11361803>). National Archives. 10 March 1970. Archived (<https://web.archive.org/web/20160107041000/http://discovery.nationalarchives.gov.uk/details/r/C11361803>) from the original on 7 January 2016. Retrieved 15 May 2015.
55. Bischofberger, Walter (1980). *Panzerwaffen : Feuerkraft und Panzerung im Vergleich* (in German). Allgemeine schweizerische Militärzeitschrift ASMZ Nr. 12/1980. p. 697.
56. "waffenHQ.de: Leopard 2" (<http://www.whq-forum.de/cms/27.0.html>) (in German). Archived (<https://web.archive.org/web/20111125204600/http://www.whq-forum.de/cms/27.0.html>) from the original on 25 November 2011. Retrieved 3 January 2012.
57. "Main Battle Tank – Leopard 2" (<http://www.fprado.com/armorsite/leo2.htm>). Fabio Prado. 12 December 2009. Archived (<https://web.archive.org/web/20100802143624/http://www.fprado.com/armorsite/leo2.htm>) from the original on 2 August 2010. Retrieved 12 May 2010.
58. Swedish Defence Materiel Administration: Leopard 2 – Stridsvagn 122 (Swedish) (<https://web.archive.org/web/20050210213646/http://www.fmv.se/WmTemplates/Page.aspx?id=1106>)
59. Lobitz, Frank (20 June 2009). *Kampfpanzer LEOPARD 2 – Entwicklung und Einsatz in der Bundeswehr* (in German). Vollert - Tankograd Publ. p. 124. ISBN 978-3936519082.
60. Scheibert, Michael (1996). *Leopard 2 A5 – Euro-Leopard 2*. Podzun-Pallas-Verlag GmbH. p. 60. ISBN 3-7909-0576-3.
61. Van Oosbree, Gerard (1998). *ARMOR Magazine article: Germany's Big Guns* (<http://www.ciar.org/ttk/mbt/armor/armor-magazine/armor-mag.1998.mj/3leo2.pdf>) (PDF). U.S. Army Armor Center. p. 29. Archived (<https://web.archive.org/web/20160107041000/http://www.ciar.org/ttk/mbt/armor/armor-magazine/armor-mag.1998.mj/3leo2.pdf>) (PDF) from the original on 7 January 2016. Retrieved 15 May 2015.
62. Foss, Christopher F. (2005). *Jane's Armour and Artillery 2005–2006*. Jane's IHS. p. 33. ISBN 978-0710626868.
63. "Germany receives first Leopard 2A7 MBT" (<https://www.janes.com/article/46904/germany-receives-first-leopard-2a7-mbt>). *janes.com*. 10 December 2014. Archived (<https://web.archive.org/web/20150501043922/http://www.janes.com/article/46904/germany-receives-first-leopard-2a7-mbt>) from the original on 1 May 2015. Retrieved 15 May 2015.
64. *LEOPARD 2A7. Y – DAS MAGAZIN DER BUNDESWEHR* (German Bundeswehr magazine). March 2015. p. 63.
65. Hilmes, Rolf (30 November 2007). *Kampfpanzer Heute und Morgen* (in German). Motorbuch Verlag. p. 217. ISBN 978-3613027930.
66. "Canadian Forces Armour — Leopard 2A6M CAN Main Battle Tank" (<https://web.archive.org/web/20100627144153/http://www.casr.ca/101-army-armour-leopard-2a6m.htm>). Canadian American Strategic Review. Archived from the original (<http://www.casr.ca/101-army-armour-leopard-2a6m.htm>) on 27 June 2010. Retrieved 11 May 2010.

67. Eshel, Tamir (23 February 2021). "Israel to Equip German Leopard Tanks with TROPHY APS" (https://defense-update.com/20210223_leopard2-trophy.html). *Defense Update*. Retrieved 26 April 2021.
68. "Battle tank Leopard 2/Kampfpanzer Leopard 2" (<https://fas.org/man/dod-101/sys/land/row/leopard2.htm>). Fas.org. 18 June 1999. Archived (<https://web.archive.org/web/20111209004055/http://www.fas.org/man/dod-101/sys/land/row/leopard2.htm>) from the original on 9 December 2011. Retrieved 3 January 2012.
69. *Jane's Ammunition Handbook* (subscription), "120 mm DM 33A1 and DM 43A1 APFSDS-T rounds (Germany)" (https://archive.today/20130103130425/http://www.janes.com/extracts/extract/jah/jah_0303.html). Retrieved 11 November 2008.
70. *Worldwide Equipment Guide (WEG)*, volume 1: ground systems (<http://cgsc.contentdm.oclc.org/cdm/ref/collection/p4013coll11/id/2089>) Archived (<https://web.archive.org/web/20150214010624/http://cgsc.contentdm.oclc.org/cdm/ref/collection/p4013coll11/id/2089>) 14 February 2015 at the Wayback Machine. US ARMY TRADOC G-2, December 2011.
71. Jerchel, Michael; Uwe Schnellbacher (1998). *Leopard 2 Main Battle Tank 1979–1998*. Oxford, United Kingdom: Osprey. p. 22. ISBN 1-85532-691-4.
72. "Rheinmetall 120 mm L/44 smoothbore gun (Germany)" (<https://web.archive.org/web/20090220223645/http://www.janes.com/extracts/extract/jaau/jaau0021.html>). *Jane's Armour & Artillery*. Archived from the original (<https://www.janes.com/extracts/extract/jaau/jaau0021.html>) on 20 February 2009. Retrieved 7 November 2008.
73. Foss, Christopher F (2006). *Jane's Tank Recognition Guide, 4th Edition*. Hammersmith, London: HarperCollins Publishers. p. 35. ISBN 978-0-00-718326-5.
74. "Lahat Leopard 2 Upgrade" (<https://web.archive.org/web/20131103052841/http://www.rheinmetall-detec.de/index.php?fid=4113&lang=3&pdb=1>). Germany: Rheinmetall Defence AG. Archived from the original (<http://www.rheinmetall-detec.de/index.php?fid=4113&lang=3&pdb=1>) on 3 November 2013. Retrieved 11 October 2010.
75. "Leopard 2 Main Battle Tank, Germany" (<http://www.army-technology.com/projects/leopard/>). ArmyTechnology.com. Archived (<https://web.archive.org/web/20151117040254/http://www.army-technology.com/projects/leopard/>) from the original on 17 November 2015. Retrieved 13 May 2010.
76. Jerchel, Michael (1998). *Leopard 2 : Main Battle Tank, 1979–1998*. Schnellbacher, Uwe. London: Osprey Military. pp. D. ISBN 9781855326910. OCLC 40544103 (<https://www.worldcat.org/oclc/40544103>).
77. Front Cover Richard Holmes; Hew Strachan; Chris Bellamy (2001). *The Oxford companion to military history* (<https://books.google.com/books?id=ro8YAAAAIAAJ&q=leopard+>). Oxford University Press. pp. 493, 902. ISBN 9780198662099. Archived (https://web.archive.org/web/20140101184545/http://books.google.com/books?ei=MQ6aTYLeIiL0QHR_-WHDA&ct=result&hl=com&id=ro8YAAAAIAAJ&dq=%22main+battle+tank%22&q=leopard+&redir_esc=y) from the original on 1 January 2014. Retrieved 4 April 2011.
78. "Krauss-Maffei Wegmann" (<https://web.archive.org/web/20100301161335/http://www.kmweg.de/10894---PRODUKTE~kettenfahrzeuge~leopard2~leopard2.html>). Archived from the original (<http://www.kmweg.de/10894---PRODUKTE~kettenfahrzeuge~leopard2~leopard2.html>) on 1 March 2010.
79. Jerchel, p. 36.
80. Jerchel, p. 38.
81. Jerchel, p. 40.
82. "Kingdom of Sweden (Modern)" (<https://tanks-encyclopedia.com/sweden-modern-nation-page/>). *Tanks Encyclopedia*. Retrieved 25 January 2023.
83. "Projekt "Stridsvagn Ny" " (http://www.ointres.se/projekt_stridsvagn_ny.htm). *www.ointres.se*. Retrieved 13 May 2022.

84. Mulcahy, Paul. "Swedish Tanks" (https://www.pmulcahy.com/tanks/swedish_tanks.html). Retrieved 25 January 2023.
85. "Leopard 2 (October 2019 update)" (https://janes.ihs.com/Janes/Display/JAA_0021-JAFV). Jane's UK Ltd. 16 October 2019. Retrieved 17 February 2020.
86. "AEV Kodiak" (<https://tanks-encyclopedia.com/modern-switzerland-germany-aev-3-kodiak/>). Online Tank Museum. 18 July 2019. Retrieved 17 February 2020.
87. "Jensby tilfreds med kostbare kampvogne" (<https://jyllands-posten.dk/indland/ECE3397964/Jensby-tilfreds-med-kostbare-kampvogne/>) (in Danish). Jyllands-Posten. 11 September 2003. Retrieved 27 January 2023.
88. "Financial Report of the Danish Defense Ministry" (http://www.folketinget.dk/img20031/udvbilag/lib2/20031_23282/20031_23282.pdf) (PDF) (in Danish). 3 August 2004. Archived (https://web.archive.org/web/20190708123048/http://webarkiv.ft.dk/img20031/udvbilag/lib2/20031_23282/20031_23282.pdf) (PDF) from the original on 8 July 2019.
89. "Greece, Krauss-Maffei Wegmann sign procurement contract for 170 Leopard 2 HEL MBTs" (<http://www.defense-aerospace.com/articles-view/release/3/17030/greece-buys-170-leopard-2-tanks.html>). *Defense Aerospace*. 20 March 2003. Archived (<https://web.archive.org/web/20160209170008/http://www.defense-aerospace.com/articles-view/release/3/17030/greece-buys-170-leopard-2-tanks.html>) from the original on 9 February 2016. Retrieved 11 January 2016.
90. "Leopard 2 Main Battle Tank, Germany" (<http://www.army-technology.com/projects/leopard/>). Army-Technology.com. Archived (<https://web.archive.org/web/20151117040254/http://www.army-technology.com/projects/leopard/>) from the original on 17 November 2015. Retrieved 11 January 2016.
91. "Greece Signs Contracts for 183 Leopard 2s, 150 Leopard 1s" (<http://www.defenseindustrydaily.com/greece-signs-contracts-for-183-leopard-2s-150-leopard-1s-0978/>). *Defense Industry Daily*. 5 August 2005. Archived (<https://web.archive.org/web/20160306012909/http://www.defenseindustrydaily.com/greece-signs-contracts-for-183-leopard-2s-150-leopard-1s-0978/>) from the original on 6 March 2016. Retrieved 11 January 2016.
92. Jerchel, p. 42.
93. Ministerie van defensie – Nieuws (http://www.mindef.nl/actueel/nieuws/2007/04/20070412_verkoop_leopard.aspx) Archived (https://web.archive.org/web/20080508210939/http://www.mindef.nl/actueel/nieuws/2007/04/20070412_verkoop_leopard.aspx) 8 May 2008 at the Wayback Machine
94. Braatz, Kurt (19 December 2018). "Krauss-Maffei Wegmann unterstützt ungarische Heeres-Modernisierung" (https://web.archive.org/web/20181220230842/https://www.kmweg.de/fileadmin/user_upload/pdfs/Press_Releases/2018/Krauss-Maffei-Wegmann-unterstuetzt-ungarische-Heeres-Modernisierung_19-12-2018.pdf) (PDF). *Krauss-Maffei Wegmann* (in German). Munich. Archived from the original (https://www.kmweg.de/fileadmin/user_upload/pdfs/Press_Releases/2018/Krauss-Maffei-Wegmann-unterstuetzt-ungarische-Heeres-Modernisierung_19-12-2018.pdf) (PDF) on 20 December 2018. Retrieved 22 December 2018.
95. Dunai, Peter (20 December 2018). "Update: Hungary orders Leopard 2 MBTs and PzH 2000 SPHs" (<https://www.janes.com/article/85369/update-hungary-orders-leopard-2-mbts-and-pzh-2000-sphs>). *IHS Jane's 360*. Budapest. Archived (<https://web.archive.org/web/20181221152550/https://www.janes.com/article/85369/update-hungary-orders-leopard-2-mbts-and-pzh-2000-sphs>) from the original on 21 December 2018. Retrieved 22 December 2018.
96. "Norway to order 54 new army tanks from Germany" (<https://www.reuters.com/world/europe/norway-order-54-new-army-tanks-germany-nrk-2023-02-03/>). *Reuters*. 3 February 2023. Retrieved 4 February 2023.
97. Treloar, Stephen (3 February 2023). "Norway to Buy 54 Leopard 2 to Replace its Aging Tank Force" (<https://www.bloomberg.com/news/articles/2023-02-03/norway-to-buy-54-leopard-2-to-replace-its-aging-tank-force>). *Bloomberg News*. Retrieved 4 February 2023.

98. "Norway's Leopard 2 tanks will be new 2A8 variant with active protection system" (<https://www.shephardmedia.com/news/landwarfareintl/configuration-is-finalised-for-norways-leopard-2-a8-tank-order/>). Shephard. Retrieved 13 June 2023.
99. "Germany to Sell 298 Leopard 2 Tanks to Turkey" (<http://www.defenseindustrydaily.com/germany-to-sell-298-leopard-2-tanks-to-turkey-01473/>). defenseindustrydaily.com. 11 November 2005. Archived (<https://web.archive.org/web/20160122093943/http://www.defenseindustrydaily.com/germany-to-sell-298-leopard-2-tanks-to-turkey-01473/>) from the original on 22 January 2016. Retrieved 11 January 2016.
100. "Asian Tiger Ordering Leopards" (<http://www.defenseindustrydaily.com/asian-tiger-ordering-leopards-02864/>). defenseindustrydaily.com. 12 December 2006. Archived (<https://web.archive.org/web/20160201033039/http://www.defenseindustrydaily.com/asian-tiger-ordering-leopards-02864/>) from the original on 1 February 2016. Retrieved 18 January 2016.
101. Hardy, James (12 November 2013). "Rheinmetall confirms Indonesian Leopard 2 contract" (<https://www.janes.com/article/29983/rheinmetall-confirms-indonesian-leopard-2-contract>). Jane's IHS. Archived (<https://web.archive.org/web/20160201002810/http://www.janes.com/article/29983/rheinmetall-confirms-indonesian-leopard-2-contract>) from the original on 1 February 2016. Retrieved 11 January 2016.
102. Foss, Christopher F. (16 June 2014). "Leopard migrates eastward" (<https://www.janes.com/article/39370/leopard-migrates-eastward-es14e2>). Jane's IHS. Archived (<https://web.archive.org/web/20160304071434/http://www.janes.com/article/39370/leopard-migrates-eastward-es14e2>) from the original on 4 March 2016. Retrieved 11 January 2016.
103. Foss, Christopher F. (21 February 2015). "Leopard ready to prowl in Qatar" (<https://www.janes.com/article/49181/leopard-ready-to-prowl-in-qatar-idx15vp>). Jane's IHS. Archived (<https://web.archive.org/web/20151219015316/http://www.janes.com/article/49181/leopard-ready-to-prowl-in-qatar-idx15vp>) from the original on 19 December 2015. Retrieved 10 January 2016.
104. Binnie, Jeremy (23 December 2015). "Qatar parades new equipment" (<https://www.janes.com/article/56846/qatar-parades-new-equipment>). Jane's IHS. Archived (<https://web.archive.org/web/20160107040959/http://www.janes.com/article/56846/qatar-parades-new-equipment>) from the original on 7 January 2016. Retrieved 11 January 2016.
105. "Desert Leopards: Germany Selling Heavy Armor to the Saudis?" (<http://www.defenseindustrydaily.com/desert-leopards-germany-selling-heavy-armor-to-the-saudis-06993/>). defenseindustrydaily.com. 13 April 2014. Archived (<https://web.archive.org/web/20160107041000/http://www.defenseindustrydaily.com/Desert-Leopards-Germany-Selling-Heavy-Armor-to-the-Saudis-06993/>) from the original on 7 January 2016. Retrieved 18 January 2016.
106. "German tank sale to Saudi Arabia likely to be cancelled: newspaper" (<https://www.reuters.com/article/us-germany-saudi-arms-idUSBREA3C07E20140413>). Reuters. reuters.com. 13 April 2014. Archived (<https://web.archive.org/web/20160107041001/http://www.reuters.com/article/us-germany-saudi-arms-idUSBREA3C07E20140413>) from the original on 7 January 2016. Retrieved 18 January 2016.
107. Ogorkiewicz, Richard (20 February 2015). *Tanks: 100 years of evolution*. Osprey Publishing. p. 185. ISBN 9781472806703.
108. *Acquisition of the ABRAMS Main Battle Tank* (https://web.archive.org/web/20110605234042/http://www.anao.gov.au/~media/Uploads/Documents/2007%2008_audit_report_1.pdf) (PDF). Department of Defence. 2007. p. 37. ISBN 978-0-642-80974-2. Archived from the original (http://www.anao.gov.au/~media/uploads/documents/2007%2008_audit_report_1.pdf) (PDF) on 5 June 2011. Retrieved 11 January 2016.
109. Suci, Peter (2 July 2023). "Video Of 'Leopard 2A4U' Main Battle Tank In Ukraine Went Viral – Did It Reveal Too Much?" (<https://www.forbes.com/sites/petersuci/2023/07/02/video-of-leopard-2a4u-main-battle-tank-in-ukraine-went-viral-did-it-reveal-too-much/>). forbes.com. Retrieved 2 September 2023.

110. Mia Jankowicz (24 January 2023). "100 Leopard tanks in 12 countries are standing by for Germany's go-ahead to be sent to Ukraine, official says" (<https://www.businessinsider.com/100-leopard-tanks-waiting-germany-permission-ukraine-scholz-2023-1?amp>). *Business Insider*. Retrieved 25 January 2023.
111. Connolly, Kate (24 January 2023). "Berlin plans to send German Leopard tanks to Ukraine, according to reports" (<https://www.theguardian.com/world/2023/jan/24/berlin-plans-to-send-german-leopard-tanks-to-ukraine-according-to-reports>). *The Guardian*. Retrieved 25 January 2023.
112. Bertrand, Natasha; Liebermann, Oren (25 January 2023). "US finalizing plans to send approximately 30 Abrams tanks to Ukraine, two US officials say" (<https://edition.cnn.com/2023/01/24/politics/us-abram-tanks-ukraine/index.html>). *CNN*. Retrieved 25 January 2023.
113. "Ukraine war: 'Frank' talks as Ukraine pushes Germany for tanks" (<https://www.bbc.com/news/world-europe-64355839>). *BBC*. 20 January 2023. Retrieved 21 January 2023.
114. Solomom, Erika (20 January 2023). "Allies Fail to Agree on Sending Tanks to Ukraine" (<https://www.nytimes.com/2023/01/20/world/europe/ukraine-germany-us-tanks.html>). *The New York Times*. Retrieved 21 January 2023.
115. "Ukraine frustrated as Germany holds back the decision on supply of tanks" (<https://www.theguardian.com/world/2023/jan/20/ukraine-germany-leopard-2-tanks-ramstein>). *The Guardian*. 20 January 2023. Retrieved 21 January 2023.
116. "Ukraine defence minister says forces will train on German tanks in Poland - VOA" (<https://www.reuters.com/world/europe/ukraine-defence-minister-says-forces-will-train-german-tanks-poland-voa-2023-01-21/>). *Reuters*. 21 January 2023. Retrieved 21 January 2023.
117. "Germany ready to let Poland send Leopard tanks to Ukraine: foreign minister" (<https://www.politico.eu/article/germany-ready-allow-poland-sending-leopard-tanks-ukraine-foreign-minister-annalena-baerbock/>). *POLITICO*. 22 January 2023. Retrieved 27 January 2023.
118. Connolly, Kate (24 January 2023). "Poland requests German permission to send Leopard tanks to Ukraine" (<https://www.theguardian.com/world/2023/jan/24/poland-requests-german-permission-to-send-leopard-tanks-to-ukraine>). *The Guardian*. Retrieved 25 January 2023.
119. "Tanks for Ukraine: Germany to send Leopard 2s and allow others to export – reports" (<https://www.bbc.com/news/world-europe-64391272>). *BBC News*. 24 January 2023. Retrieved 24 January 2023.
120. Hickmann, Christoph (24 January 2023). "(S+) Krieg in der Ukraine: Deutschland schickt Leopard-Panzer" (<https://www.spiegel.de/politik/deutschland/krieg-in-der-ukraine-deutschland-schickt-leopard-panzer-a-e2dde871-88d0-4cf5-8aae-482d58fd850f>). *Der Spiegel* (in German). ISSN 2195-1349 (<https://www.worldcat.org/issn/2195-1349>). Retrieved 24 January 2023.
121. Oltermann, Philip; Roth, Andrew (25 January 2023). "Germany announces it will supply Leopard 2 tanks to Ukraine" (<https://www.theguardian.com/world/2023/jan/25/germany-leopard-2-tanks-ukraine>). *The Guardian*. Retrieved 25 January 2023.
122. "Inside Washington's about-face on sending tanks to Ukraine" (<https://www.politico.com/news/2023/01/25/inside-washingtons-about-face-on-sending-tanks-to-ukraine-00079560>). *Politico*. 25 January 2023. Retrieved 27 January 2023.
123. Chazan, Guy; Foy, Henry (14 February 2023). "Germany chides allies for delays in delivering tanks to Ukraine" (<https://www.ft.com/content/49233164-d28b-4d7d-868c-88e642e28e0b>). *Financial Times*. Retrieved 15 February 2023.
124. "German Chancellor Olaf Scholz agrees to provide Ukraine with Leopard 2 battle tanks" (<https://amp.abc.net.au/article/101894198>). *ABC*. 25 January 2023. Retrieved 25 January 2023.
125. "Deadly Russian missiles hit Ukraine power facilities" (<https://www.france24.com/en/live-news/20230126-deadly-russian-missiles-hit-ukraine-power-facilities>). *France 24*. 26 January 2023. Retrieved 26 January 2023.

126. Verteidigungsministerium [@Verteidigungsministerium] (24 February 2023). "Germany to provide 4 additional Leopard 2A6" (<https://twitter.com/Verteidigungsministerium/status/1629136976505892865>) (Tweet). Retrieved 24 February 2023 – via Twitter.
"Verteidigungsminister Pistorius hat entschieden, weitere vier #Kampfpanzer #Leopard2 A6 aus Beständen der Bundeswehr an die #Ukraine abzugeben. Damit erhöht  die Anzahl der gelieferten Panzer von 14 auf 18 & stellt zusammen mit  &  insgesamt ein  Bataillon zur Verfügung."
127. Pitel, Laura (27 March 2023). "First German Leopard 2 tanks delivered to Ukraine" (<https://www.ft.com/content/2f3f301d-8de7-426a-94fa-8bb398ffe232>). *Financial Times*. Retrieved 27 March 2023.
128. "Kehrtwende: Deutschland soll 25 Panzer kaufen dürfen" (<https://www.srf.ch/news/schweiz/kriegsmaterial-fuer-die-ukraine-kehrtwende-deutschland-soll-25-panzer-kaufen-duerfen>) (in German). Swiss Radio and Television SRF. 27 March 2023. Retrieved 2 April 2023.
129. John Revill (27 September 2023). "Swiss to sell back Leopard II tanks to help restock after Ukraine donations" (<https://www.reuters.com/business/aerospace-defense/swiss-sell-back-leopard-ii-tanks-help-restock-after-ukraine-donations-2023-09-27/>). *Reuters*. Retrieved 27 September 2023.
130. Easton, Andrew (25 January 2023). "Poland says it is ready to send 14 Leopard tanks to Ukraine" (https://www.bbc.com/news/live/world-europe-64396659?pinched_post_locator=urn:asset:91846f35-da18-45f4-a960-7d18b6acfa59). *BBC News*. Retrieved 25 January 2023.
131. "Poland delivers first Leopard tanks to Ukraine – DW – 02/24/2023" (<https://www.dw.com/en/poland-delivers-first-leopard-tanks-to-ukraine/a-64809432>). *DW*. 24 February 2023. Retrieved 24 February 2023.
132. Aiello, Rachel (26 January 2022). "Canada sending 4 tanks to Ukraine and deploying soldiers to train, defence minister says" (<https://www.ctvnews.ca/politics/canada-sending-4-tanks-to-ukraine-and-deploying-soldiers-to-train-defence-minister-says-1.6247335>). *CTV News*. Retrieved 26 January 2023.
133. "Kyiv says new attacks are being planned by Moscow" (<https://www.euronews.com/2023/02/05/canadian-leopard-2-tanks-en-route-to-ukraine-as-troops-prepare-for-suspected-moscow-attack>). *euronews*. 5 February 2023. Retrieved 15 February 2023.
134. "Canada sending four more battle tanks, ammunition to Ukraine" (<https://www.ctvnews.ca/politics/canada-sending-four-more-battle-tanks-to-ukraine-1.6288251>). *CTV News*. 24 February 2023. Retrieved 24 February 2023.
135. "The agreement on Leopard tanks applies to the Czech Republic" (<https://mocr.army.cz/informacni-servis/zpravodajstvi/dohoda-o-tancich-leopard-pro-cr-plati-241773/>). *Ministry of Defense of Czech Republic* (in Czech). 20 January 2023. Retrieved 26 February 2023.
136. "Armáda převzala první z německých Leopardů. Zda koupí další, ukáže studie" (https://www.idnes.cz/zpravy/nato/leopard-armada-praslavice-cernochova-ukrajina-tank.A221220_202604_zpr_nato_inc) [The army took delivery of the first of the German Leopards. The study will show whether they will buy more]. *iDNES.cz* (in Czech). 21 December 2022.
137. Axe, David (24 January 2023). "Leopard 2 Tanks From All Over Europe Are Heading To Ukraine As German Resistance Collapses" (<https://www.forbes.com/sites/davidaxe/2023/01/24/leopard-2-tanks-from-all-over-europe-are-heading-to-ukraine-as-german-resistance-collapses/>). *Forbes*. Retrieved 25 January 2023.
138. Jungersen, Rasmus; Tørnqvist, Tim (19 January 2023). "'Vi sender ikke alt afsted': Danske Leopard-kampvogne bliver endnu ikke sendt til Ukraine" (<https://www.dr.dk/nyheder/politik/vi-sender-ikke-alt-afsted-danske-leopard-kampvogne-bliver-endnu-ikke-sendt-til>) [We are not sending everything': Danish Leopard tanks are not yet being sent to Ukraine]. *DR* (in Danish). Danish Broadcasting Corporation. Retrieved 28 January 2023.

139. Bugge, Mathilde (25 January 2023). "Partier åbner for at sende Danmarks bedste kampvogne til Ukraine" (<https://www.dr.dk/nyheder/politik/partier-aabner-sende-danmarks-bedste-kampvogne-til-ukraine>) [Parties open to send Denmark's best tanks to Ukraine]. *DR* (in Danish). Danish Broadcasting Corporation. Retrieved 28 January 2023.
140. "Hvis alle sender kampvogne, bliver det et gedemarked, siger militæranalytiker" (<https://nyheder.tv2.dk/udland/2023-01-31-hvis-alle-sender-kampvogne-bliver-det-et-gedemarked-siger-militaeranalytiker>) [If everyone sends tanks, it will be a goat market, military analyst says]. *TV2* (in Danish). 31 January 2023. Retrieved 3 February 2023.
141. "Ukraine to Get as Many as 178 Leopard 1 Tanks From EU Allies" (<https://www.bloomberg.com/news/articles/2023-02-07/germany-denmark-netherlands-to-send-kyiv-178-leopard-1-tanks>). *Bloomberg*. 7 February 2023. Retrieved 26 February 2023.
142. "Danmark og Nederlandene indgår samarbejde om donation af Leopard 2-kampvogne til Ukraine" (<https://www.fmn.dk/da/nyheder/2023/danmark-og-nederlandene-indgar-samarbejde-om-donation-af-leopard-2-kampvogne-til-ukraine/>). *Danish Ministry of Defense* (in Danish). 20 April 2023.
143. Sorgi, Gregorio (12 January 2023). "Finland could deliver Leopard tanks to Ukraine, its president says" (<https://www.politico.eu/article/finland-leopard-2-tanks-ukraine-sauli-niinisto-nato-defense/>). *Politico*. Retrieved 25 January 2023.
144. "Suomi luovuttaa Ukrainalle kolme Leopard 2 -raivausvaunua: "Etulinjan vehkeitä – varma taistelukentän työjuhta" " (<https://www.mtvuutiset.fi/artikkeli/suomi-luovuttaa-ukrainalle-kolme-leopard-2-raivausvaunua/8639832>). *mtvuutiset.fi* (in Finnish). 23 February 2023. Retrieved 24 February 2023.
145. "EXPLAINED: Ukraine Will Get Leopard Tanks, What Happens Next?" (<https://www.kyivpost.com/post/11499>). *Kyiv Post*. 25 January 2023. Retrieved 26 February 2023.
146. "Greece to send Leopard-2 tanks to Ukraine" (<https://greekcitytimes.com/2023/01/29/greece-to-send-leopard-2-tanks-to-ukraine/>). *Greek City Times*. 29 January 2023. Retrieved 26 February 2023.
147. Take, Sayumi; Sharp, Andrew (31 January 2023). "Greek PM slams Turkey for 'sitting on the fence' over Ukraine war" (<https://asia.nikkei.com/Editor-s-Picks/Interview/Greek-PM-slams-Turkey-for-sitting-on-the-fence-over-Ukraine-war>). *Nikkei Asia*. Retrieved 26 February 2023.
148. Axe, David (25 April 2023). "Ukrainian Tankers Would Love To Get Their Hands On Ex-Greek Leopard 2A6HELs. Politics Might Disappoint Them" (<https://www.forbes.com/sites/davidaxe/2023/04/25/ukrainian-tankers-would-love-to-get-their-hands-on-ex-greek-leopard-2a6hels-politics-might-disappoint-them/?sh=4d0d48bd2f18>). *Forbes*.
149. "Netherlands considering providing Ukraine with Leopard 2 tanks" (<https://nltimes.nl/2023/01/25/netherlands-considering-providing-ukraine-leopard-2-tanks>). *NL Times*. 25 January 2023. Retrieved 25 January 2023.
150. "Netherlands says still supports tank deliveries for Ukraine" (<https://www.reuters.com/world/europe/netherlands-says-still-supports-tank-deliveries-ukraine-2023-02-15/>). *Reuters*. 15 February 2023. Retrieved 26 February 2023.
151. "More Leopards for Ukraine: Denmark, Netherlands to send 14 tanks" (<https://www.politico.eu/article/denmark-netherlands-to-donate-14-leopard-2-tanks-to-ukraine/>). *Politico*. 20 April 2023.
152. "Norway considering sending Leopard 2 tanks to Ukraine: Media" (<https://english.alarabiya.net/News/world/2023/01/25/Norway-considering-sending-Leopard-2-tanks-to-Ukraine-Media->). *Al Arabiya News*. 25 January 2023. Retrieved 25 January 2023.

153. Gjerstad, Tore; Melgård, Marie (24 January 2023). "Opplysninger til DN: Norge kan gi Ukraina opptil åtte stridsvogner" (<https://www.dn.no/politikk/stridsvogner/krigen-i-ukraina/opplysninger-til-dn-norge-kan-gi-ukraina-opptil-atte-stridsvogner/2-1-1392690>) [Information to DN: Norway can give Ukraine up to eight tanks]. *DN* (in Norwegian). Retrieved 25 January 2023.
154. Gambino, Lauren; Smith, David (25 January 2023). "US joins Germany in sending tanks to Ukraine as Biden hails 'united' effort" (<https://www.theguardian.com/us-news/2023/jan/25/us-m1-abrams-biden-tanks-ukraine-russia-war>). *The Guardian*. Retrieved 26 January 2023.
155. "Norway details Leopard 2 donation to Ukraine" (<https://www.shephardmedia.com/news/land-warfareintl/norway-donating-tank-package-to-ukraine/>). *Shephard*. Retrieved 15 February 2023.
156. "Norway with comprehensive tanks-package to Ukraine" (<https://www.regjeringen.no/en/aktuelt/brusselnyhet/id2963053/>). *Norwegian Government*. 14 February 2023. Retrieved 2 March 2023. "The decision to donate Leopard 2-tanks and special purpose tanks has been made after close dialogue with both the Chief of Defence and other European countries."
157. Gonçalves, Francisco J. (25 January 2023). "Portugal prepara envio de quatro tanques para a Ucrânia" (<https://www.cmjornal.pt/mundo/detalhe/portugal-prepara-envio-de-quatro-tanques-para-a-ucrania>) [Portugal prepares to send four tanks to Ukraine]. *Correio da Manhã* (in Portuguese). Retrieved 25 January 2023.
158. "Portugal to send Leopard tanks to Ukraine, PM says" (<https://www.reuters.com/world/europe/portugal-send-leopard-tanks-ukraine-pm-says-2023-02-04/>). *Reuters*. 4 February 2023. Retrieved 4 February 2023.
159. "Answering The Call: Heavy Weaponry Supplied To Ukraine" (<https://www.oryxspioenkop.com/2022/04/answering-call-heavy-weaponry-supplied.html>). *Oryx*. 27 April 2023.
160. "Slovakia does not plan to supply Ukraine with Leopard 2, but ready to provide T-72 instead" (<https://www.pravda.com.ua/eng/news/2023/01/25/7386530/>). *Ukrainska Pravda*. 25 January 2023. Retrieved 15 February 2023.
161. González, Miguel (5 June 2022). "España, dispuesta a entregar a Ucrania misiles antiaéreos y carros de combate Leopard" (<https://elpais.com/espana/2022-06-05/espana-dispuesta-a-entregar-a-ucrania-misiles-antiaereos-y-carros-de-combate-leopard.html>) [Spain, ready to deliver anti-aircraft missiles and Leopard tanks to Ukraine]. *El País* (in Spanish). Retrieved 24 March 2023.
162. "Germany to boost military mission in Lithuania" (<https://www.reuters.com/world/europe/germany-scholz-pledges-boost-military-mission-lithuania-2022-06-07/2022-08-02/>). *Reuters*. 8 June 2022. Retrieved 24 March 2023.
163. "Spain mulls whether to send high-tech tanks to Ukraine" (<https://www.france24.com/en/europe/20220610-spain-mulls-whether-to-send-high-tech-tanks-to-ukraine>). *France24*. 10 June 2022. Retrieved 24 March 2023.
164. Amann, Melanie (10 June 2022). "Was wurde aus den spanischen Panzerlieferungen?" (<https://www.spiegel.de/politik/deutschland/news-olaf-scholz-russland-ukraine-waffenlieferungen-afd-a-10bbf76f-bc59-4d16-ae68-11ba716dacf1>) [What became of the Spanish tank deliveries?]. *Der Spiegel* (in German). Retrieved 24 March 2023.
165. "Spain says its mothballed German-made tanks in no fit state to send to Ukraine" (<https://www.reuters.com/world/europe/spain-says-its-mothballed-german-made-tanks-no-fit-state-send-ukraine-2022-08-02/>). *Reuters*. 2 August 2022. Retrieved 24 March 2023.
166. González, Miguel (25 January 2023). "España se suma al plan europeo para entregar tanques Leopard a Ucrania" (<https://elpais.com/espana/2023-01-25/espana-se-suma-al-plan-europeo-para-entregar-tanques-leopard-a-ucrania.html>) [Spain joins the European plan to deliver Leopard tanks to Ukraine]. *El País* (in Spanish). Retrieved 25 January 2023.

167. "Norwegen will Panzer liefern, Ramstein-Treffen ohne neue Zusagen" (<https://www.zeit.de/politik/ausland/2023-02/ukraine-ueberblick-norwegen-panzer-nato-ramstein-bachmut>) [Norway wants to deliver tanks, Ramstein meetings without new commitments]. *Zeit Online* (in German). 14 February 2023.
168. "Spain to send six 2A4 Leopard tanks to Ukraine -defence minister" (<https://www.reuters.com/world/europe/spain-send-six-2a4-leopard-tanks-ukraine-defence-minister-2023-02-22/>). *Reuters*. 22 February 2023. Retrieved 22 February 2023.
169. Fernández, Juan José (5 February 2023). "España reforzará la letalidad de sus tanques Leopard antes de enviarlos a Ucrania" (<https://www.epe.es/es/internacional/20230205/espana-reforzara-letalidad-tanques-leopard-guerra-ucrania-82519615>) [Spain will reinforce the lethality of its Leopard tanks before sending them to Ukraine]. *el Periódico de España* (in Spanish). Retrieved 23 February 2023.
170. "Politico: Spain, Finland to send Leopard tanks to Ukraine" (<https://www.politico.eu/article/finland-and-helsinki-russia-ukraine-war-to-send-three-leopard-2-tanks-to-ukraine/>). 23 February 2023. Retrieved 23 February 2023.
171. Oscarsson, Tea (25 January 2023). "Tyskland bekräftar: Leopard 2 till Ukraina" (<https://www.svd.se/a/q1kp8e/efter-leopard-2-kan-sverige-skicka-stridsvagn-122>) [Germany confirms: Leopard 2 to Ukraine]. *Svenska Dagbladet*. Retrieved 25 January 2023.
172. Pål Jonson [@PIJonson] (24 February 2023). "[The Swedish PM] and I just announced that Sweden is joining the Leopard 2 tank coalition" (<https://twitter.com/PIJonson/status/1629117599513550848>) (Tweet). Retrieved 24 February 2023 – via [Twitter](#). "Sweden will give up to 10 Leopard 2 A5 main battle tanks to Ukraine."
173. "Kosovo: Schießerei am deutschen Kontrollpunkt" (<http://www.spiegel.de/politik/ausland/0,1518,27113,00.html>) [Kosovo: shooting at German checkpoint]. *Spiegel Online* (in German). 14 June 1999. Archived (<https://web.archive.org/web/20130510014656/http://www.spiegel.de/politik/ausland/kosovo-schiesserei-am-deutschen-kontrollpunkt-a-27113.html>) from the original on 10 May 2013. Retrieved 18 January 2016.
174. Lobitz, Frank (2009). *Kampfpanzer Leopard 2. Entwicklung und Einsatz in der Bundeswehr* [Leopard main battle tank 2. Development and German Army Service] (in German). Tankograd Publishing – Verlag Jochen Vollert. p. 229 et seq. ISBN 978-3-936519-08-2.
175. Jerchel, p. 37.
176. "Tanks for the Lesson: Leopards, too, for Canada" (<https://web.archive.org/web/20080621072607/http://www.defenseindustrydaily.com/tanks-for-the-lesson-leopards-too-for-canada-03208/%23choices>). *Defense Industry Daily*. December 2006. Archived from the original (<http://www.defenseindustrydaily.com/tanks-for-the-lesson-leopards-too-for-canada-03208/#choices>) on 21 June 2008. Retrieved 8 January 2009.
177. Priestley, Stephen (8 October 2008). "In to Action – Canadian Leopard C2 tanks and the TLAV M113A3s in Combat" (<https://web.archive.org/web/20100109063605/http://casr.ca/ft-leopard-tank.htm>). Canadian American Strategic Review. Archived from the original (<http://www.casr.ca/ft-leopard-tank.htm>) on 9 January 2010. Retrieved 8 January 2009.
178. Corbett, Brian (19 September 2007). "Canada's new main battle tank—Leopard 2" (https://web.archive.org/web/20090225074757/http://www.forces.gc.ca/site/community/mapleleaf/vol_10/vol10_27/1027_07.pdf) (PDF). The Maple Leaf. Archived from the original (http://www.forces.gc.ca/site/community/mapleleaf/vol_10/vol10_27/1027_07.pdf) (PDF) on 25 February 2009. Retrieved 8 January 2009.
179. "Germany gets a thank-you from Canadian IED survivor" (<https://archive.today/20120729174700/http://www.dailyheraldtribune.com/ArticleDisplay.aspx?archive=true&e=2378441>). Grand Prairie Daily Herald-Tribune. 6 December 2007. Archived from the original (<http://www.dailyheraldtribune.com/ArticleDisplay.aspx?archive=true&e=2378441>) on 29 July 2012. Retrieved 27 October 2010.

180. Freeman, Alan (8 December 2007). "Tank hit by IED back in service: Hillier" (<https://www.the-globeandmail.com/news/world/tank-hit-by-ied-back-in-service-hillier/article20406656/>). Toronto: The Globe and Mail. Retrieved 31 July 2008.
181. Priestley, Stephen (May 2007). "The Danish Army is Prepared to Deploy Tanks to Helmand Province" (<https://web.archive.org/web/20081216170241/http://www.casr.ca/ft-leopard-2a5-denmark.htm>). Canadian American Strategic Review. Archived from the original (<http://www.casr.ca/ft-leopard-2a5-denmark.htm>) on 16 December 2008. Retrieved 9 January 2009.
182. "Danish tanks arrive in Afghanistan" (<http://www.nato.int/multi/2007/071121b-afghan-stories/video/071121b-01.wmv>). NATO. 28 November 2008. Archived (<https://web.archive.org/web/20081218111409/http://www.nato.int/multi/2007/071121b-afghan-stories/video/071121b-01.wmv>) from the original on 18 December 2008. Retrieved 15 December 2008.
183. "LEOPARD 2 bewährt sich gegen IEDs" (https://web.archive.org/web/20090111052006/http://www.europaeische-sicherheit.de/Ausgaben/2008/2008_01/Umschau/2008,01,umschau.html) (in German). Europäische Sicherheit. January 2008. Archived from the original (http://www.europaeische-sicherheit.de/Ausgaben/2008/2008_01/Umschau/2008,01,umschau.html) on 11 January 2009. Retrieved 7 January 2009.
184. Stephen, Priestley (January 2008). "Danish Leopard tanks in supporting action in Helmand Province: Can Open Government be measured by a Military Press Release?" (<https://web.archive.org/web/20080930180306/http://www.casr.ca/ft-leopard-2a5-denmark-2.htm>). Canadian American Strategic Review. Archived from the original (<http://www.casr.ca/ft-leopard-2a5-denmark-2.htm>) on 30 September 2008. Retrieved 31 July 2008.
185. Lohse, Eckart (16 March 2008). "Leopardenjagd am Hindukusch" (<https://www.faz.net/s/RubDDBDABB9457A437BAA85A49C26FB23A0/Doc~E74D84BBBF12B478BBD4C7901BB39BF59~ATpl~Ecommon~Scontent.html>) (in German). Frankfurter Allgemeine Zeitung. p. 6. Archived (<https://web.archive.org/web/20080704212849/https://www.faz.net/s/RubDDBDABB9457A437BAA85A49C26FB23A0/Doc~E74D84BBBF12B478BBD4C7901BB39BF59~ATpl~Ecommon~Scontent.html>) from the original on 4 July 2008. Retrieved 31 July 2008.
186. "Status over fredagens begivenheder – Afghanistan: Alle danske soldater er nu tilbage i deres respektive lejre i Helmand provinsen" (<https://web.archive.org/web/20091017063251/http://forsvaret.dk/HOK/Nyt%20og%20Presse/ISAF/Pages/Statusoverfredagensbegivenheder.aspx>) (in Danish). Royal Danish Army. 25 July 2008. Archived from the original (<http://forsvaret.dk/HOK/Nyt%20og%20Presse/ISAF/Pages/Statusoverfredagensbegivenheder.aspx>) on 17 October 2009. Retrieved 31 July 2008.
187. "IN PICTURES: Op Red Dagger strikes in Helmand (mod.uk)" (<https://web.archive.org/web/20090107225202/http://www.mod.uk/DefenceInternet/DefenceNews/MilitaryOperations/InPicturesOpRedDaggerStrikesInHelmand.htm>). Archived from the original (<http://www.mod.uk/DefenceInternet/DefenceNews/MilitaryOperations/InPicturesOpRedDaggerStrikesInHelmand.htm>) on 7 January 2009. Retrieved 10 January 2009.
188. "Islamic State: We captured Turkish tanks in Syria battle" (<http://www.middleeasteye.net/news/claims-capture-two-tanks-after-killing-70-turkish-troops-syria-845385252>). Middle East Eye. 22 December 2016. Archived (<https://web.archive.org/web/20161226221245/http://www.middleeasteye.net/news/claims-capture-two-tanks-after-killing-70-turkish-troops-syria-845385252>) from the original on 26 December 2016. Retrieved 21 January 2017.
189. "Typeцкие потери в технике в Сирии" (<https://web.archive.org/web/20161227172433/http://bmpd.livejournal.com/2343546.html>). BMPD (in Russian). 25 December 2016. Archived from the original (<http://bmpd.livejournal.com/2343546.html>) on 27 December 2016.

190. "Er galt als unzerstörbar: In Syrien wird ein Panzer-Mythos zerstört" (http://www.focus.de/politik/videos/schwachstelle-entdeckt-verluste-in-syrien-ein-deutscher-panzer-mythos-wird-jetzt-zerstoert_id_6487678.html). *Focus* (in German). 12 January 2017. Archived (https://web.archive.org/web/20170113110948/http://www.focus.de/politik/videos/schwachstelle-entdeckt-verluste-in-syrien-ein-deutscher-panzer-mythos-wird-jetzt-zerstoert_id_6487678.html) from the original on 13 January 2017. Retrieved 21 January 2017.
191. Leith Fadel (20 January 2017). "Turkish Army suffers heavy casualties in failed east Aleppo offensive" (<https://www.almasdarnews.com/article/turkish-army-suffers-heavy-casualties-failed-east-aleppo-offensive/>). *Al-Masdar News*. Archived (<https://web.archive.org/web/20170120205256/https://www.almasdarnews.com/article/turkish-army-suffers-heavy-casualties-failed-east-aleppo-offensive/>) from the original on 20 January 2017. Retrieved 21 January 2017.
192. "The Battle for Al-Bab: Verifying Euphrates Shield Vehicle Losses – bellingscat" (<https://www.bellingcat.com/news/mena/2017/02/12/battle-al-bab-verifying-turkish-military-vehicle-losses/>). 12 February 2017. Archived (<https://web.archive.org/web/20170215134435/https://www.bellingcat.com/news/mena/2017/02/12/battle-al-bab-verifying-turkish-military-vehicle-losses/>) from the original on 15 February 2017. Retrieved 18 February 2017.
193. "Loading..." (<http://defense-watch.com/2017/01/20/photos-destroyed-turkish-leopard-2a4-tanks-northern-syria/>) *defense-watch.com*. Archived (<https://web.archive.org/web/20170218134955/http://defense-watch.com/2017/01/20/photos-destroyed-turkish-leopard-2a4-tanks-northern-syria/>) from the original on 18 February 2017. Retrieved 18 February 2017.
194. Hegmann, Gerhard (12 January 2017). "Syrien: Leopard-2-Verluste kratzen am deutschen Panzer-Mythos" (<https://www.welt.de/wirtschaft/article161078829/IS-Kaempfer-zerstoeren-den-deutschen-Panzer-Mythos.html>). *Die Welt*. Archived (<https://web.archive.org/web/20170116180108/https://www.welt.de/wirtschaft/article161078829/IS-Kaempfer-zerstoeren-den-deutschen-Panzer-Mythos.html>) from the original on 16 January 2017. Retrieved 15 January 2017 – via www.welt.de.
195. "Germany's Leopard 2 Tank Was Considered One of the Best (Until It Went to Syria)" (<https://nationalinterest.org/blog/the-buzz/germanys-leopard-2-tank-was-considered-one-the-best-until-it-24234>). 28 January 2018. Archived (<https://web.archive.org/web/20190422033359/http://nationalinterest.org/blog/the-buzz/germanys-leopard-2-tank-was-considered-one-the-best-until-it-24234>) from the original on 22 April 2019. Retrieved 6 May 2019.
196. "Germany's Leopard 2 Tank Was Considered One of the Best (Until It Went to Syria)" (<https://nationalinterest.org/blog/buzz/turkeys-leopard-2-tanks-are-getting-crushed-syria-95396?amp>). 9 November 2019. Retrieved 21 January 2023.
197. "Türkei bestätigt der Bundesregierung Einsatz von Leopard-Panzern" (<https://web.archive.org/web/20180130091236/https://de.reuters.com/article/deutschland-t-rkei-panzer-idDEKBN1F11R2?il=0>). *DE* (in German). Archived from the original (<https://de.reuters.com/article/deutschland-t-rkei-panzer-idDEKBN1F11R2?il=0>) on 30 January 2018. Retrieved 29 January 2018.
198. "Syria war: Germany suspends upgrade to Turkey tanks" (<https://www.bbc.com/news/world-europe-42820151.amp>). 25 January 2018. Retrieved 21 January 2023.
199. "Turkey upgrades its tanks, puts Altay's turret on Leopard 2A4" (<https://bulgarianmilitary.com/amp/2022/10/18/turkey-upgrades-its-tanks-puts-altays-turret-on-leopard-2a4/>). 18 October 2022. Retrieved 21 January 2023.
200. "How One Of World's Most Powerful Main Battle Tanks 'Leopard 2' Was Decimated By ISIS In Syria" (<https://eurasianimes.com/how-one-of-worlds-most-powerful-main-battle-tanks-leopard-2-were-decimated-by-isis-in-syria/?amp>). 16 June 2021. Retrieved 21 January 2023.
201. Nikolov, Boyko (18 April 2023). "Must see: Ukrainians broke Leopard, they detached the tank turret" (<https://bulgarianmilitary.com/2023/04/18/must-see-ukrainians-broke-leopard-they-detached-the-tank-turret/>). *bulgarianmilitary*. Retrieved 2 September 2023.
202. @OSINTua (1 July 2023). "It happened. Enjoy" (<https://twitter.com/OSINTua/status/1675097032447479808>) (Tweet). Retrieved 1 July 2023 – via [Twitter](https://twitter.com).

203. "Leopard 2A4 With Reactive Armor: It Actually Exists" (https://en.defence-ua.com/weapon_and_tech/leopard_2a4_with_reactive_armor_it_actually_exists-7181.html). *en.defence-ua.com*. 1 July 2023. Retrieved 2 September 2023.
204. Malyasov, Dylan (1 July 2023). "Ukrainian soldiers adding extra protection to their Leopard 2 tanks" (<https://defence-blog.com/ukrainian-soldiers-adding-extra-protection-to-their-leopard-2-tanks/>). *defence-blog.com*. Retrieved 2 September 2023.
205. "The Russians Just Destroyed Their First Ukrainian Leopard 2 Tank" (<https://www.forbes.com/sites/davidaxe/2023/06/08/the-russians-just-destroyed-their-first-ukrainian-leopard-2-tank/?sh=2271aa5275c9>). *Forbes*. Retrieved 8 June 2023.
206. Axe, David (27 June 2023). "25 Tanks And Fighting Vehicles, Gone In A Blink: The Ukrainian Defeat Near Mala Tokmachka Was Worst Than We Thought" (<https://web.archive.org/web/20230629074507/https://gum.criteo.com/syncframe?origin=publishertag&topUrl=www.forbes.com>). *Forbes*. Archived from the original (<https://www.forbes.com/sites/davidaxe/2023/06/27/25-tanks-and-fighting-vehicles-gone-in-a-blink-the-ukrainian-defeat-near-mala-tokmachka-was-worst-than-we-thought/?sh=69525a037918>) on 29 June 2023.
207. "Ukraine only lost 5 out of 71 Leopard 2 tanks during summer in counteroffensive – Forbes" (<https://news.yahoo.com/ukraine-only-lost-5-71-121522507.html>). *News Yahoo!*. 29 August 2023. Retrieved 3 September 2023.
208. "In 13 Weeks, Ukraine Has Lost Just Five Of Its 71 Leopard 2 Tanks" (<https://www.forbes.com/sites/davidaxe/2023/08/28/in-13-weeks-ukraine-has-lost-just-five-of-its-71-leopard-2-tanks/?sh=68278c6185b1>). *Forbes*. 28 August 2023. Retrieved 3 September 2023.
209. "Russia destroyed all Leopard tanks supplied to Ukraine by Poland, Portugal: Shoigu" (<https://english.alarabiya.net/News/world/2023/07/03/Russia-destroyed-all-Leopard-tanks-supplied-to-Ukraine-by-Poland-Portugal-Shoigu#:~:text=Russian%20forces%20destroyed%2016%20German,Sergey%20Shoigu%20claimed%20on%20Monday.>). *Al Arabiya*. 3 July 2023. Retrieved 1 September 2023.
210. "Documenting Ukrainian Equipment Losses During The Russian Invasion Of Ukraine" (<https://www.oryxspioenkop.com/2022/02/attack-on-europe-documenting-ukrainian.html>). *oryxspioenkop.com*. Retrieved 2 September 2023.
211. Jerchel, Michael; Schnellbacher, Uwe (1998). *Leopard 2 Main Battle Tank 1979–1998*. New York, United States of America: Osprey. pp. 17–18. ISBN 978-1-85532-691-0.
212. "Sipri: Trade Registers" (<https://web.archive.org/web/20140423140358/http://portal.sipri.org/publications/pages/transfer/trade-register>). *sipri.org*. Archived from the original (<http://portal.sipri.org/publications/pages/transfer/trade-register>) on 23 April 2014. Retrieved 1 April 2015.
213. Liess, Stefan (22 July 2007). "Leopard 2" (<https://web.archive.org/web/20080420172440/http://kpz.axlbox.de/servlets/sfs>). *Kampfpanzer.de*. Archived from the original (<http://kpz.axlbox.de/servlets/sfs;jsessionid=18966A1AD99684AFF90F809734FBFD8D?s=rquu2hgS7kjI7CT3y&t=/contentManager/selectCatalog&i=1158408897725&b=1158408897725&l=0&e=UTF-8&ParentID=1164791614897&intro=1&CustomerID=0&startRow=0&active=no&cards=1>) on 20 April 2008. Retrieved 20 April 2008.
214. ARG. "RUAG Leopard 2 Upgrade Main Battle Tank" (http://www.military-today.com/tanks/ruag_leopard_upgrade.htm). *Military-Today.com*. Retrieved 24 June 2022.
215. "RUAG introduces a new Leopard 2 A4 armor upgrade at IAV 2013 International Armoured Vehicles | IAV 2013 News coverage report pictures video | Defence and security military exhibition 2013" (https://www.armyrecognition.com/iaav_2013_news_coverage_report_pictures_video/ruag_introduces_a_new_leopard_2_a4_armor_upgrade_at_iaav_2013_international_armoured_vehicles.html). *Armyrecognition.com*. Retrieved 24 June 2022.
216. "Zwei Schweizer Prototypen des Kampfpanzers Leopard 2" (<https://www.festungsmuseum.ch/zwei-schweizer-prototypen-des-kampfpanzers-leopard-2/>). *Militär- und Festungsmuseum Full-Reuenthal*. Retrieved 5 April 2023.

217. "Leopard 2A4 tanks for Indonesia will be fitted with the Revolution urban warfare package" (https://armyrecognition.com/may_2016_global_defense_security_news_industry/leopard_2a4_tanks_for_indonesia_will_be_fitted_with_the_revolution_urban_warfare_package_12505163.amp.html). *armyrecognition.com*. 25 May 2016. Retrieved 20 April 2022.
218. ARG. "Revolution Main Battle Tank" (<http://www.military-today.com/tanks/revolution.htm>). *Military-Today.com*. Retrieved 24 June 2022.
219. "Pz 87–140 image" (http://img144.imagevenue.com/img.php?image=13657_pz87_140_122_51lo.jpeg#). Archived (https://web.archive.org/web/20141129024533/http://img144.imagevenue.com/img.php?image=13657_pz87_140_122_51lo.jpeg) from the original on 29 November 2014. Retrieved 15 November 2014.
220. "Tanks for the Lesson: Leopards, too, for Canada" (<http://www.defenseindustrydaily.com/tanks-for-the-lesson-leopards-too-for-canada-03208/>). Archived (<https://web.archive.org/web/20080621072607/http://www.defenseindustrydaily.com/tanks-for-the-lesson-leopards-too-for-canada-03208/%23choices>) from the original on 21 June 2008. Retrieved 12 October 2010.
221. "Background — Canadian Forces Upgraded Leopard 2A4M Tanks" (<https://web.archive.org/web/20110401120025/http://www.casr.ca/bg-leopard-2a4m.htm>). *Canadian-American Strategic Review*. February 2011. Archived from the original (<http://www.casr.ca/bg-leopard-2a4m.htm>) on 1 April 2011. Retrieved 19 March 2011.
222. "Force mobility enhancement" (<https://www.canada.ca/en/departement-national-defence/services/procurement/force-mobility-enhancement.html>). 13 December 2018. Retrieved 27 November 2020.
223. Foss, Christopher. "Upgraded Leopard 2 MBT set for firing trials" (<https://www.janes.com/products/janes/defence-security-report.aspx?ID=1065929677&channel=defence&subChannel=land>). *Jane's Information Group*. Archived (<https://web.archive.org/web/20120914173954/http://www.janes.com/products/janes/defence-security-report.aspx?ID=1065929677&channel=defence&subChannel=land>) from the original on 14 September 2012. Retrieved 5 June 2011.
224. "Bakanın ağzından savunma ihalelerinde rüşvet kuşkusu" (<http://www.radikal.com.tr/Default.aspx?aType=RadikalYazar&ArticleID=1028157&Yazar=MURAT%20YETK%DDN&Date=10.11.2010&CategoryID=98>). *Radikal*. 9 November 2010. Retrieved 28 December 2010.
225. Bakanın ağzından savunma ihalelerinde rüşvet kuşkusu (30 November 2009). "Savunma ihalelerinde Rüşvet Skandalı 09 Kasım 2010 17:08" (<https://web.archive.org/web/20101111091245/http://www.aktifhaber.com/savunma-ihalelerinde-rusvet-skandalı-352580h.htm>). *Aktifhaber.com*. Archived from the original (<http://www.aktifhaber.com/savunma-ihalelerinde-rusvet-skandalı-352580h.htm>) on 11 November 2010. Retrieved 28 December 2010.
226. "Armoured breakthrough" (<https://www.defence24.pl/pancerny-przelom-leopardy-z-gliwic-ida-do-wojska-analiza>). 28 May 2020.
227. "BMC to modernize 84 Leopard 2A4s" (<https://www.savunmasanayist.com/bmc-84-adet-leopard-2a4-modernize-edecek/>). *savunmasanayist.com* (in Turkish). 7 February 2021. Retrieved 7 February 2021.
228. "Quiet Modernisation for Turkish Leopard 2 Tanks" (<https://web.archive.org/web/20201028193559/http://en.c4defence.com/Archive/quiet-modernisation-for-turkish-leopard-2-tanks/10326/1>). *c4defence.com*. Archived from the original (<http://en.c4defence.com/Archive/quiet-modernisation-for-turkish-leopard-2-tanks/10326/1>) on 28 October 2020.
229. "Turkish Army to Receive Leopard 2A4 with Roketsan T1 Reactive-Passive Armor" (<https://militaryleak.com/2020/09/18/turkish-army-to-receive-leopard-2a4-with-roketsan-t1-reactive-passive-armor/>). 18 September 2020.

230. "IDEF 2021: BMC proposes upgrade for Turkish army Leopard 2A4 MBT | IDEF 2021 News Official Online Show Daily Media Partner | Defence security exhibitions 2021 show daily news category | Defense & Security Exhibitions Archives" (https://www.armyrecognition.com/idef_2021_news_official_online_show_daily_media_partner/idef_2021_bmc_proposes_upgrade_for_leopard_2a4_mbt.html). Armyrecognition.com. 18 August 2021. Retrieved 24 June 2022.
231. weapons defence industry military technology UK. "Turkey unveils new Leopard 2A4 main battle tank upgrade fitted with Altay tank turret | weapons defence industry military technology UK | analysis focus army defence military industry army" (https://www.armyrecognition.com/weapons_defence_industry_military_technology_uk/turkey_unveils_new_leopard_2a4_main_battle_tank_upgrade_fitted_with_altay_tank_turret.html). Armyrecognition.com. Retrieved 24 June 2022.
232. "Leopard 2A4 Modernizasyonu Yeni Gelişmeler" (<https://www.savunmasanayist.com/leopard-2a4-modernizasyonu-yeni-gelistmeler/>). SavunmaSanayiST. 23 January 2020. Retrieved 24 June 2022.
233. Axe, David (2 September 2023). "30 Ukrainian Leopard 2A4 Tanks Assembled In One Field —A Reminder That Ukraine Still Has Most Of Its Leopard 2s" (<https://www.forbes.com/sites/daVIDaxe/2023/09/02/30-ukrainian-leopard-2a4-tanks-assembled-in-one-field-a-reminder-that-ukraine-still-has-most-of-its-leopard-2s/>). *forbes.com*. Retrieved 2 September 2023.
234. Huhtanen, Jarmo (25 March 2015). "Armeija ottaa vanhojen Marksman-vaunujen tykkitornit uuteen käyttöön" (<https://www.hs.fi/kotimaa/art-2000002811544.html>). *hs.fi*. Helsingin Sanomat. Archived (<https://web.archive.org/web/20170419013744/http://www.hs.fi/kotimaa/art-2000002811544.html>) from the original on 19 April 2017. Retrieved 25 March 2015.
235. "MAAVOIMIEN LIIKKUVUUDEN JA TULIVOIMAN ROLL OUT PANSSARIPRIKAATISSA 5.8.2015 – Esiteltävä kalusto" (https://web.archive.org/web/20151005155147/http://www.puolustusvoimat.fi/wcm/a2748e00495d6be1a9e0a98833fba23b/Toimintanaytoksen_materiaalitiedot_5%2B8%2B2015.pdf?MOD=AJPERES) (PDF) (in Finnish). Finnish Defence Forces. 5 August 2015. Archived from the original (http://puolustusvoimat.fi/wcm/a2748e00495d6be1a9e0a98833fba23b/Toimintanaytoksen_materiaalitiedot_5+8+2015.pdf?MOD=AJPERES) (PDF) on 5 October 2015. Retrieved 2 December 2017.
236. "Marksman" (<http://weaponsystems.net/weaponsystem/EE03%20-%20Marksman.html>). *weaponsystems.net*. Archived (<https://web.archive.org/web/20170112062530/http://www.weaponsystems.net/weaponsystem/EE03%20-%20Marksman.html>) from the original on 12 January 2017. Retrieved 2 December 2017. "The Leopard 2 chassis also provides a much increased mobility over the older T-55AM chassis, both on roads and in the field."
237. Michael Jerchel, Schnellbacher: "Leopard 2 Main Battle Tank 1979–98", Osprey Publishing, 1998, ISBN 9781782006954, section "The Improved Leopard 2"
238. "Kampvogne" (https://web.archive.org/web/20180612141030/https://www2.forsvaret.dk/viden-om/materiel/haerens_materiel/kampvogne/Pages/Kampvogne2.aspx). *forsvaret.dk* (in Danish). Archived from the original (https://www2.forsvaret.dk/viden-om/materiel/haerens_materiel/kampvogne/Pages/Kampvogne2.aspx) on 12 June 2018. Retrieved 7 June 2018.
239. "New MBT122B Evolution with Unprecedented Protection" (https://web.archive.org/web/20111008032615/http://www.ibd-deisenroth-engineering.de/news/items/MBT122B_with_new_Protection.html?file=tl_files%2Fresources%2Fcontent-pdfs%2F2010-12%20New%20MBT122B%20Evolution%20with%20Unprecedented%20Protection.pdf) (PDF). IBD Deisenroth Engineering. Archived from the original (http://www.ibd-deisenroth-engineering.de/news/items/MBT122B_with_new_Protection.html?file=tl_files/resources/content-pdfs/2010-12%20New%20MBT122B%20Evolution%20with%20Unprecedented%20Protection.pdf) (PDF) on 8 October 2011. Retrieved 28 December 2010.

240. David.B (31 May 2015). "Krauss-Maffei Wegmann Leopard 2 (1978)" (http://www.tanks-encyclopedia.com/coldwar/West_Germany/Leopard-2.php). *tanks-encyclopedia.com*. Archived (https://web.archive.org/web/20171102173045/http://www.tanks-encyclopedia.com/coldwar/West_Germany/Leopard-2.php) from the original on 2 November 2017. Retrieved 12 November 2017.
241. Hilmes, Rolf (30 November 2007). *Kampfpanzer heute und morgen: Konzepte – Systeme – Technologien* (in German). Germany: Motorbuch Verlag. p. 230. ISBN 978-3613027930.
242. "Backgrounder: Renewing the Canadian Armed Forces' Tank Capability" (<https://web.archive.org/web/20110609180039/http://www.forces.gc.ca/site/news-nouvelles/news-nouvelles-eng.asp?id=2252>). DND/CF. Archived from the original (<http://www.forces.gc.ca/site/news-nouvelles/view-news-afficher-nouvelles-eng.asp?id=2252>) on 9 June 2011. Retrieved 10 January 2008.
243. "Finland in 200 million-euro used tank deal with the Netherlands" (http://yle.fi/uutiset/finland_in_200_million-euro_used_tank_deal_with_the_netherlands/7035576). Yle. 16 January 2014. Archived (https://web.archive.org/web/20140116193831/http://yle.fi/uutiset/finland_in_200_million-euro_used_tank_deal_with_the_netherlands/7035576) from the original on 16 January 2014. Retrieved 16 January 2014.
244. *Janes Land Warfare Platforms: Armoured Fighting Vehicles - Tracked 2020-2021*. Jane's Information Group. 2020. p. 99. ISBN 9780710633446.
245. Foss, Christopher F (8 August 2007). "Canada receives first upgraded Leopard 2A6M MBT". *Jane's Defence Weekly*. **44** (32): 10. ISSN 0265-3818 (<https://www.worldcat.org/issn/0265-3818>). OCLC 210556835 (<https://www.worldcat.org/oclc/210556835>). BL Shelfmark 4646.840000.
246. "Leopard 2 HEL images and info" (<http://greekmilitary.net/greekmbtanks.htm>). Archived (<https://web.archive.org/web/20080516162723/http://greekmilitary.net/greekmbtanks.htm>) from the original on 16 May 2008. Retrieved 6 July 2008.
247. "The First of Twenty Loaned Leopard 2A6M CAN Arrive in Kandahar: A Glimpse of the CF's Future (With a Little Help From Our Friends)" (<https://web.archive.org/web/20080418075012/http://www.sfu.ca/casr/ft-leopard-2a6m-can.htm>). Canadian American Strategic Review. Archived from the original (<https://www.sfu.ca/casr/ft-leopard-2a6m-can.htm>) on 18 April 2008. Retrieved 6 July 2008.
248. "CASR Background — CF Leased & Purchased Leopard 2 A6M / 2 A4 Tanks" (<https://web.archive.org/web/20080416165407/http://www.sfu.ca/casr/bg-leopard2-netherlands.htm#2a6m>). Archived from the original (<https://www.sfu.ca/casr/bg-leopard2-netherlands.htm#2a6m>) on 16 April 2008. Retrieved 6 July 2008.
249. "CASR Background — Leopard 2 A6M / 2 A4 Tanks for the Canadian Forces?" (<https://web.archive.org/web/20080416181639/http://www.sfu.ca/casr/bg-leopard2-afghan.htm>). Canadian American Strategic Review. Archived from the original (<https://www.sfu.ca/casr/bg-leopard2-afghan.htm>) on 16 April 2008. Retrieved 6 July 2008.
250. "Letter of Interest: Tank Replacement Project" (http://www.merx.com/English/SUPPLIER_Menu.Asp?WCE=Show&TAB=1&PORTAL=MERX&State=7&id=PW-%24%24BL-225-16850&FED_ONLY=0&hcode=pdIdErlx17TWSULPr3KQEg%3d%3d). MERX Website – Government of Canada. 19 March 2008. Archived (https://web.archive.org/web/20080609232922/http://www.merx.com/English/SUPPLIER_Menu.Asp?WCE=Show&TAB=1&PORTAL=MERX&State=7&id=PW-%24%24BL-225-16850&FED_ONLY=0&hcode=pdIdErlx17TWSULPr3KQEg%3d%3d) from the original on 9 June 2008. Retrieved 20 April 2008.
251. "Tank 2000'den ALTAY'a Türkiye'nin tank serüveni" (<https://www.savunmasanayist.com/tank-2000den-altaya-turkiyenin-tank-seruveni/>). 18 February 2020.
252. Candil, Antonio (1 February 2007). "The Spanish Leopard 2E: A Magnificent Tool". *Military Technology*. Mönch Editorial Group, p. 66.

253. "Breakthrough. Polish Army Takes Delivery of Leopard 2PL Tanks - Defence24.com" (<https://defence24.com/breakthrough-polish-army-takes-delivery-of-leopard-2pl-tanks>). *defence24.com*. 29 May 2020. Retrieved 9 July 2020.
254. "Bumar Leopard 2PL MBT" (<https://bumar.gliwice.pl/en/strefa-militarna/o/modernizacja-czolgow-leopard-2a4>). *bumar.gliwice.pl* (in Polish). Retrieved 9 July 2020.
255. "Pierwsze Leopardy 2PL już w Polsce" (<http://polska-zbrojna.pl/home/articleshow/27196?t=Pierwsze-Leopardy-2PL-juz-w-Polsce#>). *Polska-zbrojna.pl*. Archived (<https://web.archive.org/web/20190624125547/http://www.polska-zbrojna.pl/home/articleshow/27196?t=Pierwsze-Leopardy-2PL-juz-w-Polsce>) from the original on 24 June 2019. Retrieved 17 January 2020.
256. pisze, Patriota (9 September 2020). "Wojsko Polskie prezentuje zmodernizowane czołgi" (<https://zbiam.pl/wojsko-polskie-prezentuje-zmodernizowane-czolgi/>). *Wydawnictwo militarne ZBIAM*.
257. "The LEOPARD 2 A7V" (http://www.kmweg.com/fileadmin/user_upload/pdfs/Press_Release_s/2016/Press_release_en_EUROSATORY_2016.pdf) (PDF). Krauss-Maffei Wegmann GmbH & Co. KG. Archived (https://web.archive.org/web/20161120213553/http://www.kmweg.com/fileadmin/user_upload/pdfs/Press_Releases/2016/Press_release_en_EUROSATORY_2016.pdf) (PDF) from the original on 20 November 2016. Retrieved 20 November 2016.
258. "MCS Mobile Camouflage System" (http://saab.com/land/signature-management/platform-integrated-systems/mcs_mobile_camouflage_system/). *Saab Solutions*. Archived (https://web.archive.org/web/20161120215408/http://saab.com/land/signature-management/platform-integrated-systems/mcs_mobile_camouflage_system/) from the original on 20 November 2016. Retrieved 20 November 2016.
259. [1] (<https://militaryleak.com/2020/02/08/royal-danish-army-receives-first-leopard-2a7-main-battle-tank/>) Royal Danish Army Receives First Leopard 2A7V Main Battle Tank.
260. Yeo, Mike (25 February 2019). "German documents reveal Singapore received more Leopard 2 tanks" (<https://www.defensenews.com/land/2019/02/21/german-documents-reveal-singapore-received-more-leopard-2-tanks/>). *Defense News*. Retrieved 6 August 2021.
261. Yeo, Mike (20 March 2018). "Singapore denies acquisition of new Leopard tank variant" (<https://www.defensenews.com/global/asia-pacific/2018/03/20/singapore-denies-acquisition-of-new-leopard-tank-variant/>). *Defense News*. Retrieved 6 August 2021.
262. "Singapore allegedly acquires Leopard 2A7 tanks – Shephard Media" (<https://www.shephardmedia.com/news/landwarfareintl/singapore-allegedly-acquires-leopard-2a7-tanks/>). *www.shephardmedia.com*. Retrieved 6 August 2021.
263. "Tyske Leopard 2A7 blir Norges nye stridsvogn" (<https://www.tv2.no/nyheter/innenriks/tyske-leopard-2a7-blir-norges-nye-stridsvogn/15467364/>) (in Norwegian). TV2. 3 February 2023. Retrieved 3 February 2023.
264. KMW Press release (http://www.kmweg.de/2922-bD1lbiZuZXdzX2lkPTQzMdAx~rechte-navi~pressemitteilungen~news_detail.html) Archived (https://web.archive.org/web/20100627160134/http://www.kmweg.de/2922-bD1lbiZuZXdzX2lkPTQzMdAx~rechte-navi~pressemitteilungen~news_detail.html) 27 June 2010 at the *Wayback Machine* about the Leopard 2A7 +.
265. "Product Information – KMW" (<http://www.kmweg.com/home/tracked-vehicles/main-battle-tanks/leopard-2-a7/product-information.html>). *www.kmweg.com*. Archived (<https://web.archive.org/web/20170516212527/http://www.kmweg.com/home/tracked-vehicles/main-battle-tanks/leopard-2-a7/product-information.html>) from the original on 16 May 2017. Retrieved 8 March 2017.
266. Geiger, Waldemar; Heiming, Gerhard (14 April 2023). "Neue Kampfpanzer – Bundeswehr soll Leopard 2 A8 erhalten" (<https://esut.de/2023/04/fachbeitraege/41311/neue-kampfpanzer-bundeswehr-soll-leopard-2-a8-erhalten/>). *Europäische Sicherheit & Technik*. Mittler Report Verlag GmbH. Retrieved 15 April 2023.

267. *Italian Parliament website*. 12 July 2023 https://www.camera.it/leg19/824?tipo=A&anno=2023&mese=07&giorno=12&view=filtered_scheda&commissione=04#/ (https://www.camera.it/leg19/824?tipo=A&anno=2023&mese=07&giorno=12&view=filtered_scheda&commissione=04#/). `{{cite web}}: Missing or empty |title= (help)`
268. <https://www.oryxspioenkop.com/2023/08/high-on-defence-listing-dutch-arms.html>
269. "Norges nye stridsvogner blir Leopard 2A8 med israeldsignet aktiv beskyttelse" (<https://www.tu.no/artikler/norges-nye-stridsvogner-blir-leopard-2a8-med-israelskdesignet-aktiv-beskyttelse/532729>) (in Norwegian). TU.no. 17 June 2023. Retrieved 17 June 2023.
270. "Prototype of the LEGUAN Assault Bridge Arrives as Scheduled" (http://www.baainbw.de/portal/a/baain/start/aktuell/allena!/ut/p/z1/hY_RC4lwEMb_I29ONHtULBNERcNyLzF02MI2GUt66I9vl-pNuocP7vvufscBgTMQQRc-Us2loJPPpOxJc4jA_5niLcVLtMMq8Q-vXcYnKFEMlp38jxMRopSIEDRPQGcZmneFBAwTlwJxeCqataiY0NzoqqqVyZqn0ZJOHUiZx-AAAdcpMYYe97yn1FVbBrUuwFSRbXFniJC33-dmlvn4buSsUwsUr20ceY7_uwKPzxDdw28pM!) (Press release). BAAINBW. 6 July 2017. Archived (https://web.archive.org/web/20180722184749/http://www.baainbw.de/portal/a/baain/start/aktuell/allena!/ut/p/z1/hY_RC4lwEMb_I29ONHtULBNERcNyLzF02MI2GUt66I9vl-pNuocP7vvufscBgTMQQRc-Us2loJPPpOxJc4jA_5niLcVLtMMq8Q-vXcYnKFEMlp38jxMRopSIEDRPQGcZmneFBAwTlwJxeCqataiY0NzoqqqVyZqn0ZJOHUiZx-AAAdcpMYYe97yn1FVbBrUuwFSRbXFniJC33-dmlvn4buSsUwsUr20ceY7_uwKPzxDdw28pM!) from the original on 22 July 2018. Retrieved 25 July 2017.
271. *Ejercito De Tierra*, p. 58, Antonio J. Candil Muñoz, Tankograd Publishing, 2010.
272. "Finland improves mobility systems" (<https://web.archive.org/web/20080915061230/http://www.janes.com/extract/jdw2007/jdw34244.html>). *Jane's Defence Weekly*. 1 October 2007. Archived from the original (<https://www.janes.com/extract/jdw2007/jdw34244.html>) on 15 September 2008. Retrieved 8 January 2009.
273. "Patria Land & Armament Heavy Mine Breaching Vehicle (Finland)" (<https://www.janes.com/extracts/extract/jmvl/jmvla084.html>). *Jane's Military Vehicles and Logistics*. 9 January 2009. Retrieved 17 January 2009.
274. "ARV/AEV WISENT 2" (<http://www.ffg-flensburg.de/en/heavy-tracked-vehicles/arv-aev-wisent-2/>). Flensburger Fahrzeugbau. Archived (<https://web.archive.org/web/20150926025418/http://www.ffg-flensburg.de/en/heavy-tracked-vehicles/arv-aev-wisent-2/>) from the original on 26 September 2015. Retrieved 31 December 2014.
275. Lobitz, Frank (2009). "Technical Data – Leopard 2 MBT". *Leopard 2 Main Battle Tank Development and German Army Service*. Germany Erlangen: Tankograd Publishing. p. 315. ISBN 978-3-936519-08-2.
276. "Welser Bundesheer reaktiviert Panzer, anstatt sie als Ersatzteillager zu nützen" (<https://www.nachrichten.at/oberoesterreich/wels/Welser-Bundesheer-reaktiviert-Panzer-anstatt-sie-als-Ersatzteillager-zu-nuetzen;art67,2637276>). *Oberösterreichische Nachrichten*. 31 July 2017. Retrieved 23 February 2023.
277. "Canadian donations and military support to Ukraine" (<https://www.canada.ca/en/departments-national-defence/campaigns/canadian-military-support-to-ukraine.html>). *Government of Canada*. 18 April 2023.
278. "How Canada sabotaged its own fleet of tanks | National Post" (<https://nationalpost.com/opinion/adam-zivo-how-canada-sabotaged-its-own-fleet-of-tanks>).
279. "Is Canada not sending its tanks to Ukraine because they're broken? | National Post" (<https://nationalpost.com/opinion/is-canada-not-sending-its-tanks-to-ukraine-because-theyre-broken>).

280. KMW delivers first LEOPARD 2 A6M to Canada (http://www.epicos.com/epicos/portal/media-type/html/user/anon/page/default.psml/js_panename/News+Information+Article+View;jsessionid=3BB03398A96F7A60F18A1F4E04FADE1D.tomcat5?articleid=81090&showfull=false). Archived (https://web.archive.org/web/20081227110710/http://www.epicos.com/epicos/portal/media-type/html/user/anon/page/default.psml/js_panename/News%2BInformation%2BArticle%2BView%3Bjsessionid%3D3BB03398A96F7A60F18A1F4E04FADE1D.tomcat5?articleid=81090&showfull=false) 27 December 2008 at the [Wayback Machine](#)
281. "Canadian Armed Forces Armour — Armoured Recovery Vehicle 3 (Büffel)" (<https://web.archive.org/web/20080414202920/http://www.sfu.ca/casr/101-army-armour-arv3-buffel.htm>). Archived from the original (<https://www.sfu.ca/casr/101-army-armour-arv3-buffel.htm>) on 14 April 2008. Retrieved 27 August 2008.
282. "Sale of surplus Leopard 2 Battle Tanks to Canada" (<https://web.archive.org/web/20110706221210/http://www.vtg.admin.ch/internet/vtg/en/home/dokumentation/news/newsdetail.37618.nsb.html>). Confédération Suisse. 2 October 2011. Archived from the original (<http://www.vtg.admin.ch/internet/vtg/en/home/dokumentation/news/newsdetail.37618.nsb.html>) on 6 July 2011. Retrieved 20 February 2008.
283. "Canada sending four more battle tanks, ammunition to Ukraine" (<https://www.ctvnews.ca/politics/canada-sending-four-more-battle-tanks-to-ukraine-1.6288251>). CTV News. 24 February 2023. Retrieved 26 February 2023.
284. "Ribbon-cutting ceremony signals start of production at FFG Canada" (https://www2.gnb.ca/content/gnb/en/news/news_release.2014.06.0769.html). Government of New Brunswick. 27 June 2014. Retrieved 7 March 2023.
285. "Chile gets 140 leopards II" (https://www.bundeswehr.de/de/organisation/heer?yw_contentURL=%2FC1256F870054206E%2FW278YGGV502INFODE%2Fcontent.jsp). German Army. Archived (https://web.archive.org/web/20080530180843/http://www.deutschesheer.de/portal/a/heer/kcxml/04_Sj9SPykssy0xPLMnMz0vM0Y_QjzKLNzSLtzAIA8IB2c76kXDRoJRUFW99X4_83FT9AP2C3lhyR0dFRQBALId7/delta/base64xml/L2dJQSEvUUt3QS80SVVFLzZfMTZfOEkw?yw_contentURL=%2FC1256F870054206E%2FW278YGGV502INFODE%2Fcontent.jsp) from the original on 30 May 2008. Retrieved 6 July 2008.
286. "More Leopard-2s For Sale, Sold" (<https://www.defenseindustrydaily.com/more-leopard-2s-for-sale-sold-04316/>). *Defense Industry Daily*. 27 November 2007. Retrieved 15 January 2020.
287. "Armáda ČR se zajímá o španělské tanky Leopard 2A4" (<http://www.armadninoviny.cz/armada-cr-se-zajima-o-spanelske-tanky-leopard-2a4.html>) [The Czech Army is interested in Spanish Leopard 2A4 tanks]. *Armadni noviny* (in Czech). 17 July 2016. Archived (<https://web.archive.org/web/20160817080537/http://www.armadninoviny.cz/armada-cr-se-zajima-o-spanelske-tanky-leopard-2a4.html>) from the original on 17 August 2016. Retrieved 11 August 2016.
288. "Polská armáda sílí a zajímá se o budoucí tanky Leopard 3" (<http://www.armadninoviny.cz/polska-armada-sili-a-zajima-se-o-budouci-tanky-leopard-3.html>) [The Polish Army is growing stronger and interested in future Leopard 3 tanks]. *Armadni noviny* (in Czech). 4 August 2016. Archived (<https://web.archive.org/web/20160828143624/http://www.armadninoviny.cz/polska-armada-sili-a-zajima-se-o-budouci-tanky-leopard-3.html>) from the original on 28 August 2016. Retrieved 24 August 2016.
289. "Česko dostane od Německa 15 starších tanků Leopard, dalších až 50 nových si koupí - Novinky.cz" (<https://www.novinky.cz/domaci/clanek/cesko-dostane-od-nemecka-15-starsich-tanku-leopard-dalsich-az-50-novych-si-koupi-40397448>). *www.novinky.cz*. Retrieved 18 May 2022.
290. "Leopard 2 Ringtausch mit Tschechien Deutschland stellt 15 Panzer zur Verfuegung" (<https://www.welt.de/politik/deutschland/article238832999/Leopard-2-Ringtausch-mit-Tschechien-Deutschland-stellt-15-Panzer-zur-Verfuegung.html>). *Die Welt*. 18 May 2022. Retrieved 18 May 2022.

291. "En kampvogn rulles ud, og med den, en ny tilgang til krigsførelse" (<https://www.forsvaret.dk/da/organisation/haeren/jyske-dragonregiment/nyheder/2019/en-kampvogn-rulles-ud-og-med-den-en-ny-tilgang-til-krigsforelse/>). *forsvaret.dk/* (in Danish). 31 October 2019. Retrieved 31 October 2019.
292. "Alle danske kampvogne er opgraderet" (<https://www.forsvaret.dk/da/nyheder/2023/alle-danske-kampvogne-er-opgraderet/>). *forsvaret.dk/* (in Danish). 23 September 2023. Retrieved 10 October 2023.
293. "Fakta: Hvad er det særlige ved Leopard 2-kampvognen?" (<https://politiken.dk/udland/art9183856/Hvad-er-det-s%C3%A6rlige-ved-Leopard-2-kampvognen>) (in Danish). Politiken. 24 January 2023. Retrieved 27 January 2023.
294. Rasmussen, P.E. (5 June 2018). "Forældede broslagningsvogne bliver opgraderet tidligere end ventet" (<https://olfi.dk/2018/06/05/foraeldede-broslagningsvogne-opgraderet-tidligere-ventet/>). OLFI. Retrieved 7 January 2023.
295. Lobitz, Frank (2022). *Gesamtwerk Leopard 2*. Tankograd Publishing. ISBN 9783936519532.
296. Materielanskaffelsesplanen (<https://www.fmi.dk/globalassets/fmi/dokumenter/2023/fsv-materielanskaffelsesplan.pdf>) (PDF) (Report). Forsvarsministeriet.
297. "Combat Aircraft" (<http://www.deagel.com/equipment/Main-Battle-Tanks2003.aspx>). *www.deagel.com*. Archived (<https://web.archive.org/web/20110403005052/http://www.deagel.com/equipment/Main-Battle-Tanks2003.aspx>) from the original on 3 April 2011. Retrieved 3 January 2011.
298. Turun Sanomat: *Pioneerien uudet ajoneuvot panevat vauhtia maavoimiin* (21.9.2007) (<http://www.turunsanomat.fi/kotimaa/?ts=1,3:1002:0:0,4:2:0:1:2007-09-21,104:2:487211,1:0:0:0:0:>)
299. Puolustusvoimille uutta pioneerikalustoa (<http://www.ess.fi/?article=161971>) Archived (<https://web.archive.org/web/20080530160630/http://www.ess.fi/?article=161971>) 30 May 2008 at the Wayback Machine, *Etelä-Suomen Sanomat* (20 September 2007)
300. "Combat Aircraft" (<http://www.deagel.com/equipment/Main-Battle-Tanks2009.aspx>). *www.deagel.com*. Archived (<https://web.archive.org/web/20110403004646/http://www.deagel.com/equipment/Main-Battle-Tanks2009.aspx>) from the original on 3 April 2011. Retrieved 3 January 2011.
301. "Suomi ostaa käytettyjä panssarivaunuja 200 miljoonalla eurolla." (<http://www.hs.fi/kotimaa/Suomi+ostaa+k%C3%A4ytettyj%C3%A4+panssarivaunuja+200+miljoonalla+eurolla/a1389841933576>) Archived (<https://web.archive.org/web/20140116183949/http://www.hs.fi/kotimaa/Suomi+ostaa+k%C3%A4ytettyj%C3%A4+panssarivaunuja+200+miljoonalla+eurolla/a1389841933576>) 16 January 2014 at the Wayback Machine. (2014/01/16) Helsingin Sanomat newspaper
302. "Suomen uudet Leopard-tankit täydessä toiminnassa" (http://www.iltalehti.fi/uutiset/2015080520136327_uu.shtml) Archived (https://web.archive.org/web/20150807000401/http://www.iltalehti.fi/uutiset/2015080520136327_uu.shtml) 7 August 2015 at the Wayback Machine. (2015/08/5) Iltalehti newspaper
303. Gebauer, Matthias (21 January 2023). "(S+) Waffenhilfe für die Ukraine: Diese Leopard-Panzer könnte die Bundeswehr abgeben" (<https://www.spiegel.de/politik/deutschland/ukraine-krieg-diese-leopard-panzer-koennte-die-bundeswehr-abgeben-a5df41a86-3110-4152-82c9-02afb552d5dd>). *Der Spiegel* (in German). ISSN 2195-1349 (<https://www.worldcat.org/issn/2195-1349>). Retrieved 22 January 2023.
304. "Bericht zur materiellen Einsatzbereitschaft der Hauptwaffensysteme der Bundeswehr l/2021" (<https://www.bmvg.de/resource/blob/5087620/334537aade84aac453057b3260ed28ff/20210531-download-bericht-materielle-einsatzbereitschaft-data.pdf>) (PDF). bmvg. Retrieved 1 July 2021.

305. Troianovski, Anton (10 March 2015). "Calls in Germany Grow to Reverse Cuts to Army" (<https://www.atlanticcouncil.org/blogs/natosource/calls-in-germany-grow-to-reverse-cuts-to-army/>). *Atlantic Council*. Retrieved 27 January 2022.
306. [2] (<http://www.defensenews.com/story/defense/land/vehicles/2015/04/18/germany-to-buy-back-tanks-amid-russia-threat/25879281/>) Defence News. 18 April 2015.
307. Glaw, Julian (26 May 2023). "Auslieferung ab 2025: Bundeswehr bestellt 18 Leopard-2-Panzer bei Krauss-Maffei Wegmann" (https://rp-online.de/politik/deutschland/bundeswehr-18-neue-leopard-2-panzer-ab-2025_aid-91109397). *RP ONLINE* (in German). Retrieved 4 June 2023.
308. "Greece ready to spend €2 billion to modernize its Leopard 1 and 2 heavy tanks" (<https://www.meta-defense.fr/en/2022/05/11/greece-ready-to-pay-2-mde-to-modernize-its-leopard-1-and-2-heavy-tanks/>). 11 May 2022.
309. "Teljes a "létszám" " (<https://honvedelem.hu/hirek/teljes-a-letszam.html>). December 2020.
310. Nassauer, Otfried; Matthias Gebauer (7 May 2013). "Umstrittener Rüstungsdeal: Berlin genehmigt Verkauf von 164 Panzern nach Indonesien" (<https://www.spiegel.de/politik/ausland/leopard-2-regierung-genehmigt-verkauf-von-panzern-nach-indonesien-a-898650.html>). Spiegel Online. Retrieved 10 May 2013.
311. Kostrad, Yonkav 1 (9 January 2023). "Batallyon Kavaleri 1 Kostrad on Instagram: "Cavalry Fight and Win Battle in any clime and place !!! " We choose Cavalry because we are proud and honor " " (<https://www.instagram.com/p/CnL2v1QLxiN/>). *Instagram*. Retrieved 3 February 2023.
312. Kostrad, Penerangan (7 December 2019). "Mobile Training Team Senjata Ohio Ordnance Work 2019 Yonkav 1 Kostrad" (<https://web.archive.org/web/20230116073159/https://kostrad.mil.id/security-pussansiad/pages/banned.php>). *kostrad.mil.id* (in Indonesian). Archived from the original (<https://kostrad.mil.id/security-pussansiad/pages/banned.php>) on 16 January 2023. Retrieved 3 February 2023.
313. "www.defensie.nl" (https://web.archive.org/web/20110807071458/http://www.defensie.nl/actueel/nieuws/2011/04/08/46180709/Defensie_hard_getroffen_door_bezuinigingen_video). Archived from the original (http://www.defensie.nl/actueel/nieuws/2011/04/08/46180709/Defensie_hard_getroffen_door_bezuinigingen_video) on 7 August 2011.
314. "dvhn.nl" (<https://web.archive.org/web/20110928093526/http://www.dvhn.nl/nieuws/drenthe/article6727686.ece/Laatste-schot-uit-Leopardtank>). Archived from the original (<http://www.dvhn.nl/nieuws/drenthe/article6727686.ece/Laatste-schot-uit-Leopardtank>) on 28 September 2011.
315. "Indonesia Pulls Out of Dutch Tanks Deal" (<https://web.archive.org/web/20120705084042/http://www.thejakartaglobe.com/home/indonesia-pulls-out-of-dutch-tanks-deal/528362>). *thejakartaglobe.com*. Archived from the original (<http://www.thejakartaglobe.com/home/indonesia-pulls-out-of-dutch-tanks-deal/528362>) on 5 July 2012. Retrieved 4 July 2012.
316. Peruvian Tank Contenders (<http://www.army-technology.com/news/newsrussia-t90s-battle-tank-peruvian-army>) Archived (<https://web.archive.org/web/20130609154729/http://www.army-technology.com/news/newsrussia-t90s-battle-tank-peruvian-army>) 9 June 2013 at the Wayback Machine – Army-Technology.com, 17 May 2013.
317. Peru; Future main battle tank projects lags on despite criticism (http://dmilt.com/index.php?option=com_content&view=article&id=7865:peru-future-main-battle-tank-projects-lags-on-despite-criticism&catid=35:latin-america&Itemid=58) Archived (https://web.archive.org/web/20130928033434/http://dmilt.com/index.php?option=com_content&view=article&id=7865%3Aperu-future-main-battle-tank-projects-lags-on-despite-criticism&catid=35%3Alatin-america&Itemid=58) 28 September 2013 at the Wayback Machine – Dmilt.com, 2 September 2013.

318. News Desk (19 January 2014). "Finland is Buying Dutch Leopards for €200 Million" (http://defense-update.com/20140119_finland_buys_dutch_leopards.html). Archived (https://web.archive.org/web/20141111075601/http://defense-update.com/20140119_finland_buys_dutch_leopards.html) from the original on 11 November 2014. Retrieved 15 November 2014.
319. [3] (<http://925.nl/archief/2015/01/13/koop-een-tank-nu-het-nog-kan/>) Archived (<https://web.archive.org/web/20150113184530/http://925.nl/archief/2015/01/13/koop-een-tank-nu-het-nog-kan/>) 13 January 2015 at the Wayback Machine – 925.nl, 13 January 2015.
320. Felstead, Peter (21 February 2023). "Norway Firms Up Order for 54 Leopard 2A7 MBTs" (<http://euro-sd.com/2023/02/news/29748/norway-firms-up-order-for-54-leopard-2a7-mbts/>). *European Security and Defence*.
321. Persen, Kjell; Talnes, Stål; Bjørnson, Harald; Hågensen, Finn-Ove (3 February 2023). "Tyske Leopard 2A7 blir Norges nye stridsvogn" (<https://www.tv2.no/nyheter/innenriks/tyske-leopard-2a7-blir-norges-nye-stridsvogn/15467364/>) (in Norwegian). TV2.no. Retrieved 2 March 2023.
322. "Norway's Leopard 2 tanks will be new 2A8 variant with active protection system | Shephard" (<https://www.shephardmedia.com/news/landwarfareintl/configuration-is-finalised-for-norways-leopard-2a8-tank-order/>).
323. "Her er Forsvarets aller nyeste anskaffelse" (<https://www.tu.no/artikler/her-er-forsvarets-aller-nyeste-anskaffelse/223384>). Teknisk Ukeblad. 19 May 2015. Archived (<https://web.archive.org/web/20181111215346/https://www.tu.no/artikler/her-er-forsvarets-aller-nyeste-anskaffelse/223384>) from the original on 11 November 2018. Retrieved 23 December 2018.
324. "Forsvaret får flere ingeniør- og bergingspanservogner" (<https://www.fma.no/aktuelt-og-media/2020/forsvaret-far-flere-ingeniør-og-bergingspanservogner>). Forsvarsmateriell. Retrieved 13 July 2023.
325. "Hæren får ingeniørpanservogn" (<https://forsvaret.no/forsvarsmateriell/presserom/forsvarsmateriell-m%C3%B8ter-studentene/h%C3%A6ren-f%C3%A5r-pansret-ingeni%C3%B8rpanservogn>). Forsvarsmateriell. Archived (<https://web.archive.org/web/20181224023654/https://forsvaret.no/forsvarsmateriell/presserom/forsvarsmateriell-m%C3%B8ter-studentene/h%C3%A6ren-f%C3%A5r-pansret-ingeni%C3%B8rpanservogn>) from the original on 24 December 2018. Retrieved 23 December 2018.
326. "Norway procures LEGUAN bridgelayers" (<https://www.monch.com/mpg/news/land/5189-norway-leguan-bridgelayers.html>). Mönch Publishing Group. 22 March 2019. Archived (<https://archive.today/20190322141049/https://www.monch.com/mpg/news/land/5189-norway-leguan-bridgelayers.html>) from the original on 22 March 2019. Retrieved 22 March 2019.
327. "Norway Procures Leguan Bridge Layer Systems For Leopard 2 Tank" (<http://www.defenseworld.net/news/24515>). *Defense World*. 22 March 2019. Archived (<https://archive.today/20190322141100/http://www.defenseworld.net/news/24515>) from the original on 22 March 2019. Retrieved 22 March 2019.
328. "Last Leopard 2A5 Tanks Reach Poland" (<https://defence24.com/last-leopard-2a5-tanks-reach-poland-photos>). December 2015. Retrieved 23 February 2023.
329. "Poland Starts Training of Ukrainian Leopard 2 Tank Crews" (<https://defence24.com/armed-forces/poland-starts-training-of-ukrainian-leopard-2-tank-crews>). 14 February 2023. Retrieved 23 February 2023.
330. "Więcej Leopardów do modernizacji. Podpisano aneks [News Defence24.pl] – Defence24" (<https://www.defence24.pl/wiecej-leopardow-do-modernizacji-podpisano-aneks-news-defence24pl>). Archived (<https://web.archive.org/web/20180713140859/https://www.defence24.pl/wiecej-leopardow-do-modernizacji-podpisano-aneks-news-defence24pl>) from the original on 13 July 2018. Retrieved 14 July 2018.

331. Sabak, Juliusz (7 July 2022). "Leopard 2PL - Rosną opóźnienia i koszty. PGZ: Wojsko dostaje "praktycznie nowe czołgi" [WYWIAD]" (<https://defence24.pl/przemysl/leopard-2pl-ros-na-opoznienia-i-koszty-pgz-wojsko-dostaje-praktycznie-nowe-czolgi-wywiad>) [Leopard 2PL - Delays and costs are increasing. PGZ: The army gets "practically new tanks"]. *Defence 24* (in Polish).
332. "Leopardy 2 A5 dotarły" (<https://gdziwojsko.wordpress.com/2014/05/16/leopardy-2-a5-dotarly/>). 16 May 2014. Archived (<https://web.archive.org/web/20140904102927/http://gdziwojsko.wordpress.com/2014/05/16/leopardy-2-a5-dotarly/>) from the original on 4 September 2014. Retrieved 15 November 2014.
333. "Premiera Leoparda 2PL" (http://www.altair.com.pl/e-report/view?article_id=820). Archived (https://web.archive.org/web/20161223062802/http://www.altair.com.pl/e-report/view?article_id=820) from the original on 23 December 2016.
334. "UM DIA NO GRUPO DE CARROS DE COMBATE | Operacional" (<http://www.operacional.pt/um-dia-no-grupo-de-carros-de-combate/>) (in European Portuguese). Retrieved 26 April 2020.
335. "Exército pressiona ministro" (<https://www.dn.pt/portugal/exercito-pressiona-ministro-1551191.html>). *www.dn.pt* (in European Portuguese). 23 April 2010. Retrieved 14 November 2022.
336. "Militares com quase cinco mil milhões para armamento em 12 anos" (<https://www.dn.pt/edicao-do-dia/21-set-2018/militares-com-quase-cinco-mil-milhoes-para-armamento-em-12-anos-9881154.html>). *www.dn.pt* (in European Portuguese). 21 September 2018. Retrieved 14 November 2022.
337. "Portugal considers adding active protection systems to tanks" (<https://www.janes.com/defence-news/news-detail/portugal-considers-adding-active-protection-systems-to-tanks>). *Janes.com*. Retrieved 14 November 2022.
338. Qatar Orders 24 PzH 2000 Self-Propelled Howitzers and 62 Leopard 2 A7+ Main Battle Tanks (http://www.deagel.com/news/Qatar-Orders-24-PzH-2000-Self-Propelled-Howitzers-and-62-Leopard-2-A7-Main-Battle-Tanks_n000011422.aspx) Archived (https://web.archive.org/web/20130502234153/http://www.deagel.com/news/Qatar-Orders-24-PzH-2000-Self-Propelled-Howitzers-and-62-Leopard-2-A7-Main-Battle-Tanks_n000011422.aspx) 2 May 2013 at the Wayback Machine – Deagel.com, 18 April 2013.
339. Qatar; Army orders Leopard 2 PzH 2000 from Germany (http://dmilt.com/index.php?option=com_content&view=article&id=6338:qatar-army-orders-leopard-2-pzh-2000-from-germany&catid=3:asia&Itemid=56) Archived (https://web.archive.org/web/20131103110744/http://dmilt.com/index.php?option=com_content&view=article&id=6338%3Aqatar-army-orders-leopard-2-pzh-2000-from-germany&catid=3%3Aasia&Itemid=56) 3 November 2013 at the Wayback Machine – Dmilt.com, 22 April 2013.
340. Qatar; Rheinmetall awarded \$635 Million for Leo-2A7, Pz-2000 support contract (http://dmilt.com/index.php?option=com_content&view=article&id=7090:qatar-rheinmetall-awarded-635-million-for-leo-2a7-pz-2000-support-contract&catid=3:asia&Itemid=56) Archived (https://web.archive.org/web/20131103110937/http://dmilt.com/index.php?option=com_content&view=article&id=7090%3Aqatar-rheinmetall-awarded-635-million-for-leo-2a7-pz-2000-support-contract&catid=3%3Aasia&Itemid=56) 3 November 2013 at the Wayback Machine – Dmilt.com, 20 June 2013.
341. "Qatar parades new equipment" (<https://www.janes.com/article/56846/qatar-parades-new-equipment>). Archived (<https://web.archive.org/web/20160107040959/http://www.janes.com/article/56846/qatar-parades-new-equipment>) from the original on 7 January 2016. Retrieved 27 December 2015.
342. "German documents reveal Singapore received more Leopard 2 tanks" (<https://www.defensenews.com/land/2019/02/21/german-documents-reveal-singapore-received-more-leopard-2-tanks/>). 25 February 2019.

343. "DefenseNews.com: Singapore to buy refurbished Leopard tanks from Germany" (<https://archive.today/20121208145029/http://www.defensenews.com/story.php?F=2416170&C=asiapa>). Archived from the original (<http://www.defensenews.com/story.php?F=2416170&C=asiapa>) on 8 December 2012. Retrieved 18 February 2007.
344. "Trade Registers" (https://armstrade.sipri.org/armstrade/page/trade_register.php). *armstrade.sipri.org*. Retrieved 6 August 2021.
345. "Reply to Media Queries on SAF's Acquisition of Leopard 2A7 Tanks" (https://www.mindef.gov.sg/web/portal/mindef/news-and-events/latest-releases/article-detail/2018/march/20mar18_mq3/!ut/p/z1/jZBbC4JAEIV_Uezsell9VAAtTvGVqti-yBKmQl0R66Ne3RdCDpM7TDHznzMxB_DOWltfxRI3ysu5bfXhxmakHDrbUHmQShkmAwwsiTxFDNwUqqi0wcID7KNZSBe6MeKAHRdSyAiYFPE1ujhTxmwTj8DsHl7d2mBSIAMvuWXiPV8rDZ1e-1QTgBrKG_4cKneg2iwVjR3SdzDZh1j-gV-kWU2FUDkqK5-xLCTJsA006Wv-iZN86dn6k7tvADYrPGr/dz/d5/L2dBISEvZ0FBI_S9nQSEh/). *www.mindef.gov.sg*. Retrieved 21 July 2022.
346. "Bergepanzer 3 Buffel Armored Recovery Vehicle | Military-Today.com" (<http://www.military-today.com/engineering/buffel.htm>). *www.military-today.com*. Retrieved 4 September 2022.
347. "Leopard 2 Armoured Engineer Vehicle" ([https://aoh22.sg/our-army-platforms/leopard-2-armoured-engineer-vehicle-\(l2aev\)](https://aoh22.sg/our-army-platforms/leopard-2-armoured-engineer-vehicle-(l2aev))). *Army Open House 2022*. Retrieved 3 June 2022.
348. "SIPRI arms transfer database" (http://armstrade.sipri.org/armstrade/page/trade_register.php). Stockholm International Peace Research Institute. 6 November 2013. Retrieved 1 May 2016.
349. "Armáda získa nemecké tanky Leopard, Naď urobil na papieri lepší obchod než Česi - Dennikn.sk" (<https://dennikn.sk/2980924/armada-ziska-nemecke-tanky-leopard-nad-urobil-na-papieri-lepsi-obchod-nez-cesi/>). *www.dennikn.sk*. 23 August 2022.
350. Sprenger, Sebastian (23 August 2022). "Germany, Slovakia sign tank swap deal to arm Ukraine" (<https://www.defensenews.com/global/europe/2022/08/23/germany-slovakia-sign-tank-swap-deal-to-arm-ukraine/>). *Defense News*.
351. "Slovensko si prevzalo prvý z 15 tankov Leopard od Nemecka - Sme.sk" (<https://www.sme.sk/minuta/23098350/slovensko-si-prevzalo-prvy-z-15-tankov-leopard-od-nemecka>). 19 December 2022.
352. "Slovenská armáda si prevzala tretí tank Leopard" (<https://dennikn.sk/minuta/3405862/>). *Denník N* (in Slovak). 2 June 2023. Retrieved 2 June 2023.
353. "Spain Calculated How Many Leopard 2 Available And What Could Be Transferred to Ukraine | Defense Express" (https://en.defence-ua.com/events/spain_calculated_how_many_leopard_2_available_and_what_could_be_transferred_to_ukraine-5521.html). *en.defence-ua.com*. Retrieved 2 March 2023.
354. "EPE.com: España reforzará la letalidad de sus tanques Leopard antes de enviarlos a Ucrania" (<https://www.epe.es/es/internacional/20230205/espana-reforzara-letalidad-tanques-leopard-guerra-ucrania-82519615>) (in Spanish). 5 February 2023. Retrieved 23 February 2023.
355. "Los tanques Leopard que España mandará a Ucrania serán hasta 40 de los viejos 2A4" (<https://www.epe.es/es/internacional/20230125/tanques-leopard-espana-mandara-ucrania-viejos-81973890>). *El Periódico de España* (in Spanish). 25 January 2023.
356. "Spain repairs Leopard 2 tanks for Ukraine" (<https://mil.in.ua/en/news/spain-repairs-leopard-2-tanks-for-ukraine/>). *Militarnyi*. Retrieved 2 March 2023.
357. "Six Spanish Leopard tanks to leave to Ukraine in second half of April" (<https://apnews.com/article/ukraine-russia-war-security-aid-billions-e02b957dc15fcb1004bf54a2232c1f5e>). *Reuters*. 5 April 2023. Retrieved 13 April 2023.
358. "Spain Finalizes Buy of 108 Leopard 2A4 Tanks" (<http://www.defenseindustrydaily.com/spain-finalizes-buy-of-108-leopard-2A4-tanks-01155/>). *Defense Industry Daily*. Retrieved 2 March 2023.

359. "Stridsvagn Leopard 2" (<https://web.archive.org/web/20140410111225/http://www.fmv.se/sv/Projekt/Stridsvagn-121-och-122-Leopard-2/>) (in Swedish). Archived from the original (<http://www.fmv.se/sv/Projekt/Stridsvagn-121-och-122-Leopard-2/>) on 10 April 2014. Retrieved 22 September 2014. "Idag har de 160 äldre varianterna av Leopard 2 – Stridsvagn 121 – fasats ut och leasingkontraktet med Tyskland har lösts under 2011."
360. "Strv 121 / Strv 122" (https://web.archive.org/web/20171201200349/http://www.ointres.se/strv_121_och_strv_122.htm#RENO_Strv_122). *ointres.se*. Archived from the original (http://ointres.se/strv_121_och_strv_122.htm#RENO_Strv_122) on 1 December 2017. Retrieved 24 November 2017.
361. "Strv 122 Tank" (<http://fighting-vehicles.com/strv-122-tank/>). *fighting-vehicles.com*. Archived (<https://web.archive.org/web/20171127213438/http://fighting-vehicles.com/strv-122-tank/>) from the original on 27 November 2017. Retrieved 25 November 2017. "42 Strv 122 remain in active service."
362. "Sale of surplus Leopard 2 Battle Tanks to the German manufacturer" (<https://www.admin.ch/gov/en/start/documentation/media-releases.msg-id-36295.html>). *The Federal Council*. Retrieved 2 March 2023.
363. "Acquisitions – L'armée devrait recevoir 771 millions pour ses achats" (<https://www.24heures.ch/l-armee-devrait-recevoir-771-millions-pour-ses-achats-120449868912>). *24 heures* (in French). 4 June 2014. Retrieved 2 March 2023.
364. "Germany seeks to buy mothballed Swiss Leopard 2 tanks" (<https://apnews.com/article/germany-switzerland-leopard-tanks-nato-ukraine-ee66fd4d5ae07c12cd0658fefcc60311>). *Associated Press*. 3 March 2023. Retrieved 3 March 2023.
365. "Rheinmetall-Chef Papperger über das geplante Panzer-Werk in der Ukraine: "Wir müssen sie befähigen, sich selbst zu verteidigen" " (<https://www.nzz.ch/wirtschaft/rheinmetall-chef-papperger-ueber-das-geplante-panzer-werk-in-der-ukraine-wir-muessen-sie-befaeihigen-sich-selbst-zu-verteidigen-ld.1729818?reduced=true>). *Neue Zürcher Zeitung* (in German). 15 March 2023. Retrieved 16 March 2023.
366. "Swiss commission recommends selling 25 tanks back to Germany" (<https://www.swissinfo.ch/eng/politics/swiss-commission-recommends-selling-25-tanks-back-to-germany/48401162>). *Swiss Info*. 29 March 2023.
367. "Le Conseil federal adopte le programme d'armement 2001" (https://www.admin.ch/cp/f/3b41a8a5_1@fwsrv.g.bfi.admin.ch.html). *www.admin.ch*. Retrieved 2 March 2023.
368. Swiss Government (24 May 2006). "Message regarding the military acquisition material 2006" (<https://www.news.admin.ch/newsd/message/attachments/2812.pdf>) (PDF). *www.admin.ch*. Retrieved 2 March 2023.
369. "Attack On Europe: Documenting Ukrainian Equipment Losses During The Russian Invasion Of Ukraine" (<https://www.oryxspioenkop.com/2022/02/attack-on-europe-documenting-ukrainian.html>). *Oryx*. Retrieved 29 June 2023.
370. "The Russians Just Destroyed Their First Ukrainian Leopard 2 Tank" (<https://www.forbes.com/sites/davidaxe/2023/06/08/the-russians-just-destroyed-their-first-ukrainian-leopard-2-tank/?sh=7e7a57f875c9>). *Forbes*.
371. "Poland delivers first Leopard tanks to Ukraine – DW – 02/24/2023" (<https://www.dw.com/en/poland-delivers-first-leopard-tanks-to-ukraine/a-64809432>). *dw.com*. DW. 24 February 2023. Retrieved 24 February 2023.
372. Norwegian Armed Forces/Forsvaret [@Forsvaret_no] (20 March 2023). "Ukraine has now received the Norwegian tanks #leopard 2 and training of Ukrainian personnel is underway" (https://twitter.com/Forsvaret_no/status/1637798905109676032) (Tweet). Retrieved 20 March 2023 – via *Twitter*. "#Ukraine has now received the Norwegian tanks #leopard 2 and training of Ukrainian personnel is underway. See photos from the transportation. #StandWithUkraine."

373. "Robles asiste a la reunión del Grupo de Contacto para la Defensa de Ucrania y a la reunión ministerial de la OTAN" (<https://www.defensa.gob.es/gabinete/notasPrensa/2023/06/DGC-230615-ministerial-bruselas.html>). *Ministerio de Defensa* (in Spanish). 15 June 2023.
374. "Germany to provide Ukraine with 250 armored vehicles, including 100 Marder vehicles and 150 Leopard tanks" (<https://www.riotimesonline.com/brazil-news/new-multipolar-world-order/germany-to-provide-ukraine-with-250-armored-vehicles-including-100-marder-vehicles-and-150-leopard-tanks/>). *Rio Times*. 6 March 2023. Retrieved 7 March 2023.
375. "Rheinmetall will Panzerfabrik in der Ukraine aufbauen" (<https://www.spiegel.de/wirtschaft/unternehmen/fuer-panther-produktion-rheinmetall-will-panzer-fabrik-in-der-ukraine-aufbauen-a-8c1a27cf-9be8-42ad-ba51-d0b526c05719>). *Der Spiegel* (in German). 4 March 2023. Retrieved 7 March 2023.
376. "Rheinmetall could deliver 139 Leopard tanks to Ukraine - RND" (<https://www.reuters.com/world/europe/rheinmetall-could-deliver-139-leopard-tanks-ukraine-rnd-2023-01-23/>). *Reuters*. 23 January 2023.
377. Verteidigungsministerium [@Verteidigungsministerium] (24 February 2023). "Germany to provide 4 additional Leopard 2A6" (<https://twitter.com/Verteidigungsministerium/status/1629136976505892865>) (Tweet). Retrieved 24 February 2023 – via [Twitter](#).
"Verteidigungsminister Pistorius hat entschieden, weitere vier #Kampfpanzer #Leopard2 A6 aus Beständen der Bundeswehr an die #Ukraine abzugeben. Damit erhöht  die Anzahl der gelieferten Panzer von 14 auf 18 & stellt zusammen mit  &  insgesamt ein  Bataillon zur Verfügung."
378. "Deutsche Leopard-Panzer in der Ukraine angekommen" (<https://www.spiegel.de/politik/deutschland/deutsche-leopard-panzer-in-der-ukraine-angekommen-a-a275a157-d33a-4df0-a43e-bd57a0a7b6f9>). *Der Spiegel* (in German). 27 March 2023.
379. Conta Oficial do Gabinete da Ministra da Defesa Nacional [@defesa_pt] (27 March 2023). "Apoio militar à Ucrânia" (https://twitter.com/defesa_pt/status/1640416494235406338) (Tweet) (in Portuguese) – via [Twitter](#). "Já estão na #Ucrânia  os 3 carros de combate #Leopard 2A6 doados por #Portugal , entregues conforme definido com os nossos parceiros. Portugal continua a apoiar a resistência ucraniana contra a ilegal invasão da Rússia .
380. "Pentagon leaks: Greece could be sending Leopard 2A6HELs tanks to Ukraine" (<https://greekcitytimes.com/2023/04/26/pentagon-leaks-greece-leopard/>). *Greek City Times*. 26 April 2023.
381. Anita Anand [@AnitaAnandMP] (18 March 2023). "More Canadian military aid is en route to Ukraine" (<https://twitter.com/AnitaAnandMP/status/1637109663413026817>) (Tweet). Retrieved 20 March 2023 – via [Twitter](#). "More Canadian military aid is en route to Ukraine. We will #StandWithUkraine for as long as it takes."
382. "Military support for Ukraine" (<https://www.bundesregierung.de/breg-en/news/military-support-ukraine-2054992>). Press of the German Federal Government. 6 April 2023. Retrieved 6 April 2023.
383. "Spain, Finland to send Leopard tanks to Ukraine" (<https://www.politico.eu/article/finland-helsinki-russia-ukraine-war-to-send-three-leopard-2-tanks-to-ukraine/>). 23 February 2023.
384. "Suomi luovuttaa Ukrainalle kolme Leopard-vaunua lisää – Kaikkonen: Ukraina pyytänyt keskusteluja Hornetin luovuttamisesta" (<https://www.mtvuutiset.fi/artikkeli/kaikkonen-suomi-luovuttaa-ukrainalle-kolme-leopard-panssarivaunua-lisaa/8660930>). *mtvuutiset.fi* (in Finnish). 23 March 2023.
385. Axe, David. "The Ukrainian Army Has Already Lost Half Of Its Unique Leopard 2R Breaching Vehicles" (<https://www.forbes.com/sites/davidaxe/2023/06/11/the-ukrainian-army-has-already-lost-half-of-its-unique-leopard-2r-breaching-vehicles/>). *Forbes*. Retrieved 13 June 2023.
386. "WISENT 2 Armoured Support Vehicle" (<https://www.army-technology.com/projects/wisent-2-armoured-support-vehicle/>). 11 April 2016.

387. *Difesa, In arrivo 250 carri armati Leopard 2A7?* (<https://agenparl.eu/2023/03/03/difesa-in-arrivo-250-carri-armati-leopard-2a7/>) (in Italian), Agenparl, 3 March 2023
388. *Italy plans to buy Leopard 2 tanks to boost ground forces* (<https://www.reuters.com/business/aerospace-defense/italy-plans-buy-leopard-2-tanks-boost-ground-forces-2023-07-13/>), Reuters, 13 July 2023
389. *Cosa sono i carri armati Leopard 2 che l'Italia comprerà* (<https://www.wired.it/article/leopard-2-a8-carri-armati-italia-2024/>) (in Italian), Wired Italia, 13 July 2023
390. "Lithuania sends Germany letter of intent on Leopard tank acquisition" (<https://www.lrt.lt/en/news-in-english/19/2046014/lithuania-sends-germany-letter-of-intent-on-leopard-tank-acquisition>). LRT. 29 July 2023. Retrieved 29 July 2023.
391. Adamowski, Jaroslaw (15 March 2023). "Lithuania wants 54 new tanks in tweak to land forces layout" (<https://www.defensenews.com/global/europe/2023/03/15/lithuania-wants-54-new-tanks-in-tweak-to-land-forces-layout/>). *Defense News*. Retrieved 25 July 2023.
392. ES&T Redaktion (25 July 2023). "Leopard 2 – Litauen plant Beschaffung deutscher Kampfpanzer" (<https://esut.de/2023/07/meldungen/43458/leopard-2-litauen-plant-beschaffung-g-deutscher-kampfpanzer/>). *esut.de* (in German). Retrieved 25 July 2023.
393. "2,2 милиарда евро даваме за танкове и самолети "Made in USA" " (<https://web.archive.org/web/20150626114207/http://7plus7.net/2015/02/12/22-%D0%BC%D0%B8%D0%BB%D0%B8%D0%B0%D1%80%D0%B4%D0%B0-%D0%B5%D0%B2%D1%80%D0%BE-%D0%B4%D0%B0%D0%B2%D0%B0%D0%BC%D0%B5-%D0%B7%D0%B0-%D1%82%D0%B0%D0%BD%D0%BA%D0%BE%D0%B2%D0%B5-%D0%B8-%D1%81%D0%B0%D0%BC/>) [€ 2.2 billion for tanks "Made in USA"] (in Bulgarian). 7+7. 14 June 2015. Archived from the original (<http://7plus7.net/2015/02/12/22-%D0%BC%D0%B8%D0%BB%D0%B8%D0%B0%D1%80%D0%B4%D0%B0-%D0%B5%D0%B2%D1%80%D0%BE-%D0%B4%D0%B0%D0%B2%D0%B0%D0%BC%D0%B5-%D0%B7%D0%B0-%D1%82%D0%B0%D0%BD%D0%BA%D0%BE%D0%B2%D0%B5-%D0%B8-%D1%81%D0%B0%D0%BC/>) on 26 June 2015. Retrieved 14 June 2015.
394. *Signing of agreement on procurement of German PZH 2000 howitzers* (<https://web.archive.org/web/20180629235749/https://www.morh.hr/en/news/press-releases/10704-signing-of-agreement-on-procurement-of-german-pzh-2000-howitzers.html>) (press release), MORH, archived from the original (<https://www.morh.hr/en/news/press-releases/10704-signing-of-agreement-on-procurement-of-german-pzh-2000-howitzers.html>) on 29 June 2018, retrieved 6 March 2017
395. *Armata Română negociază achiziționarea de tancuri Leopard 2* (<https://www.digi24.ro/stiri/actualitate/armata-romana-negociaza-achizitionarea-de-tancuri-leopard-1158807>), digi24.ro, 12 July 2019
396. *Romania Lays Out Intent to Buy M1 Abrams MBTs* (<https://euro-sd.com/2023/03/news/29967/29967/>), European Security and Defense, 8 March 2023
397. "Milliardengeschäft: Deutsche Kampfpanzer nach Saudi-Arabien?" (http://www.focus.de/politik/deutschland/milliardengeschaeft-deutsche-kampfpanzer-nach-saudi-arabien_aid_642586.html) [Billionaire affair: German combat tank for Saudi Arabia?]. *Focus* (in German). 3 July 2011. Archived (https://web.archive.org/web/20120124031549/http://www.focus.de/politik/deutschland/milliardengeschaeft-deutsche-kampfpanzer-nach-saudi-arabien_aid_642586.html) from the original on 24 January 2012. Retrieved 5 July 2012.
398. "Germany Wants to Supply Battle Tanks to Saudi Arabia" (<http://www.spiegel.de/international/germany/growth-in-arms-exports-germany-wants-to-supply-battle-tanks-to-saudi-arabia-a-772177.html>). *Der Spiegel*. 4 July 2011. Archived (<https://web.archive.org/web/20120904060256/http://www.spiegel.de/international/germany/growth-in-arms-exports-germany-wants-to-supply-battle-tanks-to-saudi-arabia-a-772177.html>) from the original on 4 September 2012. Retrieved 5 July 2012.

399. Kundnani, Hans (9 July 2011). "Germany's contribution to the Arab spring: arms sales" (<http://www.theguardian.com/commentisfree/2011/jul/09/germany-arms-sale-saudi-arabia>). *The Guardian*. London. Archived (<https://web.archive.org/web/20160927232041/https://www.theguardian.com/commentisfree/2011/jul/09/germany-arms-sale-saudi-arabia>) from the original on 27 September 2016. Retrieved 12 December 2016.
400. "Tank Exports to Saudi Arabian Signal German Policy Shift" (<http://www.spiegel.de/international/world/the-merkel-doctrine-tank-exports-to-saudi-arabian-signal-german-policy-shift-a-791380.html>). *Der Spiegel*. 14 October 2011. Archived (<https://web.archive.org/web/20120626233408/http://www.spiegel.de/international/world/the-merkel-doctrine-tank-exports-to-saudi-arabian-signal-german-policy-shift-a-791380.html>) from the original on 26 June 2012. Retrieved 5 July 2012.
401. "KMW-Miteigentümer gegen Panzerausfuhr nach Saudi-Arabien" (<https://www.tagesschau.de/inland/ruestungsexport100.html>) [KMW co-owner against tank export to Saudi Arabia]. *Tagesschau.de* (in German). 20 June 2012. Archived (<https://web.archive.org/web/20131102152457/http://www.tagesschau.de/inland/ruestungsexport100.html>) from the original on 2 November 2013. Retrieved 2 September 2017.
402. "Desert Leopards: Germany Selling Heavy Armor to the Saudis?" (<http://www.defenseindustrydaily.com/Desert-Leopards-Germany-Selling-Heavy-Armor-to-the-Saudis-06993>). *Defense Industry Daily*. 26 June 2012. Archived (<https://web.archive.org/web/20120701023811/http://www.defenseindustrydaily.com/Desert-Leopards-Germany-Selling-Heavy-Armor-to-the-Saudis-06993>) from the original on 1 July 2012. Retrieved 5 July 2012.
403. "German tank sale to Saudi Arabia likely to be cancelled: newspaper" (<https://www.reuters.com/article/us-germany-saudi-arms-idUSBREA3C07E20140413>) Archived (<https://web.archive.org/web/20150924195805/http://www.reuters.com/article/2014/04/13/us-germany-saudi-arms-idUSBREA3C07E20140413>) 24 September 2015 at the Wayback Machine, *Reuters*, 13 April 2014.
404. "Leopardy nie dla Arabii Saudyjskiej? Niemcy torpedują doktrynę Merkel – Defence24" (<http://www.defence24.pl/leopardy-nie-dla-arabii-saudyjskiej-niemcy-torpeduja-doktryne-merkel>). 26 March 2015.
405. "السعودية "غير" مهتمة بشراء دبابت ليوبارد الألمانية | DW | 31 May 2016" (<https://www.dw.com/a/r/%D8%A7%D9%84%D8%B3%D8%B9%D9%88%D8%AF%D9%8A%D8%A9-%D8%BA%D9%8A%D8%B1-%D9%85%D9%87%D8%AA%D9%85%D8%A9-%D8%A8%D8%B4%D8%B1%D8%A7%D8%A1-%D8%AF%D8%A8%D8%A7%D8%A8%D8%A7%D8%AA-%D9%84%D9%8A%D9%88%D8%A8%D8%A7%D8%B1%D8%AF-%D8%A7%D9%84%D8%A3%D9%84%D9%85%D8%A7%D9%86%D9%8A%D8%A9/a-19295253>) [Saudi Arabia is "not" interested in buying German Leopard tanks]. *DW.COM* (in Arabic). Retrieved 12 April 2023.
406. Hans, Hans (31 May 2016). "Rüstungsexporte aus Deutschland: Saudi-Arabien will keine Leopard-Panzer mehr" (<https://www.tagesspiegel.de/politik/saudi-arabien-will-keine-leopard-panzer-mehr-3726445.html>). *Tagesspiegel Online* (in German). ISSN 1865-2263 (<https://www.worldcat.org/issn/1865-2263>). Retrieved 12 April 2023.

Bibliography

- Jerchel, Michael; Schnellbacher, Uwe (1998). *Leopard 2 Main Battle Tank 1979–1998*. London: Osprey Publishing. ISBN 978-1-85532-691-0. OCLC 40544103 (<https://www.worldcat.org/oclc/40544103>).
- Krapke, Paul W (June 2004). *Leopard 2 sein Werden und seine Leistung [Leopard 2 becoming and achievement]* (in German). Books on Demand. ISBN 978-3-83341425-1.
- Scheibert, Michael (1996). *Leopard 2 A5 Euro-Leopard* (in German). Wölfersheim-Berstadt: Pudzun-Pallas-Verlag. ISBN 3-7909-0576-3.

- Scheibert, Michael; Schneider, Wolfgang (1986). *Leopard 2 Ein Spitzenprodukt deutscher Wehrtechnik* [*Leopard 2 a top product of German military technology*] (in German). Dorheim: Pudzun-Pallas-Verlag. ISBN 3-7909-0265-9.

External links

- [Leopard 2 Revolution \(http://www.military-today.com/tanks/revolution.htm\)](http://www.military-today.com/tanks/revolution.htm)
 - [Army Vehicles.DK Leopard 2A5 \(http://www.armyvehicles.dk/leopard2a5.htm\)](http://www.armyvehicles.dk/leopard2a5.htm)
 - [Army Vehicles.DK Leopard 2A4 \(http://www.armyvehicles.dk/leopard2a4.htm\)](http://www.armyvehicles.dk/leopard2a4.htm)
 - [Military Periscope – Leopard 2 main battle tank \(https://web.archive.org/web/20060708071204/http://www.militaryperiscope.com/mdb-smpl/weapons/gcv/tanks/w0001000.shtml\)](https://web.archive.org/web/20060708071204/http://www.militaryperiscope.com/mdb-smpl/weapons/gcv/tanks/w0001000.shtml)
 - [Fprado.com – Kampfpanzer Leopard 2 \(http://www.fprado.com/armorsite/leo2.htm\)](http://www.fprado.com/armorsite/leo2.htm)
 - [Leopard 2A5 \(http://www.enemyforces.com/tanks/leopard2a5.htm\)](http://www.enemyforces.com/tanks/leopard2a5.htm) [Archived \(https://web.archive.org/web/20060430224010/http://www.enemyforces.com/tanks/leopard2a5.htm\)](https://web.archive.org/web/20060430224010/http://www.enemyforces.com/tanks/leopard2a5.htm) 30 April 2006 at the [Wayback Machine](#)
 - [A Finnish Army Leopard 2A4 firing from a hull-down position \(https://web.archive.org/web/20060625181452/http://www.edu.lahti.fi/~aroinila/leo/asemaan2.wmv\)](https://web.archive.org/web/20060625181452/http://www.edu.lahti.fi/~aroinila/leo/asemaan2.wmv)
 - [A Finnish Leopard 2A4 firing, videoed from inside the turret \(https://web.archive.org/web/20060625181443/http://www.edu.lahti.fi/~aroinila/leo/loader.wmv\)](https://web.archive.org/web/20060625181443/http://www.edu.lahti.fi/~aroinila/leo/loader.wmv)
 - [Danish Leopard 2A5 \(https://web.archive.org/web/20070823184305/http://www.danskpanser.dk/Baeltekoretojer/Leopard2A5.htm\)](https://web.archive.org/web/20070823184305/http://www.danskpanser.dk/Baeltekoretojer/Leopard2A5.htm)
 - [Singapore Leopard 2A4 MBT \(http://www.mindef.gov.sg/weapons/leopard\)](http://www.mindef.gov.sg/weapons/leopard)
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